SR-HC series

Industrial Robot

Robot controller TS3000
Robot controller TS3000E
Robot controller TS3100
Robot controller TS3100E

INSTRUCTION MANUAL

WAFER TRANSFER ROBOT

TRANSPORTATION AND INSTALLATION MANUAL

Notice

- Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
- Before operating the industrial robot, read through and completely understand this manual.
- After reading through this manual, keep it nearby for future reference.

Rev.4
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Shibaura Machine

This Instruction Manual applies to the following robots:

SR Series: SR-404HC, SR-405HC-L1550-R/L, SR-405HC-L1530S-R/L, SR-624HC

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Preface

This manual describes the basic specifications of the industrial robot and controller, and how to unpack and install them. Specifically, it describes how to unpack the shipment containing the equipment, how to install the equipment, how to connect wiring and air piping, and how to attach tools. Be sure to look through this manual before unpacking the shipment.
Before beginning the work according to this manual, read through the Safety Manual so that you can understand the safety measures.

This manual is divided into the following five parts:

Section 1	Specifications This section describes the basic specifications and names of respective units of the robot and controller.
Section 2-3	Transportation This section describes how to remove the robot and controller from their boxes and how to transport them to the installation site. This section also discusses how to temporarily store the equipment after unpacking the shipment.
Section 4-7	Installation This section discusses the equipment installation environment, space requirements, and how to install the equipment.
Section 8-9	System Connections This section describes how to connect the robot, controller and peripheral equipment.
Section 10	Hand Interface This section describes how to connect the tool to the robot arm and how to connect pipes and wires to the tool.

Precautions on Safety

Important information on the robot and controller is noted in the instruction manual to prevent injury to the user and persons nearby, prevent damage to assets and to ensure correct use. Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
 **DANGER**	This means that "incorrect handling will lead to fatalities or major injuries".
 **WARNING**	This means that "incorrect handling will lead to fatalities or serious injuries."
 **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)".

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long term treatment.

*2) Physical damage refers to major fires due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). The details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). The details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger and caution. The details are indicated with pictures or words in or near the symbol.

CAUTION

- Always read through the Safety Manual provided separately before starting actual work to ensure safety work covering from the robot installation to operation.

[Installation and transportation]

Always observe the following items to safely use the robot.

DANGER	
 Prohibited	• DO NOT install or operate if any parts are damaged or missing. Doing so could lead to electric shocks, fires or faults. • DO NOT install the robot where it may be subject to fluids such as water. Doing so could lead to electric shocks, fires or faults. • Do not place the robot near combustible matters. Doing so could lead to fires if the matter ignites due to a fault, etc.
 Mandatory	• Always secure the robot with the attached clamps before transporting it. Failure to do so could lead to injuries if the arm moves when the robot is suspended. • Wire the robot after installation. Wiring the robot before installation could lead to electric shocks or injuries. • Always use the power voltage and power capacity designated by SHIBAURA MACHINE. Failure to do so could lead to device faults or fires. • Always use the designated power cable. Using a cable other than that designated could lead to fires or faults.
 Always ground	• Completely connect the grounding cable. Failure to do so could lead to electric shocks or fires if a fault or fault current occurs. Noise could lead to malfunction. Also, it could cause misoperation by noise.

 **CAUTION**	
 Prohibited	• NEVER lift the robot by the arm 2 cover. Doing so will apply an excessive force on the robot's mechanism section and could lead to faults. • For the controller, secure the ample space for air vent. Heating of controller could lead to malfunction.
 Mandatory	• When lifting the robot, lift it up slowly. The robot will tilt slightly, so lifting it up suddenly could be hazardous. • When storing the robot, secure it to the base. The robot will be unstable if just set down, and it could tilt over.

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1. Specifications

1.1 Correspondence between the Robots and the Robot Controllers

Table 1.1 shows the correspondence between the robots and the robot controllers.

Table 1.1 Correspondence between the Robots and the Robot Controllers

		Robot	
		SR-404HC SR-624HC	SR-405HC-L1550-R/L SR-405HC-L1530S-R/L
Robot controller	TS3000	○	
	TS3000E	○	
	TS3100		○
	TS3100E		○

○: Supported

1.2 Robot Configuration Diagram(SR-404HC/SR-624HC and TS3000/TS3000E)

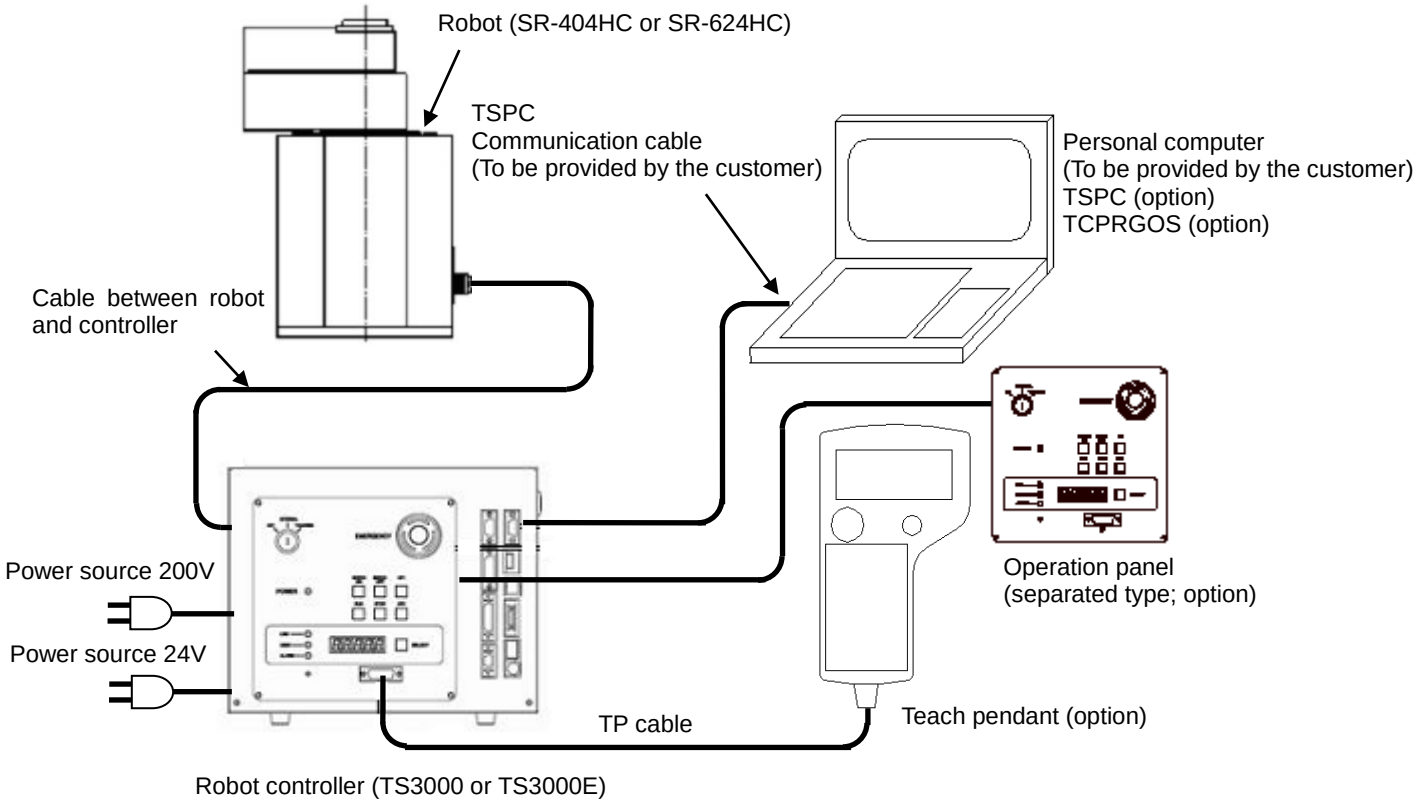


Fig. 1.1 Robot Configuration (SR-404HC/SR-624HC and TS3000/TS3000E)

For connection of the 200-volt power source, see "8.1.2 Connecting the Power Cable "ACIN"". For connection of 24-volt power source, see "8.1.7 Connecting the External Input/Output Power Cable "EMS"".

1.3 Robot Configuration Diagram(SR-405HC-L1550-R/L / SR-405HC-L1530S-R/L and TS3100 / TS3100E)

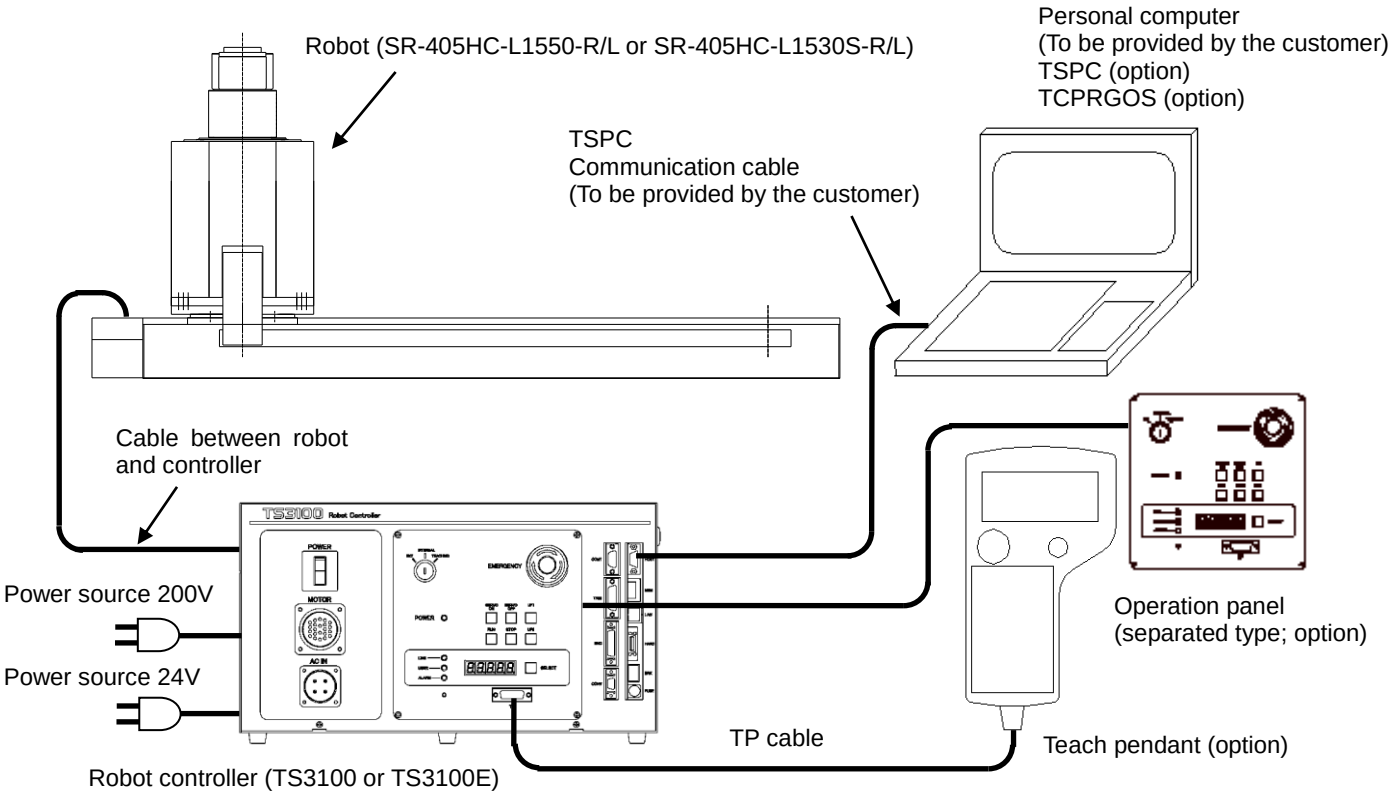


Fig. 1.2 Robot Configuration (SR-405HC-L1550-R/L / SR-405HC-L1530S-R/L and TS3100/TS3100E)

For connection of the 200-volt power source, see "9.1.2 Connecting the Power Cable "ACIN"". For connection of 24-volt power source, see "9.1.7 Connecting the External Input/Output Power Cable "EMS"".

1.4 Name of Each Part (SR-404HC and SR-624HC)

Fig. 1.3 shows the name of each part of the robot (SR-404HC and SR-624HC). Figure shows the SR-404HC.

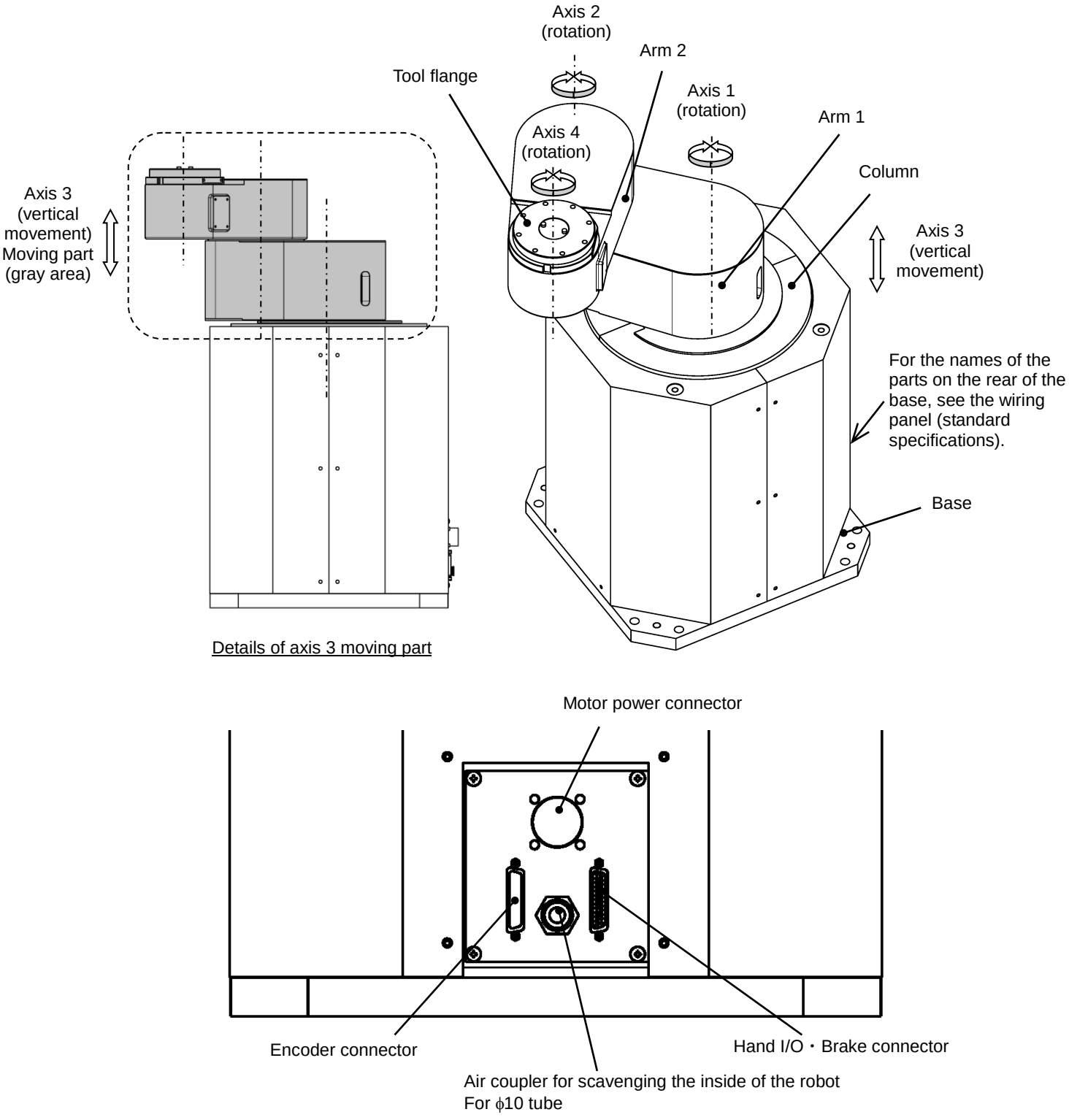


Fig. 1.3 Name of each part (SR-404HC and SR-624HC)

1.5 Name of Each Part (SR-405HC-L1550-R/L and SR-405HC-L1530S-RL)

Fig. 1.4 shows the name of each part of the robot (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L).
Figure shows the SR-405HC-L1550-R or SR-405HC-L1530S-R.

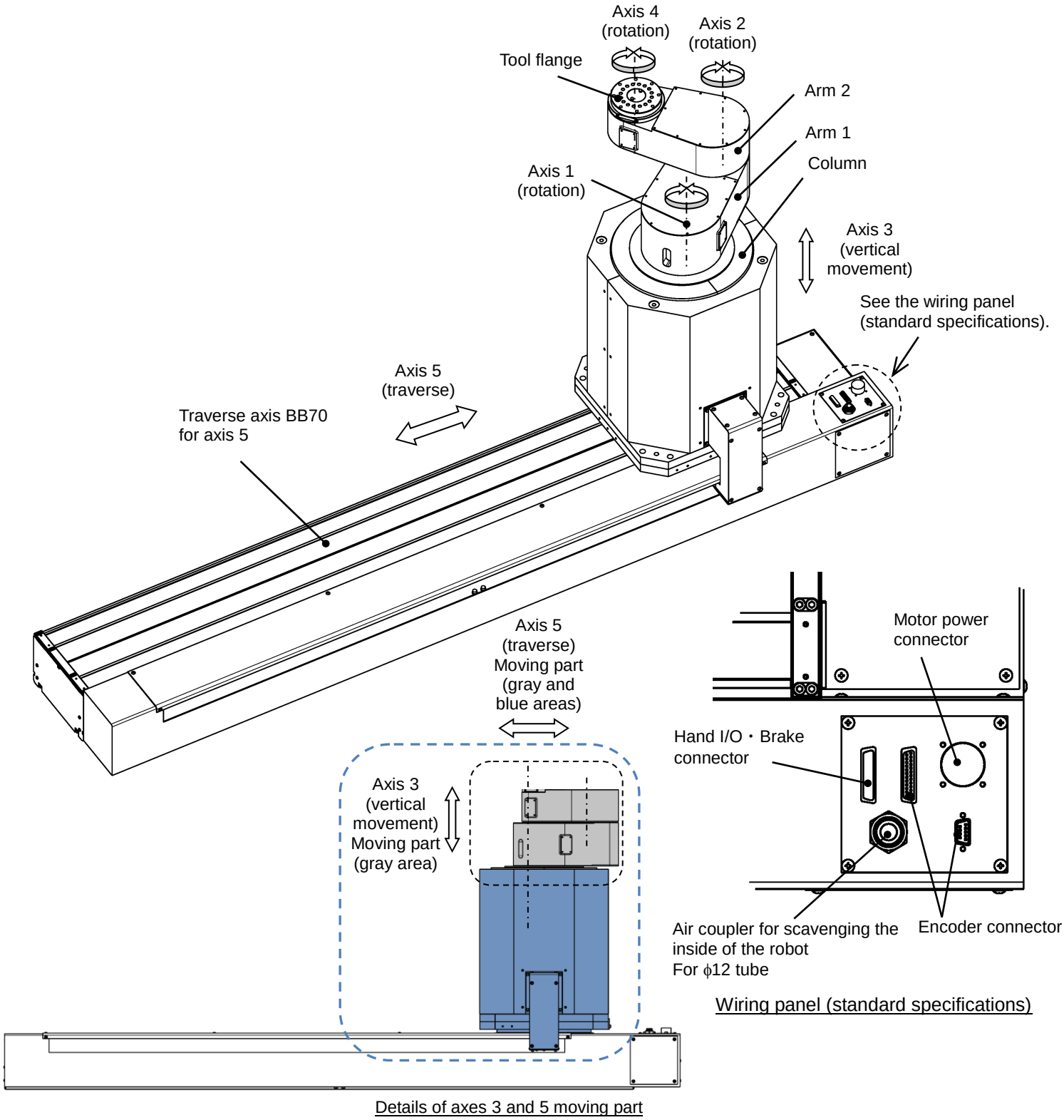


Fig. 1.4 Name of each part (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

1.6 Specifications Table (SR-404HC and SR-624HC)

Table shows specifications of the robot (SR-404HC and SR-624HC).

Item		Specifications	
Structure		Horizontal multi-articulation	
Model		SR-404HC	SR-624HC
Applicable controller		TS3000, TS3000E	
Mass of robot body		200 kg	220 kg
No. of controlled axes		4	
Arm length		400 mm (200 mm + 200 mm)	620 mm (310 mm + 310 mm)
Motor capacity / Current limit	Axis 1	600 W / 17.8Ao-p	
	Axis 2	400 W / 13.5Ao-p	
	Axis 3	400 W / 13.1Ao-p	
	Axis 4	400 W / 13.5Ao-p	
Operating range	Axis 1	±230 deg	±205 deg
	Axis 2	±160 deg	
	Axis 3	0 to 100 mm	
	Axis 4	±230 deg	
Maximum speed	Axis 1	274 deg/s	
	Axis 2	240 deg/s	
	Axis 3	200 mm/s	
	Axis 4	240 deg/s	
Maximum payload mass (*1)		20 kg	35 kg
Repeatability (*2)	X,Y	±0.05 mm	
	Z	±0.05 mm	
	C	±0.1 deg	
Drive system		By means of AC servo motors	
Position detection method		Absolute	
Paint color (*3)		Misty white	
Power supply capacity		2.6 kVA	
Degree of cleanliness (*4)		Class 100 (Equivalent to clean class 5 defined by ISO) (Particle size: 0.3 μm)	

*1: Including hand mass.
*2: The positioning repeatability of one-directional movement for 30 times of tests at a constant ambient temperature of 20°C. Absolute position accuracy is not ensured. Trajectory accuracy is not ensured.
*3: The tone may be different among production lots, but it does not affect the product quality.
*4: Degree of cleanness class 100 is defined by Fed.Std.209D
The scavenging flow rate is about 250 liters per minute. The clean room down flow current rate is 0.4m/s.

! CAUTION

- Micro vibration may occur depending on the robot posture. If micro vibration occurs, reduce the acceleration speed to use the robot.

1.7 Specifications Table (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

Table shows specifications of the robot (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L).

Item		Specifications	
Structure		Horizontal multi-articulation with traverse axis	
Model		SR-405HC-L1550-R/L	SR-405HC-L1530S-R/L
Applicable controller		TS3100、TS3100E	
Mass of robot body		290 kg	
No. of controlled axes		5	
Arm length		400 mm (200 mm + 200 mm)	
Motor capacity / Current limit	Axis 1	600 W / 17.8Ao-p	
	Axis 2	400 W / 13.5Ao-p	
	Axis 3	400 W / 13.1Ao-p	
	Axis 4	400 W / 13.5Ao-p	
	Axis 5	800 W / 19.5Ao-p	
Operating range	Axis 1	±230 deg	
	Axis 2	±160 deg	
	Axis 3	0 to 100 mm	
	Axis 4	±230 deg	
	Axis 5	0 to 1550 mm	0 to 1530 mm
Maximum speed	Axis 1	274 deg/s	
	Axis 2	240 deg/s	
	Axis 3	200 mm/s	
	Axis 4	240 deg/s	
	Axis 5	300 mm/s	180 mm/s
Maximum payload mass (*1)		20 kg	
Repeatability (*2)	X,Y	±0.05 mm	
	Z	±0.05 mm	
	C	±0.1 deg	
	T	±0.05 mm	
Drive system		By means of AC servo motors	
Position detection method		Absolute	
Paint color (*3)		Misty white	
Power supply capacity		3.8 kVA	
Degree of cleanliness (*4)		Class 100 (Equivalent to clean class 5 defined by ISO) (Particle size: 0.3 μm)	

*1: Including hand mass.
*2: The positioning repeatability of one-directional movement for 30 times of tests at a constant ambient temperature of 20°C. Absolute position accuracy is not ensured. Trajectory accuracy is not ensured.
*3: The tone may be different among production lots, but it does not affect the product quality.
*4: Degree of cleanness class 100 is defined by Fed.Std.209D
The scavenging flow rate is about 250 liters per minute. The clean room down flow current rate is 0.4m/s.



CAUTION

- Micro vibration may occur depending on the robot posture. If micro vibration occurs, reduce the acceleration speed to use the robot.

2. Transportation (SR-404HC/SR-624HC and TS3000/TS3000E)

2.1 Unpacking

The robot and controller are shipped in wooden frames and corrugated cardboards. Fig. 2.1 shows each packaging state.
Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.
After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.
For the controller accessories, see the accessory list packed with the controller.

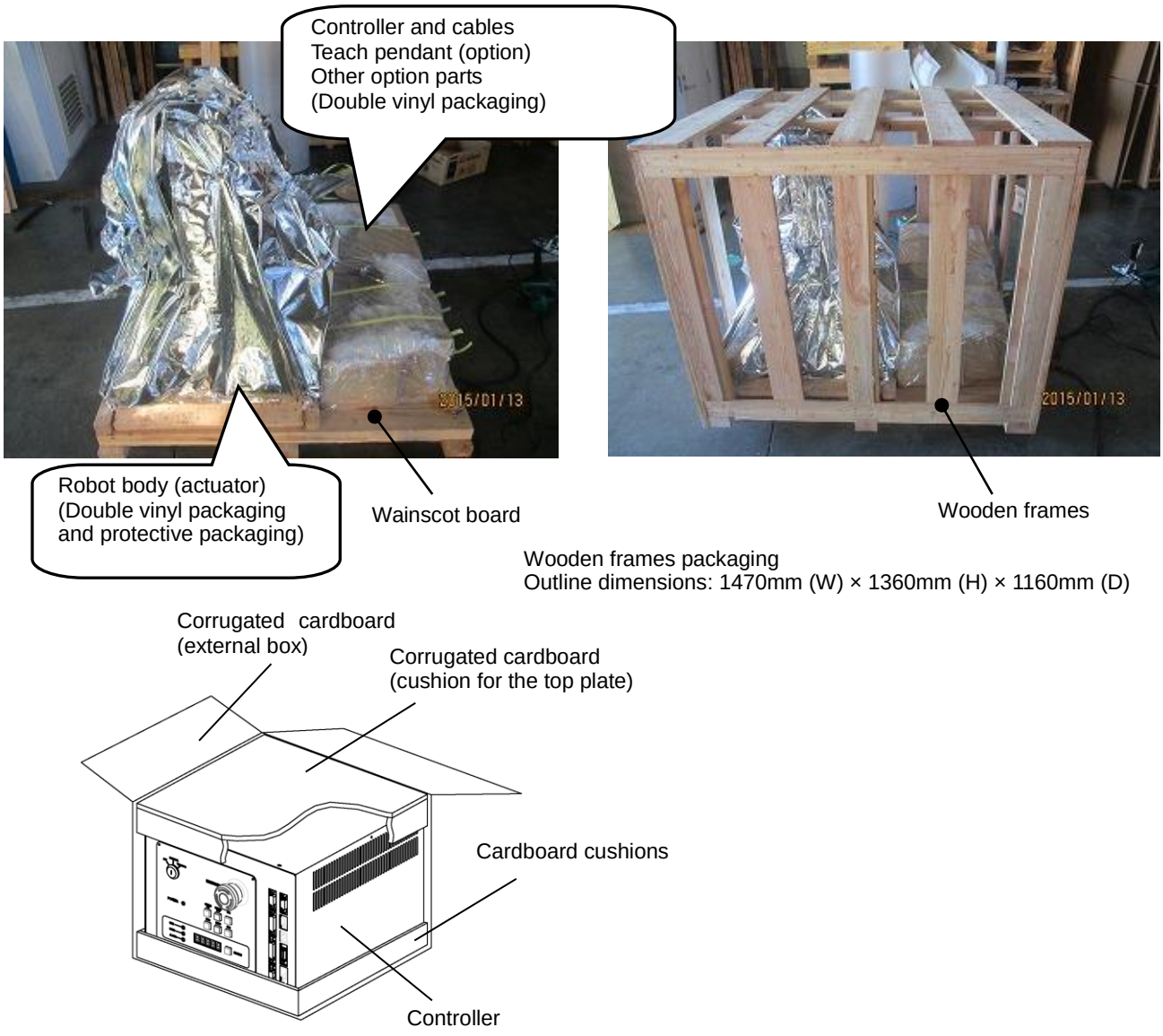


Fig. 2.1 Packaging state (SR-404HC/SR-624HC and TS3000/TS3000E)



DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden frames, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

2.2 Transportation

Move the robot and controller very carefully. Make sure that no excessive impact or vibration is exerted on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all nuts and screws completely.
If the equipment is to be moved to a location some distance from where it was unpacked, reposition the cushions as they were and put the equipment back into the corrugated cardboards.

2.2.1 Mass and Dimensions

The mass and outer dimentions of the robot are shown in Figs. 2.2 and 2.3.
For the controller mass and external dimensions, see "6.1 External Dimensions".

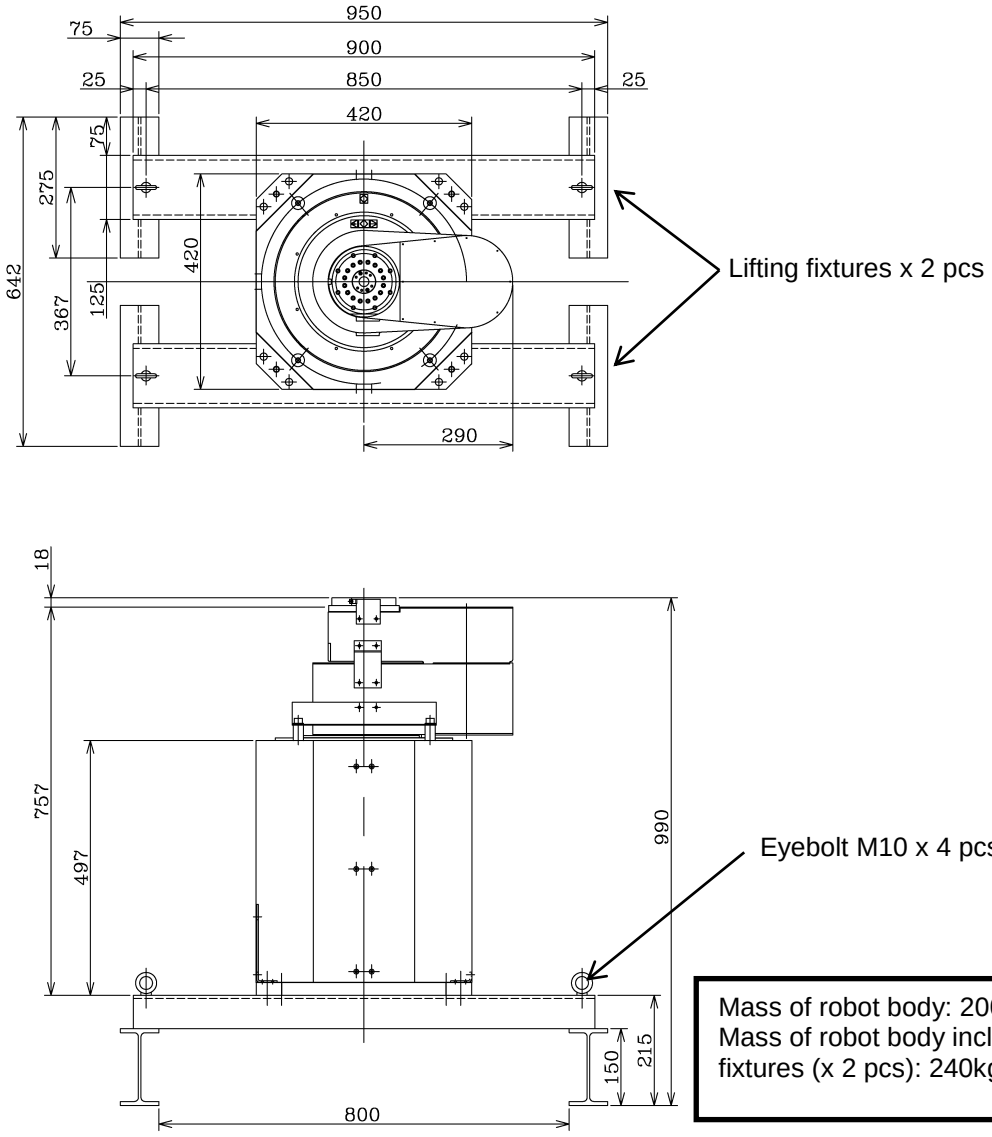


Fig. 2.2 Outer dimensions at transport (SR-404HC)

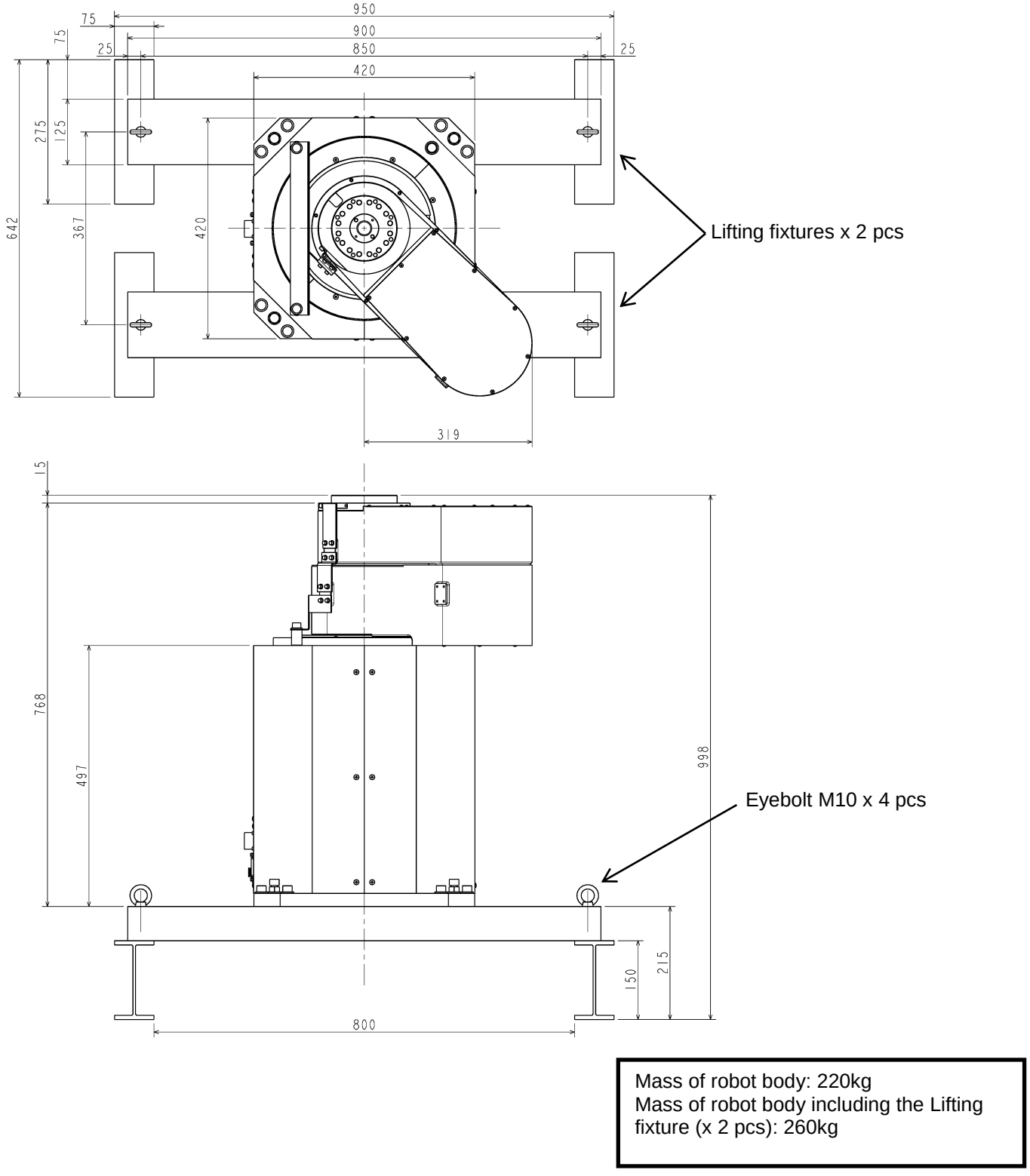


Fig. 2.3 Outer dimensions at transport (SR-624HC)

2.2.2 Transporting the Robot

When transporting the robot, observe the following procedures, fully confirming the cautions.

DANGER

- When transporting the robot, be sure to use the attached lifting fixtures.
- Be sure to secure the arm with the attached clamp before transporting the robot. Otherwise, a hazardous situation may be caused because the arm will move when the robot is suspended.

When transporting the robot by a crane, observe the following procedures.

・Case of transporting the robot by using the lifting fixtures.

1) Make sure that the Lifting fixtures and arm clamps are secured.
2) Pass wire ropes through the eyebolts of the lifting fixtures. To assure safety, be sure to use four (4) wire ropes. If there is a fear that the wire rope will contact the robot body due to the length of the wire rope, use a cushion.
3) Lift up the robot carefully, using the crane.

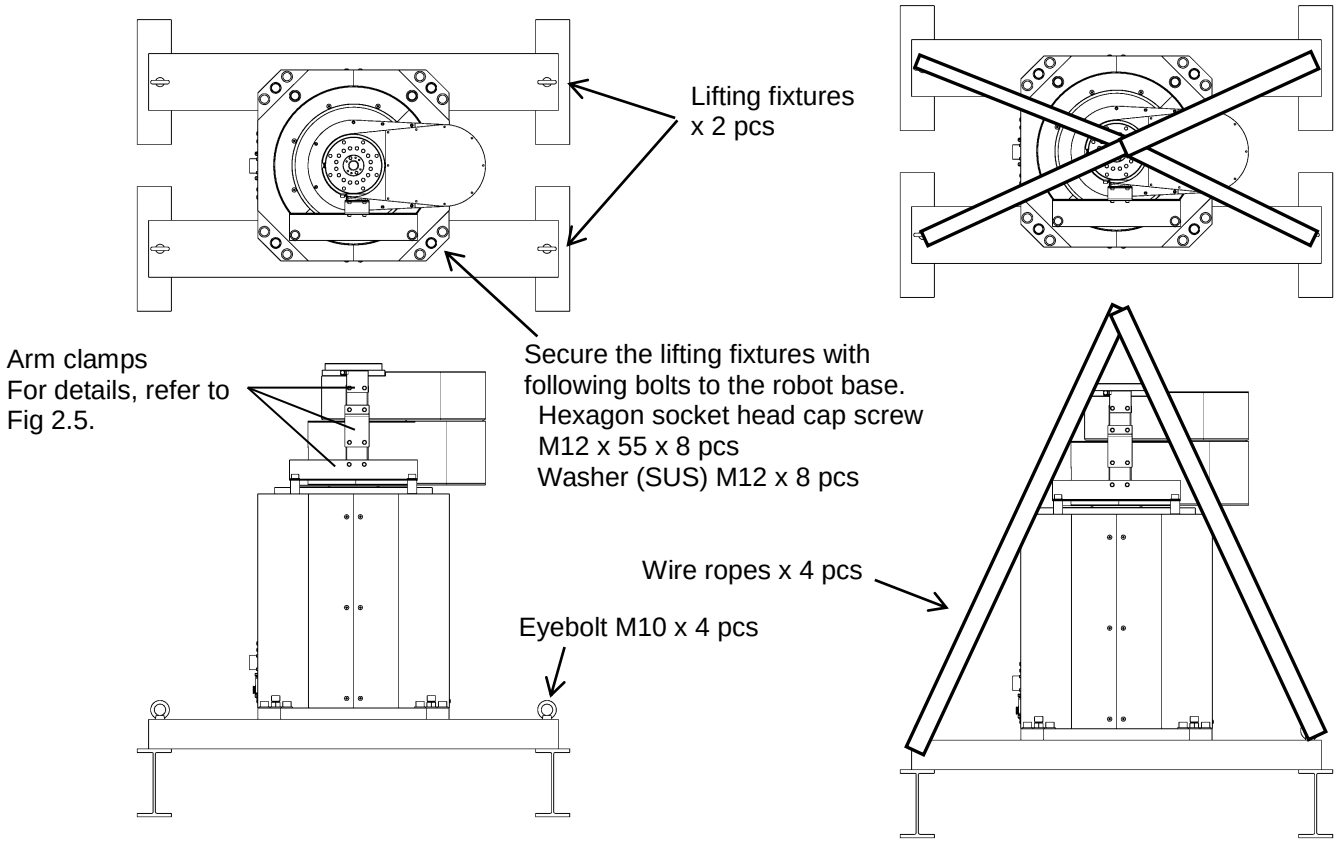


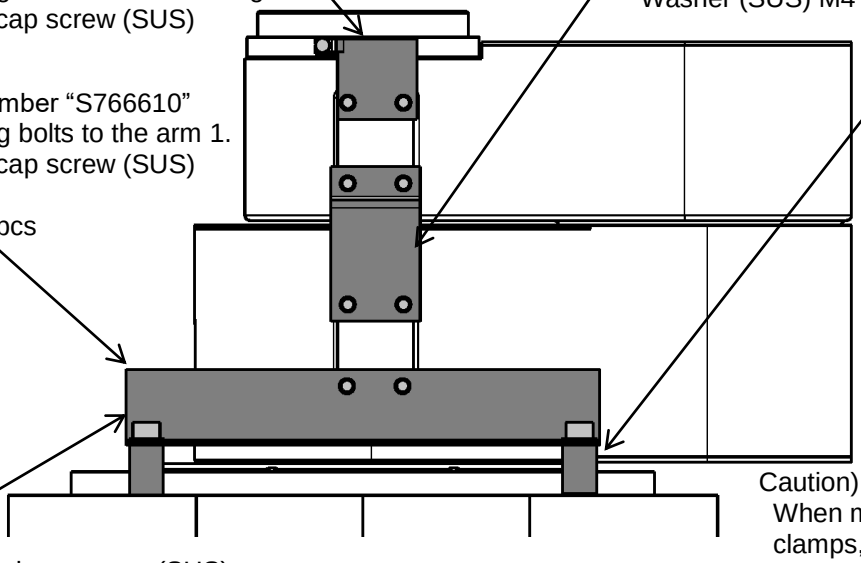
Fig. 2.4 Lifting posture of the robot by the lifting fixtures (SR-404HC and SR-624HC)

Clamp no.3, drawing number “S766612”
Secure with the following bolts to the arm 2.
Hexagon socket head cap screw (SUS)
M4 x 8 x 2 pcs
Washer (SUS) M4 x 2 pcs
Secure with the following bolts to the tool flange.
Hexagon socket head cap screw (SUS)
M5 x 10 x 1 pcs

Clamp no.2, drawing number “S766611”
Secure with the following bolts to the arm 1 and arm 2.
Hexagon socket head cap screw (SUS)
M4 x 8 x 4 pcs
Washer (SUS) M4 x 4 pcs

Clamp no.1, drawing number “S766610”
Secure with the following bolts to the arm 1.
Hexagon socket head cap screw (SUS)
M4 x 8 x 2 pcs
Washer (SUS) M4 x 2 pcs

Collar NCLA 12-30-31
(Maker Misumi) x 2 pcs



Hexagon socket head cap screw (SUS)
M10 x 45 x 2 pcs
Washer (SUS) M10 x 2 pcs

Caution)
When mounting the arm clamps, move each axes of the robot to the following positon before mounting.
Axis 1: 0deg
Axis 2: 0deg
Axis 3: 9mm
Axis 4: 0deg

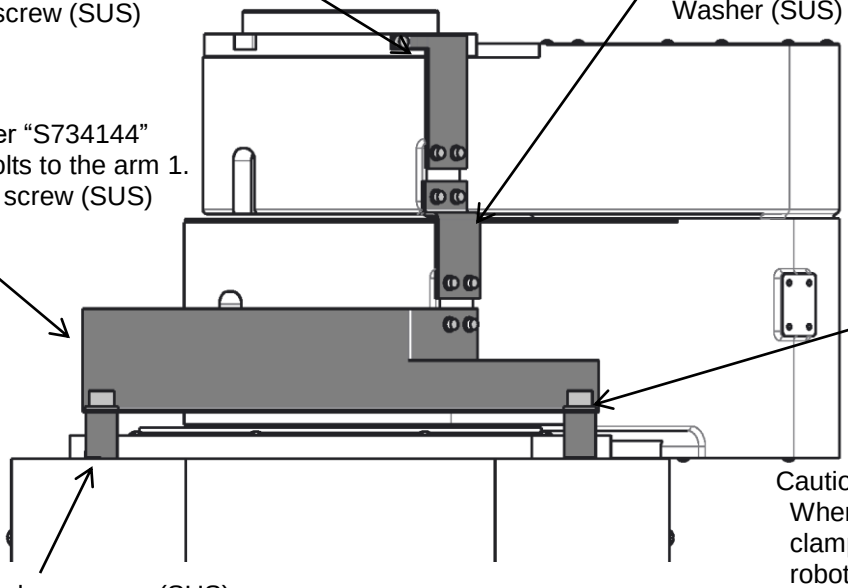
For SR-404HC

Clamp no.3, drawing number “S734068”
Secure with the following bolts to the arm 2.
Hexagon socket head cap screw (SUS)
M4 x 8 x 2 pcs
Washer (SUS) M4 x 2 pcs
Secure with the following bolts to the tool flange.
Hexagon socket head cap screw (SUS)
M5 x 10 x 1 pcs

Clamp no.2, drawing number “S734067”
Secure with the following bolts to the arm 1 and arm 2.
Hexagon socket head cap screw (SUS)
M4 x 8 x 4 pcs
Washer (SUS) M4 x 4 pcs

Clamp no.1, drawing number “S734144”
Secure with the following bolts to the arm 1.
Hexagon socket head cap screw (SUS)
M4 x 8 x 2 pcs
Washer (SUS) M4 x 2 pcs

Collar NCLA 12-30-31
(Maker Misumi) x 2 pcs



Hexagon socket head cap screw (SUS)
M10 x 45 x 2 pcs
Washer (SUS) M10 x 2 pcs

Caution)
When mounting the arm clamps, move each axes of the robot to the following positon before mounting.
Axis 1: -45deg
Axis 2: 0deg
Axis 3: 10mm
Axis 4: 0deg

For SR-624HC

Fig. 2.5 Arm clamps (SR-404HC and SR-624HC)

・Case of transporting the robot only (removing the lifting fixtures).

1) Make sure that the arm are secured with the arm clamps.
2) Remove the hexagon bolts and washers, then mount the M10 eyebolts.
   If the arm clamp no.1 is mounted to the robot, remove the clamp and mount the M10 eyebolts.
3) Pass wire ropes through the eyebolts. To assure safety, be sure to use four (4) wire ropes. If there is a fear that the wire rope will contact the robot body due to the length of the wire rope, use a cushion.
4) Lift up the robot carefully, using the crane.
5) After the robot has been installed, remove the arm clamps and eyebolts.
6) Mount the attached hexagon bolts and washers to the four (4) screw holes.

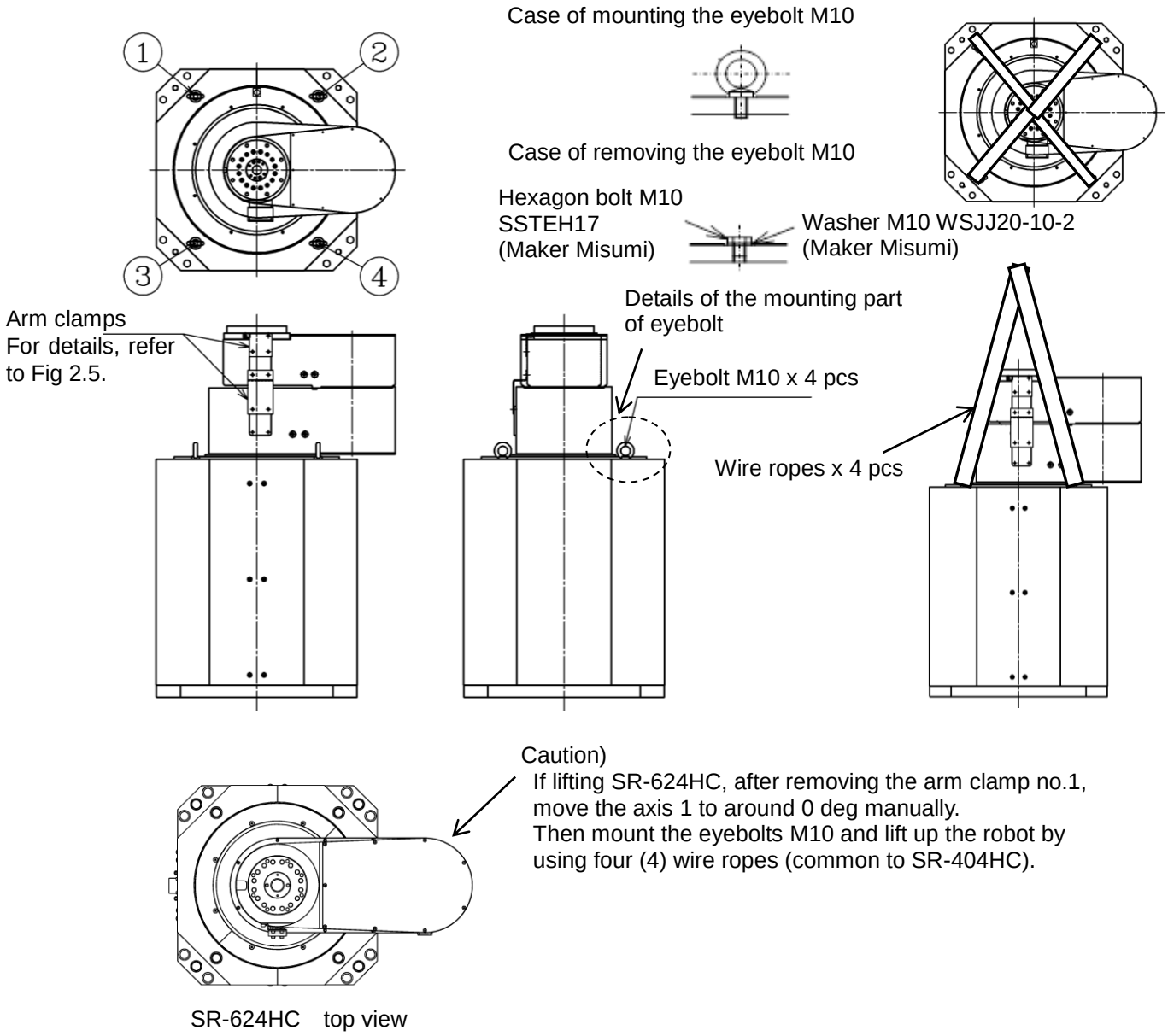


Fig. 2.6 Lifting posture of the robot (lifting the robot only) (SR-404HC and SR-624HC)

! CAUTION

- Use the wires which can well withstand the robot mass.
- The robot will tilt more or less when lifted. Carefully lift it up.
- When raising and lowering the robot, take careful precautions not to exert any excessive impact on the robot.
- When placing the robot on the floor, etc., take utmost care not to have your hands or feet caught under the robot.

2.2.3 Transporting the Controller

Disconnect all cables and teach pendant before transporting the controller.

! DANGER

- When placing the controller on the floor, etc., make sure not to have your hands or feet caught.

2.3 Storage

Avoid storing the robot and controller for long periods of time after unpacking them. If this is unavoidable, however, strictly observe the following precautions for storage.

2.3.1 Storage Precautions for the Robot

! CAUTION

- Secure the base completely to prevent the robot from falling over.
- Avoid direct sunshine and high temperature/humidity when the robot is stored. Otherwise, the parts may deteriorate.
- Seal the robot in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture.
- When the robot motor drive cable was disconnected from the controller, connect the short-circuit connector to the same cable.
- Before starting an operation, perform running completely.

2.3.2 Storage Precautions for the Controller

! CAUTION

- Keep the controller out of direct sunlight. Otherwise, the controller interior will be excessively heated up, causing a trouble.
- Seal the controller in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture.

3. Transportation (SR-405HC-L1550-R/L / SR-405HC-L1530S-R/L and TS3100/TS3100E)

3.1 Unpacking

The robot and controller are shipped separately in wooden frames and corrugated cardboards. Fig. 3.1 shows each packaging state.
Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.
After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.
For the controller accessories, see the accessory list packed with the controller.

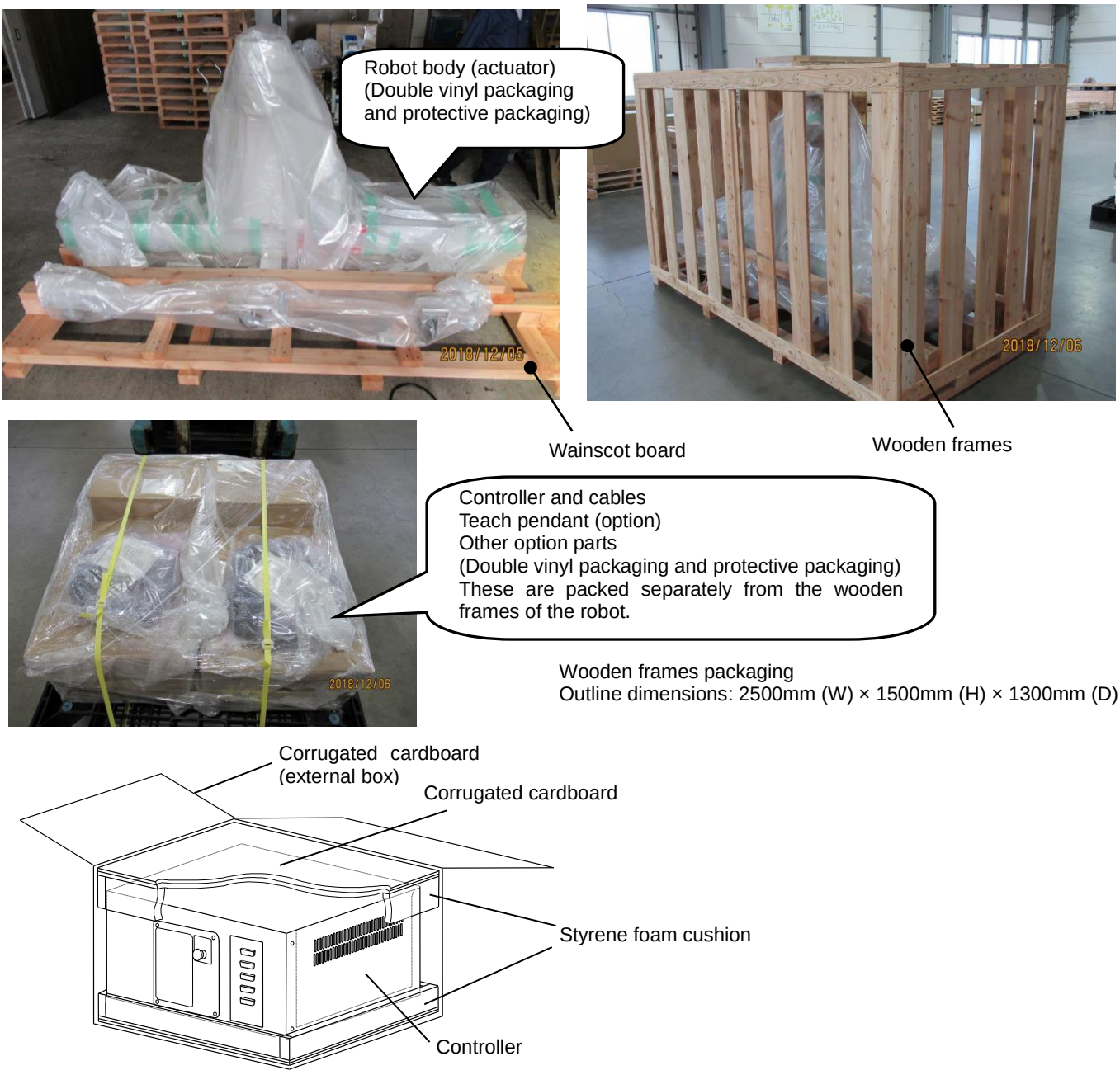


Fig. 3.1 Packaging state (SR-405HC-L1550-R/L / SR-405HC-L1530S-R/L and TS3100/TS3100E)



DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden frames, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

3.2 Transportation

Move the robot and controller very carefully. Make sure that no excessive impact or vibration is exerted on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all nuts and screws completely.
If the equipment is to be moved to a location some distance from where it was unpacked, reposition the cushions as they were and put the equipment back into the corrugated cardboards.

3.2.1 Mass and Dimensions

The mass and outer dimentions of the robot are shown in Figs. 3.2 and 3.3.
For the controller mass and external dimensions, see "7.1 External Dimensions".

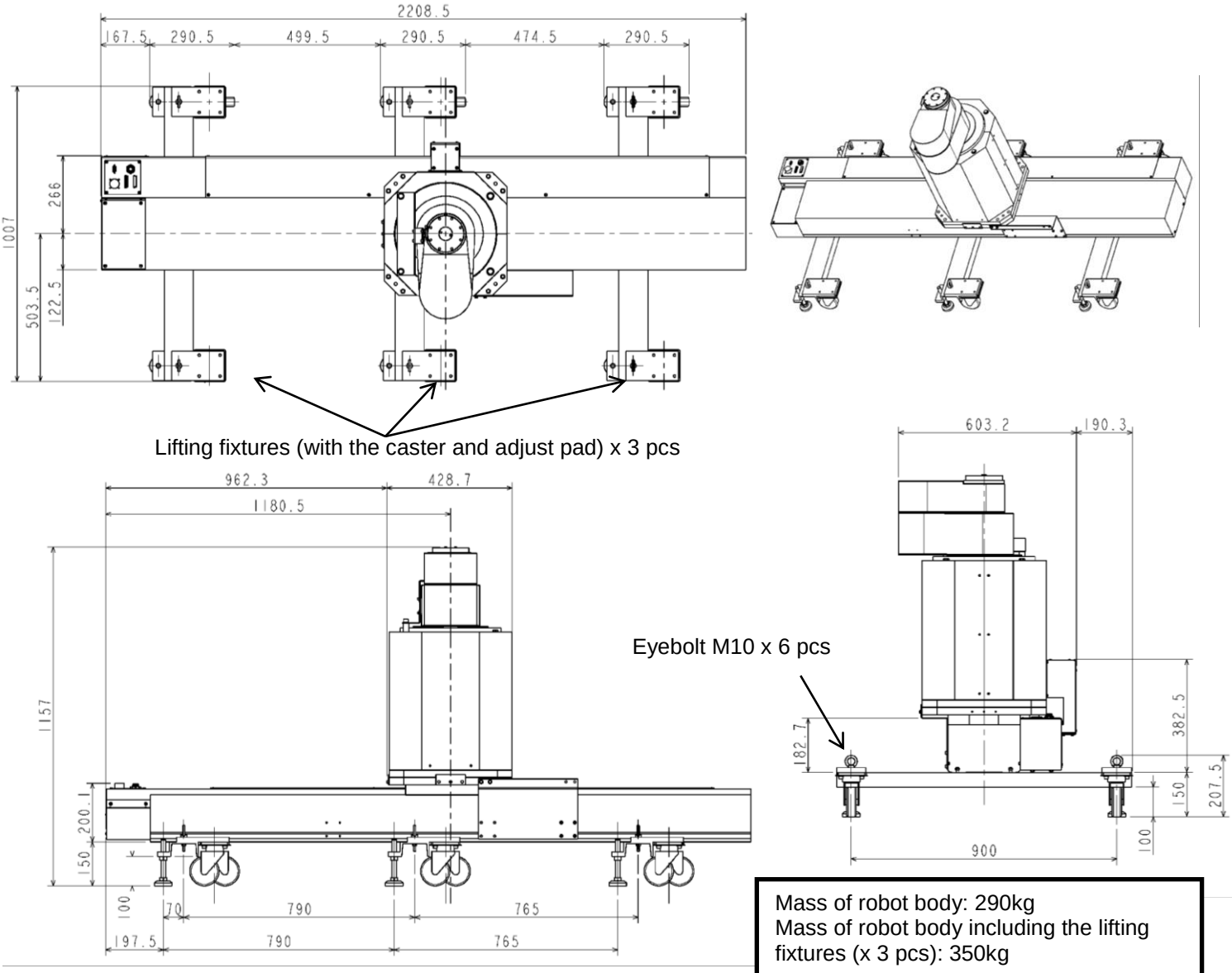
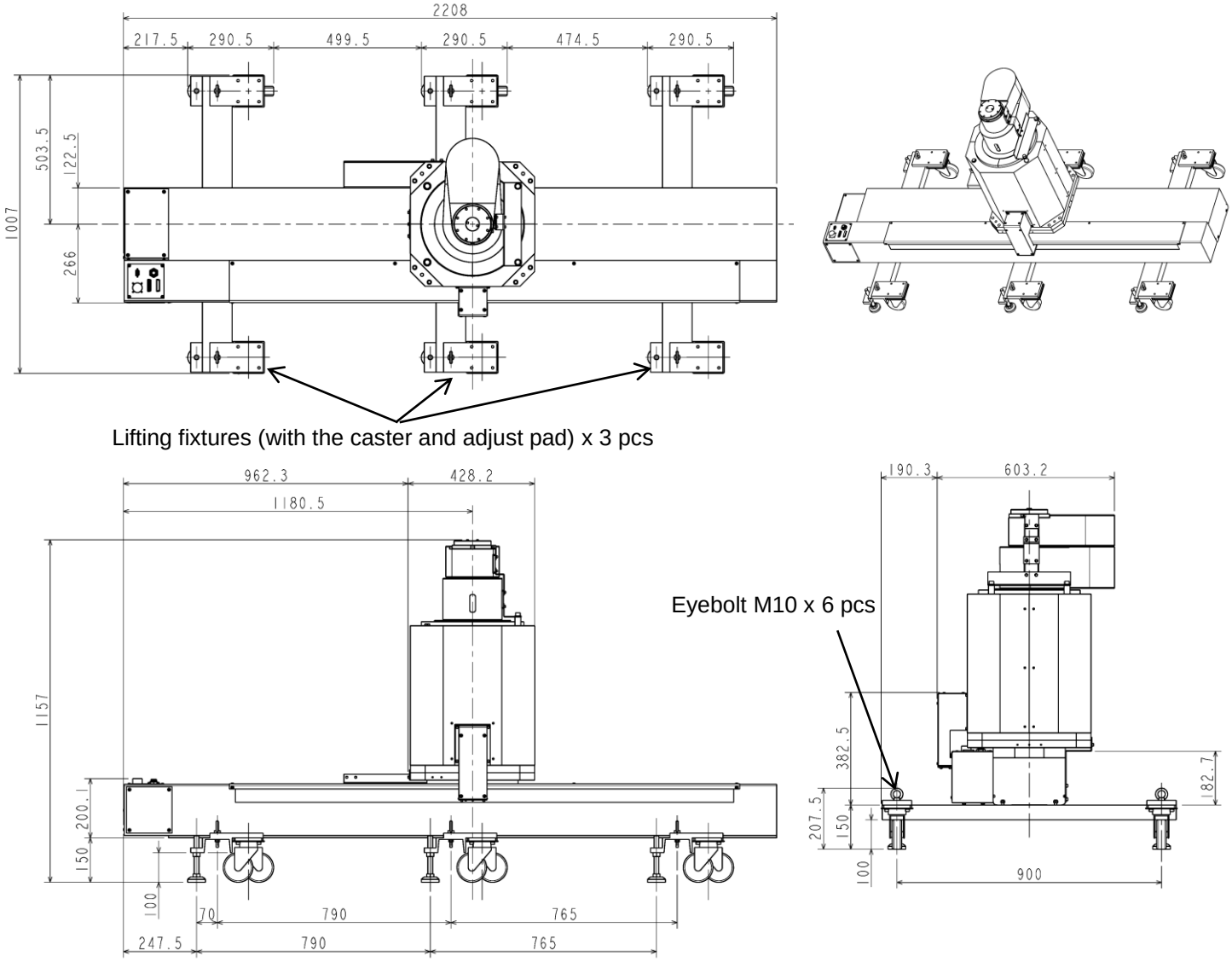


Fig. 3.2 Outer dimensions at transport (SR-405HC-L1550-R and SR-405HC-L1530S-R)



Mass of robot body: 290kg
Mass of robot body including the lifting fixtures (x 3 pcs): 350kg

Fig. 3.3 Outer dimensions at transport (SR-405HC-L1550-L and SR-405HC-L1530S-L)

3.2.2 Transporting the Robot

When transporting the robot, observe the following procedures, fully confirming the cautions.

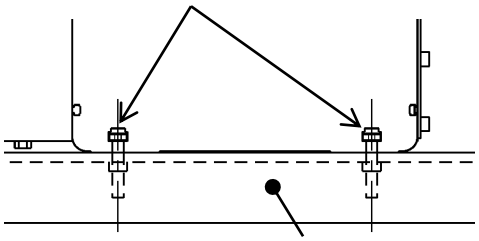
! DANGER

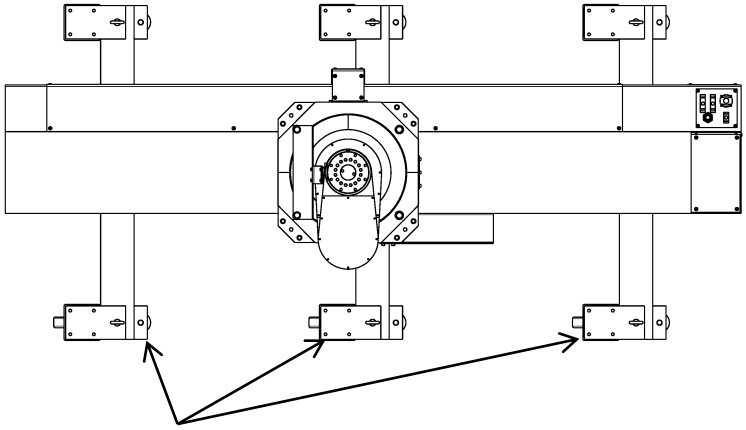
- When transporting the robot, be sure to use the attached lifting fixtures.
- Be sure to secure the arm with the attached clamp before transporting the robot. Otherwise, a hazardous situation may be caused because the arm will move when the robot is suspended.

When transporting the robot by a crane, observe the following procedures.

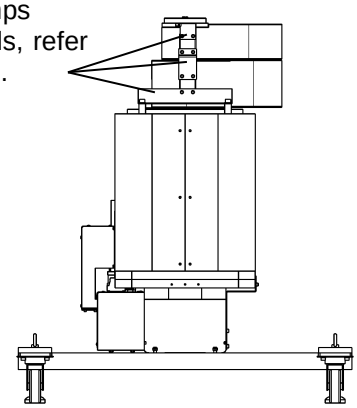
1) Make sure that the traverse axis (i.e., actuator) is secured with the clamp and the short- circuit connector that the arm is secured with the clamps.
   Arm clamps are common to SR-404HC (refer to Fig 2.5).
2) Mount the lifting fixtures to the bottom of the traverse axis.
3) Pass wire ropes through the eyebolts. To assure safety, be sure to use four (4) wire ropes. If there is a fear that the wire rope will contact the robot body due to the length of the wire rope, use a cushion.
4) Lift up the robot carefully, using the crane.
5) After the robot has been installed, remove the arm clamps and lifting fixtures.

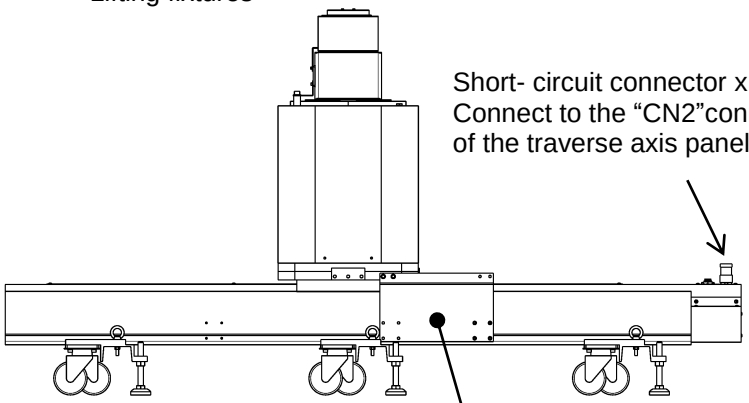
Mount the lifting fixtures to the traverse axis by using T-slots (for hexagon bolt M8) of the bottom of the traverse axis.
Per one lifting fixture:
Oval head bolt (SUS) M8 x 40 x 2 pcs
Hexagon nut M8 x 2 pcs
Washer (SUS) M8 x 2 pcs





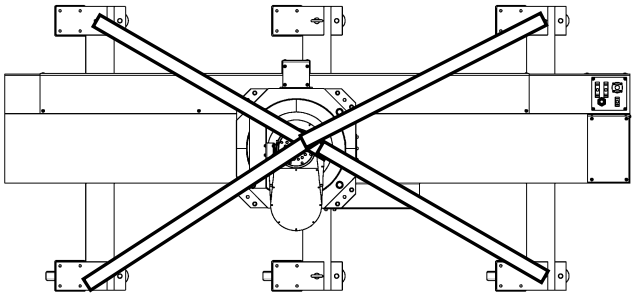
Arm clamps
For details, refer to Fig 2.5.

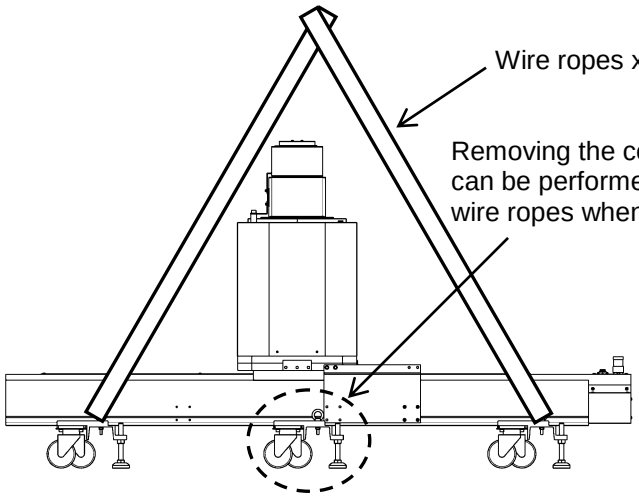




Short- circuit connector x 1 pcs
Connect to the “CN2”connector of the traverse axis panel.

Actuator clamp, drawing number “S753092”
Secure with the following bolts to the base.
Hexagon socket head cap screw (SUS)
M5 x 10 x 2 pcs
Washer (SUS) M5 x 2 pcs
Secure with the following bolts to the side of traverse axis.
Hexagon socket head cap screw (SUS)
M6 x 12 x 4 pcs





Caution)
When mounting the actuator clamp and arm clamps, move each axes of the robot to the following positon before mounting.
Axis 1: 0deg
Axis 2: 0deg
Axis 3: 9mm
Axis 4: 0deg
Axis 5: 734mm (common to “-R” and “-L”)

Fig. 3.4 Lifting posture of the robot (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

! CAUTION

- Use the wires which can well withstand the robot mass.
- The robot will tilt more or less when lifted. Carefully lift it up.
- When raising and lowering the robot, take careful precautions not to exert any excessive impact on the robot.
- When placing the robot on the floor, etc., take utmost care not to have your hands or feet caught under the robot.

3.2.3 Transporting the Controller

Disconnect all cables and teach pendant before transporting the controller.

! DANGER

- When placing the controller on the floor, etc., make sure not to have your hands or feet caught.

3.3 Storage

Avoid storing the robot and controller for long periods of time after unpacking them. If this is unavoidable, however, strictly observe the following precautions for storage.

3.3.1 Storage Precautions for the Robot

! CAUTION

- Secure the base completely to prevent the robot from falling over.
- Avoid direct sunshine and high temperature/humidity when the robot is stored. Otherwise, the parts may deteriorate.
- Seal the robot in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture.
- When the robot motor drive cable was disconnected from the controller, connect the short-circuit connector to the same cable.
- Before starting an operation, perform running completely.

3.3.2 Storage Precautions for the Controller

! CAUTION

- Keep the controller out of direct sunlight. Otherwise, the controller interior will be excessively heated up, causing a trouble.
- Seal the controller in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture.

4. Robot Installation (SR-404HC and SR-624HC)

4.1 Installation Environment

Table 4.1 shows the environmental conditions for the location in which the robot and controller are to be installed.

Table 4.1 Environmental conditions for robot and controller

Item	Specifications
Temperature	In operation: 0 to 40°C In storage: –10 to 50°C
Humidity	20 to 90% (Non-condensing) DO NOT install the robot where it may be subject to fluids such as water.
Altitude	1000 m or less
Vibration	In operation: 0.98 m/s^2 or less
Dust	No inductive dust should exist. Consult with SHIBAURA MACHINE first if you wish to use the robot and controller in a dusty environment.
Gas	No corrosive or combustible gas should exist.
Degrees of protection	IEC60529 IP20(controller only)
Overvoltage category	IEC60664-1 ClassⅢ(controller only)
Protection against electric shock	IEC61140 Class Ⅰ (controller only)
Pollution Degree	IEC60664-1 Pollution Degree 3(controller only)
Sunlight	The robot and controller should not be exposed to direct sunlight.
Power noise	A heavy noise source should not exist nearby.
Magnetic field	A heavy magnetic field source should not exist nearby.
Other ambient environment	Must be free from iron powder, oil, salt, and organic solvents. Must not be exposed to water.

! DANGER

- Do not place the robot or controller near combustible.
  Doing so could lead to fires if it ignites due to a fault, etc.

! CAUTION

- When starting fast-movement operation in low-temperature environment, errors may occur because of the increased torque. In operating the robot in low-temperature environment, start by making low-speed continuous operation for a several minutes to soften the grease before starting high-speed operations.

4.2 Installation

Before installing the robot, you should plan a layout, fully considering the working envelope (or operating range), coordinate system and space for maintenance.

4.2.1 External Dimensions

Figs. 4.1 and 4.2 show the external view drawing of the standard robot.
The external view of the robot may vary depending on the robot specifications.
Be sure to see the external view of the robot attached to the specifications.

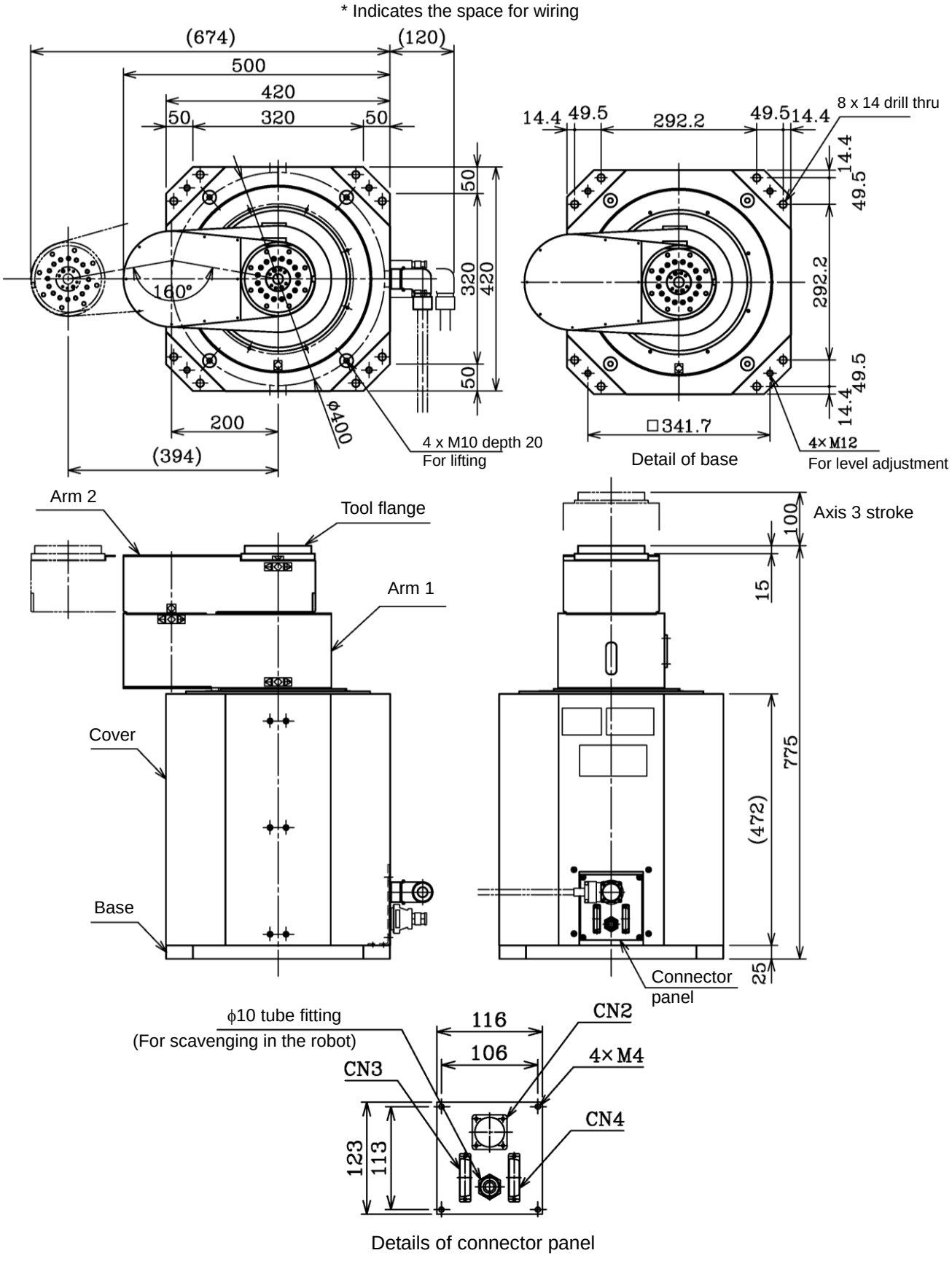
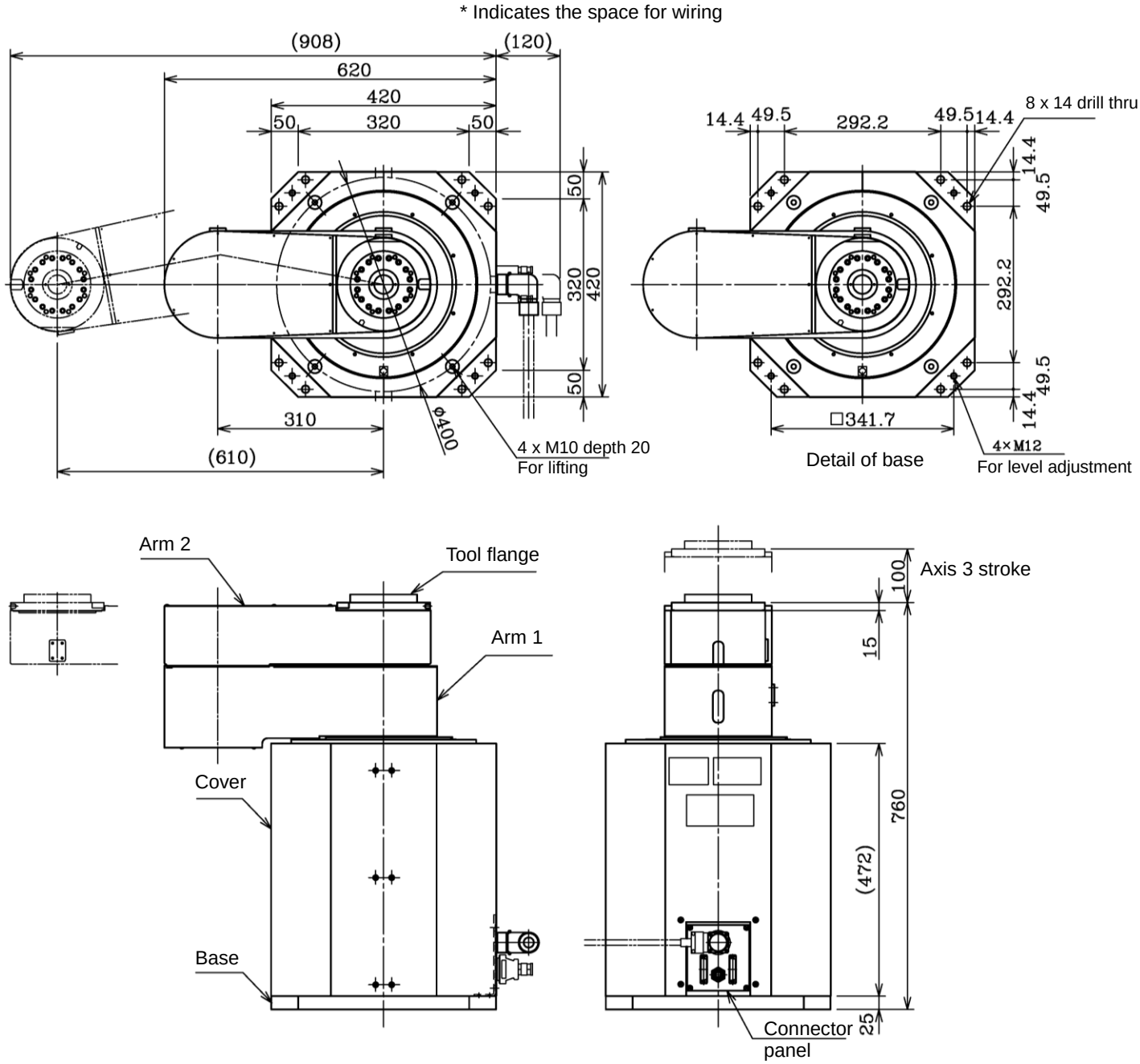


Fig. 4.1 External view (SR-404HC)

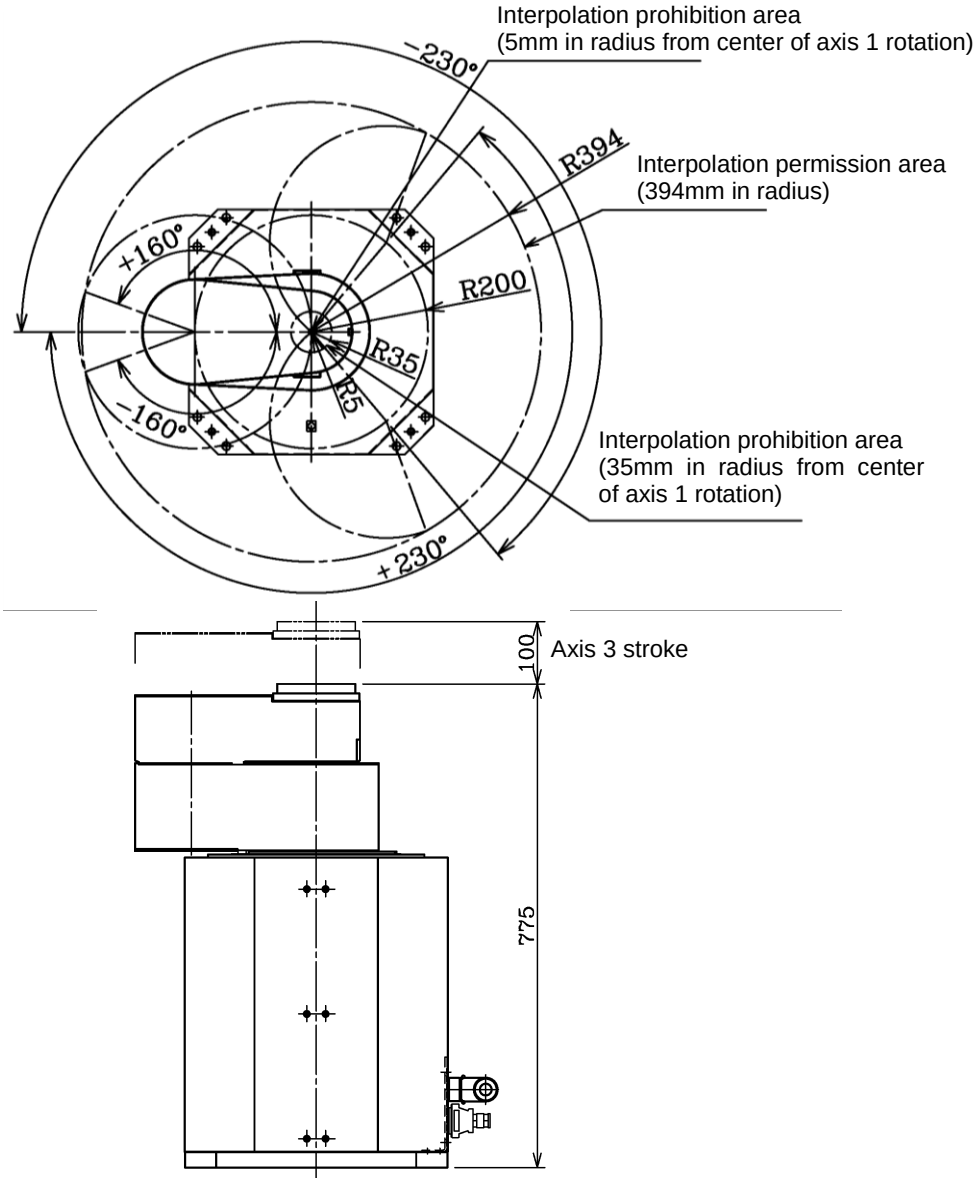


Details of connector panel

Fig. 4.2 External view (SR-624HC)

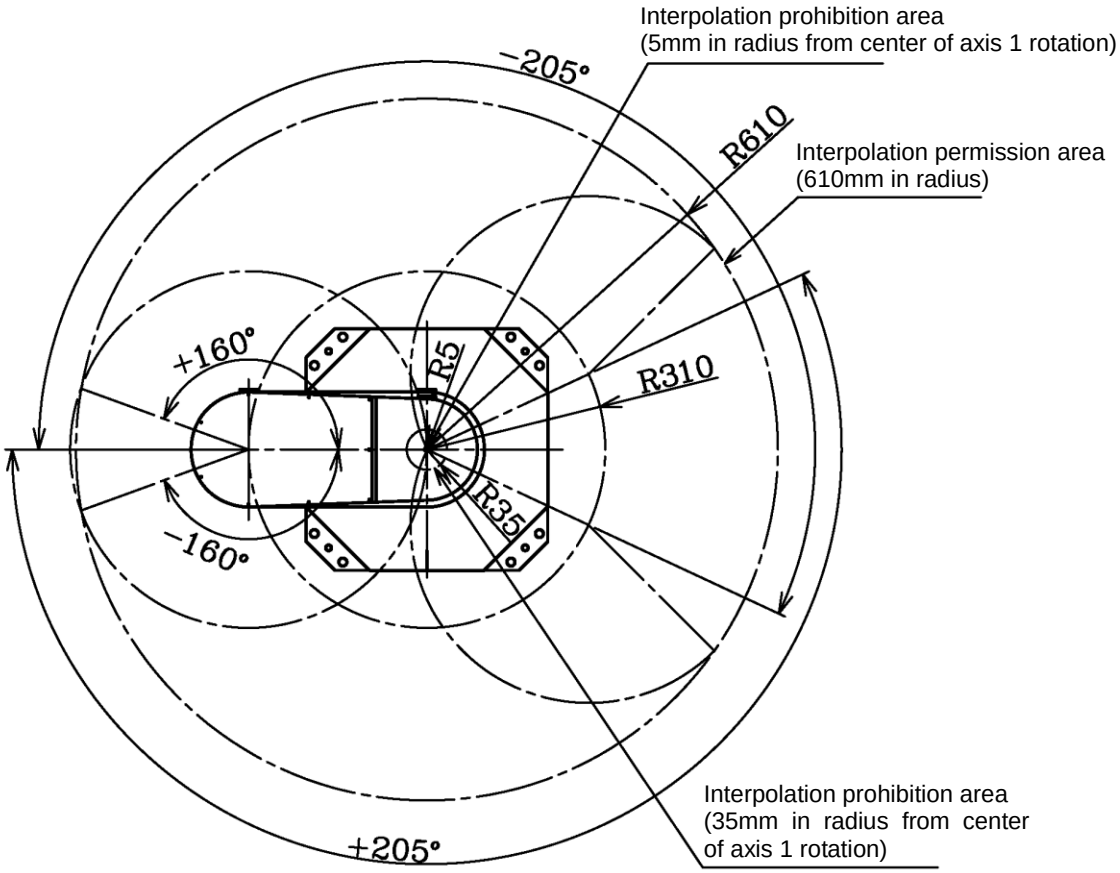
4.2.2 Working envelope

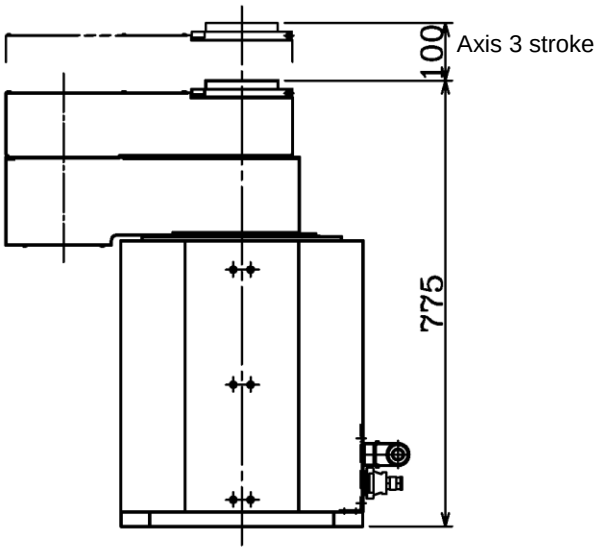
Fig. 4.3 and 4.4 show the working envelope of the standard robot.
The working envelope of the robot may vary depending on the robot specifications.
Be sure to see the external view of the robot attached to the specifications.
Axes 1, 2 and 3 can operate within the working envelope. To prevent the robot from moving out of the working envelope by misoperation, the robot is equipped with mechanical stoppers outside the working envelope. Additionally, soft limits that can be set by the user are provided. For further information, see the “Instruction Manual: ST80725 TS3000 User Parameters” provided separately.



Axis 1: Angle where arm 1 motion is possible : -230° to 0° to +230°
Axis 2: Angle where arm 2 motion is possible : -160° to 0° to +160°
Axis 3: Working envelope : 0mm to 100mm
Axis 4: Angle where tool flange motion is possible : -230° to 0° to +230°

Fig. 4.3 Working envelope (SR-404HC)





Axis 1: Angle where arm 1 motion is possible : -205° to 0° to +205°
Axis 2: Angle where arm 2 motion is possible : -160° to 0° to +160°
Axis 3: Working envelope : 0mm to 100mm
Axis 4: Angle where tool flange motion is possible : -230° to 0° to +230°

Fig. 4.4 Working envelope (SR-624HC)

4.2.3 Changing Software Limits

There are the following two types of methods to change software limits.

[1] Change the setting values of the "User Parameter File (file name: USER.PAR)." There are software limits that can be set by the customer. For more details, refer to the item of [U14] SOFTWARE LIMIT in the "Instruction Manual: ST80725 TS3000 User Parameters" separately available.

[2] By manipulating the teaching pendant, change software limits in the utility mode "J-LIM."
For more details, refer to "10.8 Joint Limit Setting [J-LIM]" in Chapter 10, "Utilities" in the "Instruction Manual: ST80720 TS3000 Operations" separately available.

Note that if software limits are changed using the above two types of methods, the factory preset values of the software limits set up in the "User Parameter File (file name: USER.PAR)" will be overwritten and saved.

Thus, **be sure to create a backup of the "User Parameter File" before changing software limits** so as to get a grasp of the factory preset values of the software limits in the "User Parameter File."

Also, when software limits were changed, be sure to turn OFF and then ON the power.
If the power is not turned OFF and then ON again, changes to the user parameters will not take effect.

4.2.4 Coordinate System

The coordinate system of the robot is exemplified in Fig. 4.5.
The coordinate system is common between the SR-404HC and SR-624HC.
Figure shows the SR-404HC.

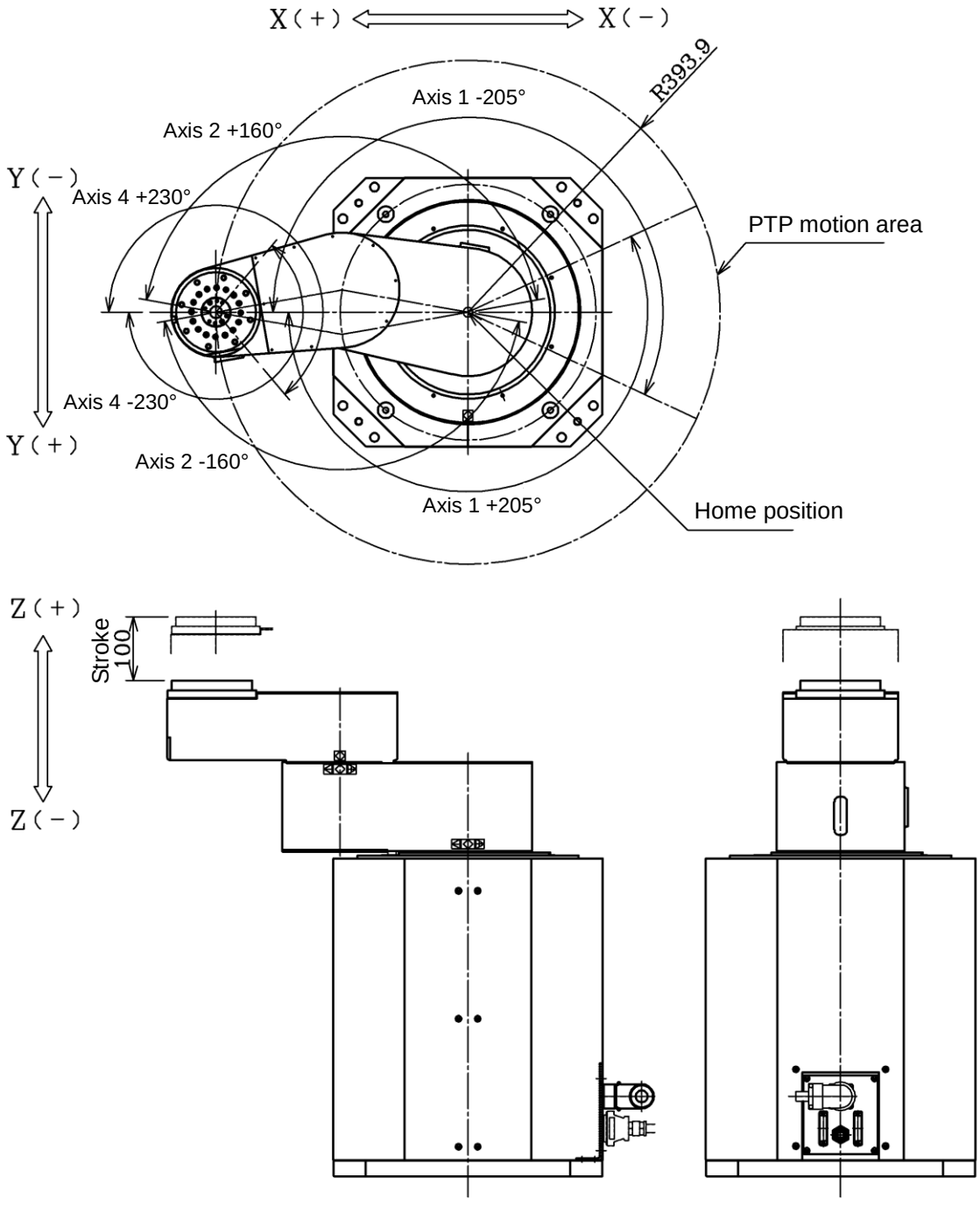


Fig. 4.5 Coordinate system of the robot

4.2.5 Installing the Robot

The robot is secured, using the mounting holes on the base (eight places). Use M12 hexagon socket head cap screws.
The robot base set areas should be rigid enough and their surfaces should be machined with flatness accuracy as small as 0.2 or less.

! CAUTION

- The robot will suddenly accelerate and decelerate during operation. When installing it on a frame, make sure that the frame has sufficient strength and rigidity.

  If the robot is installed on a frame that does not have sufficient rigidity, vibration will occur while the robot is operating, and could lead to faults.

  When installing the robot on the floor, secure the robot with anchor bolts, etc.
- Install the robot on a level place. Failure to do so could lead to a drop in performance or faults.

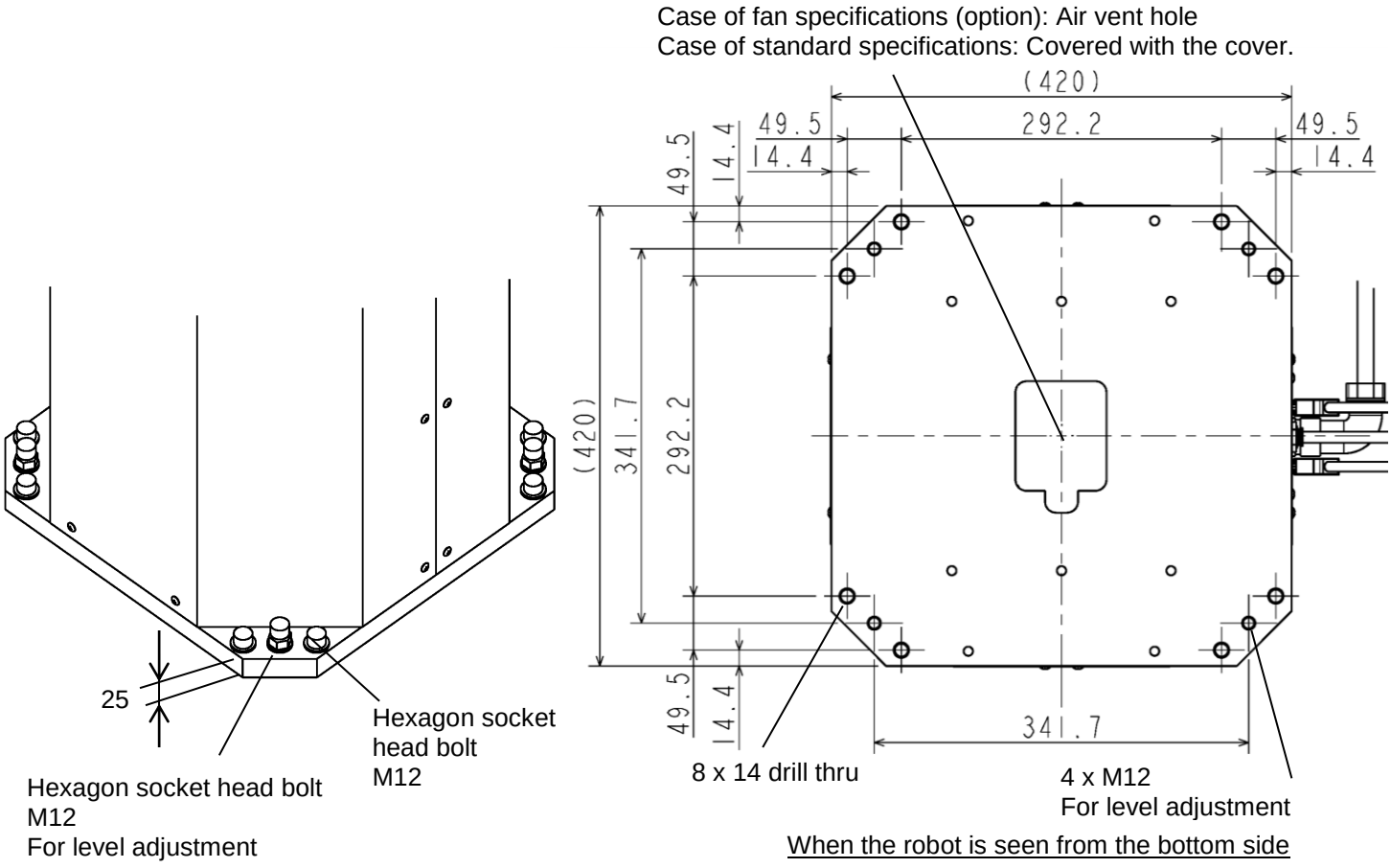


Fig. 4.6 Installation method (SR-404HC and SR-624HC)

4.3 Precautions for Handling the Teach Pendant

Be careful of the following matters when handling the teach pendant.

! CAUTION

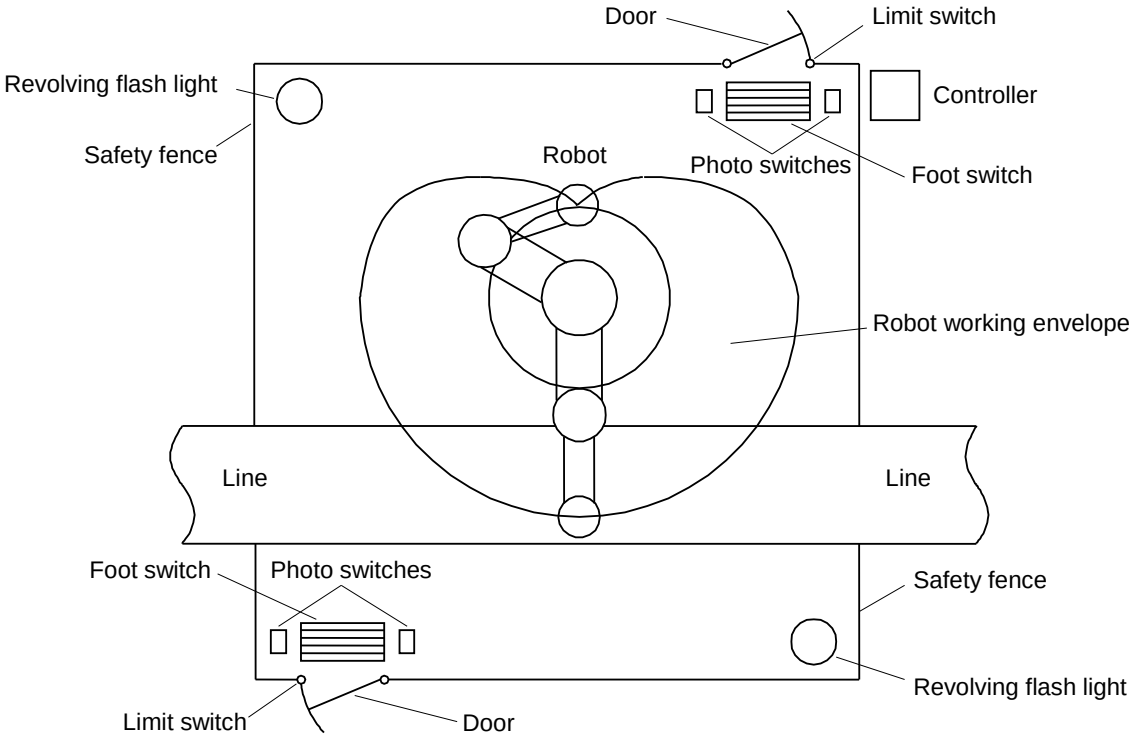
- DO NOT drop the teach pendant or hit it against anything.
- DO NOT pull the cable running from the teach pendant.
- DO NOT press the switches on the teach pendant with anything sharp (like the tip of a knife, pencil, ball-point pen, etc.).
- DO NOT place or use the teach pendant near open flames.
- DO NOT leave the teach pendant in direct sunlight for a long period of time.

4.4 Safety Measures

a) When installing the robot, provide sufficient space to carry out the work safely.

b) Clarify the hazard zone, and provide a safety fence so that other persons cannot enter the zone easily. The hazard zone is the zone near the robot's working space where a hazardous state could occur if a person enters.

c) Provide limit switches, photo switches, foot switch, etc., at the entrance of the safety fence to provide an emergency stop function that will stop the robot if a person enters the hazard zone. The emergency stop function should be an electrically independent normal close contact b (closed in normal operation) with compulsive opening function and must not be automatically recovered.



d) The controller should be installed at a place outside the hazard zone where the operator can view the robot movement.

5. Robot Installation (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

5.1 Installation Environment

For mounting environment, see "4.1 Installation Environment".

5.2 Installation

Before installing the robot, you should plan a layout, fully considering the working envelope (or operating range), coordinate system and space for maintenance.

5.2.1 External Dimensions

Figs. 5.1 to 5.4 show the external dimensions of the robot meeting the standard specifications.
The external view of the robot may vary depending on the robot specifications.
Be sure to see the external view of the robot attached to the specifications.

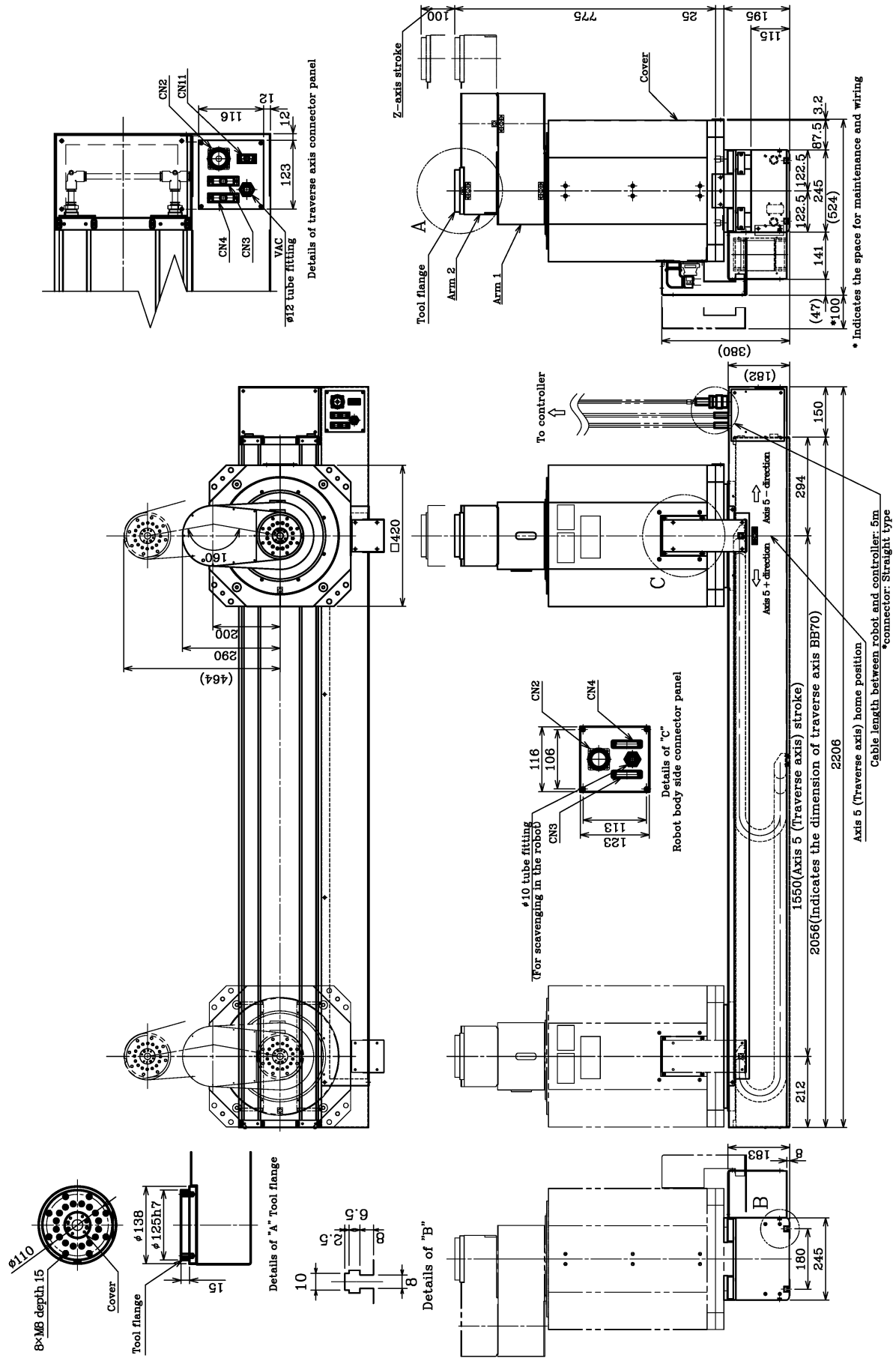


Fig. 5.1 External view (SR-405HC-L1550-R)

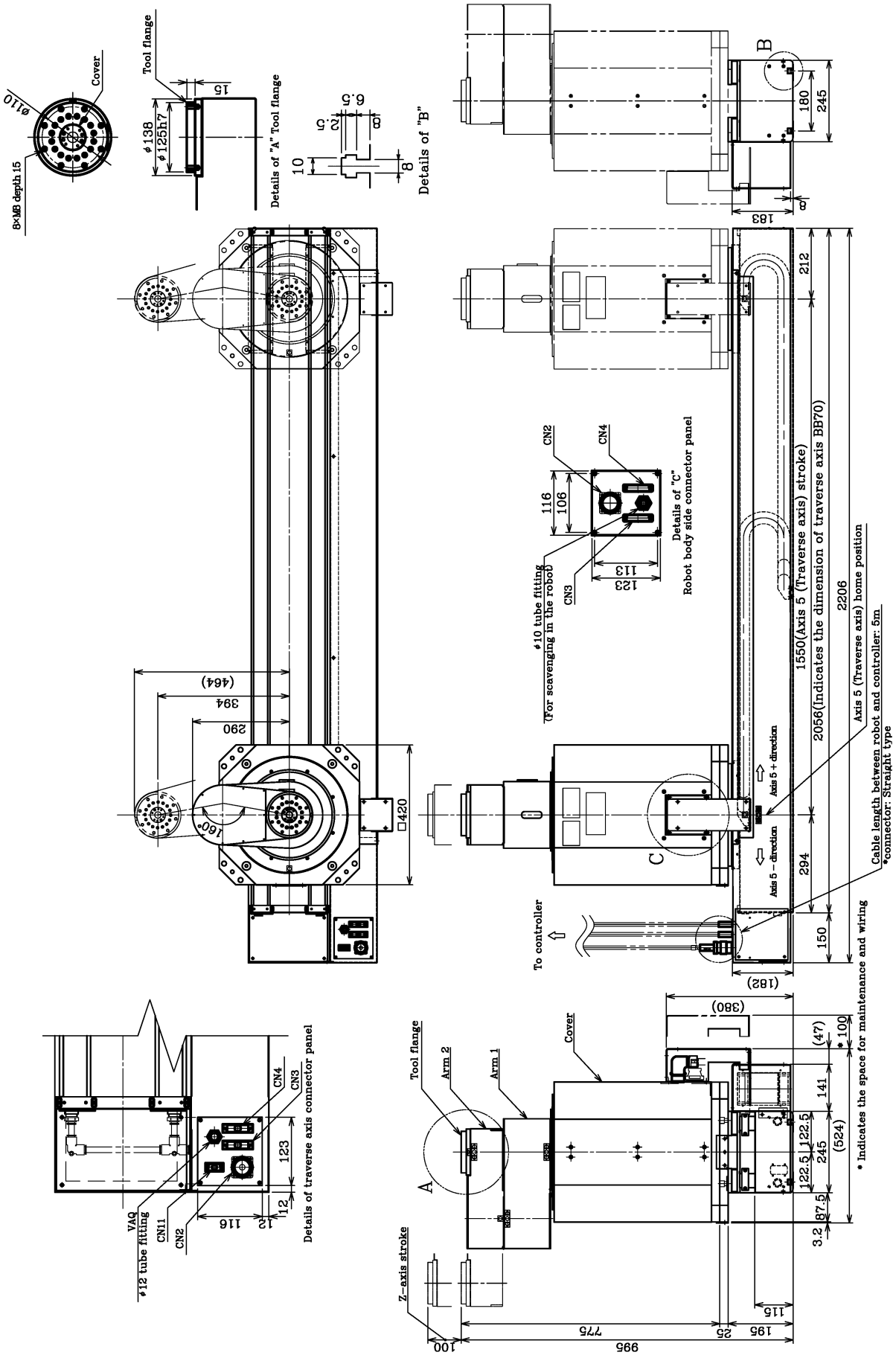


Fig. 5.2 External view (SR-405HC-L1550-L)

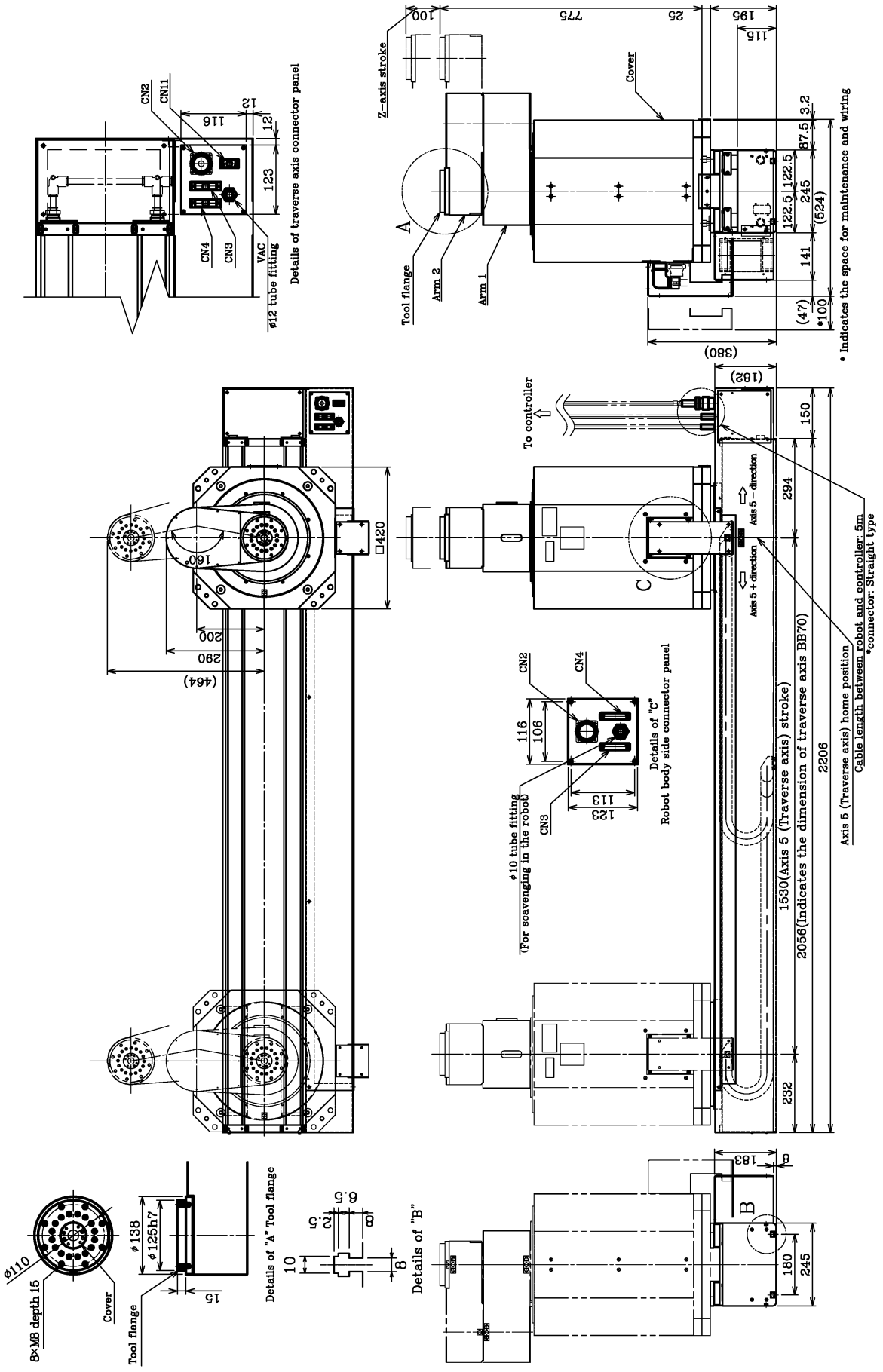


Fig. 5.3 External view (SR-405HC-L1530S-R)

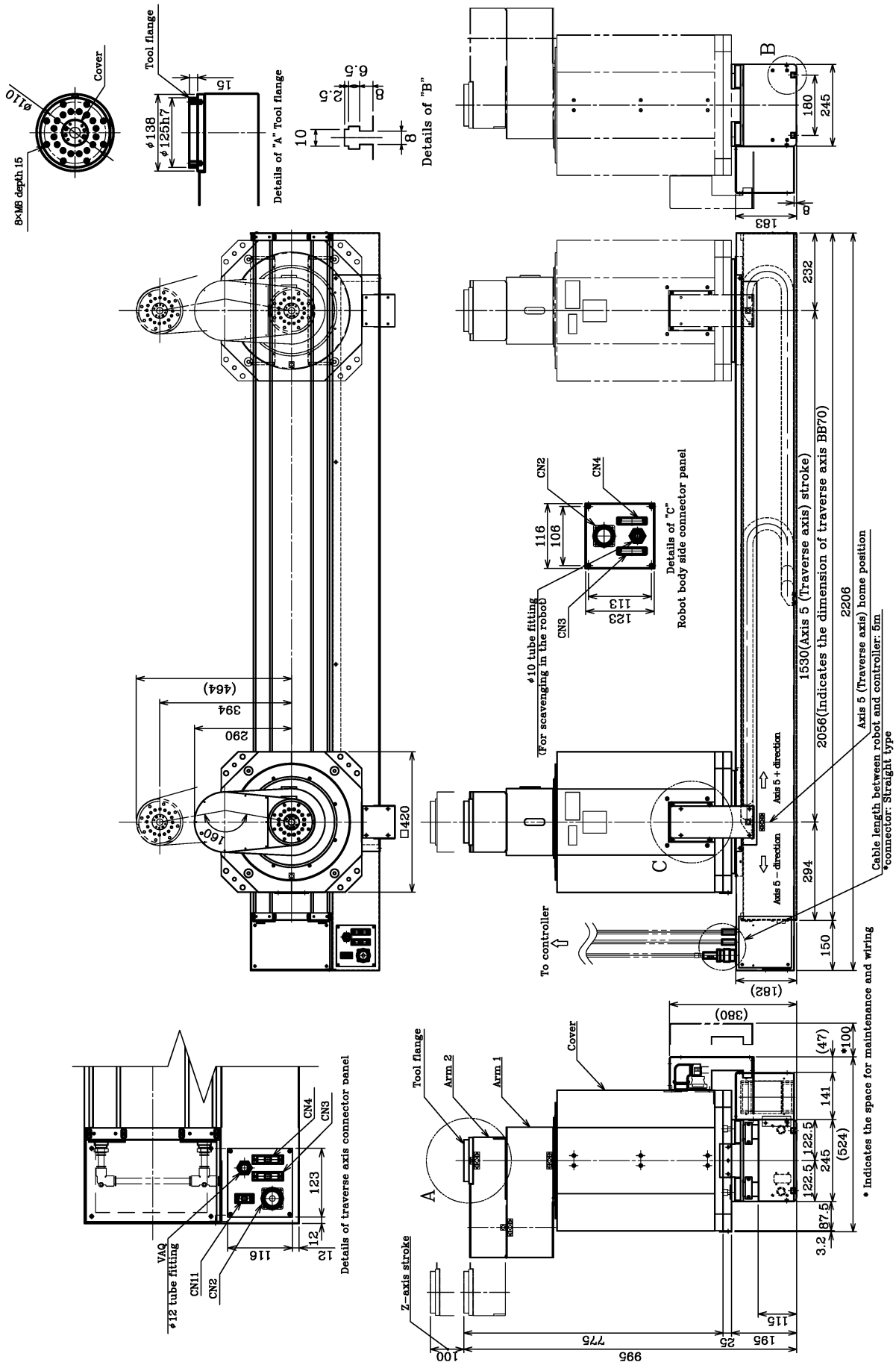
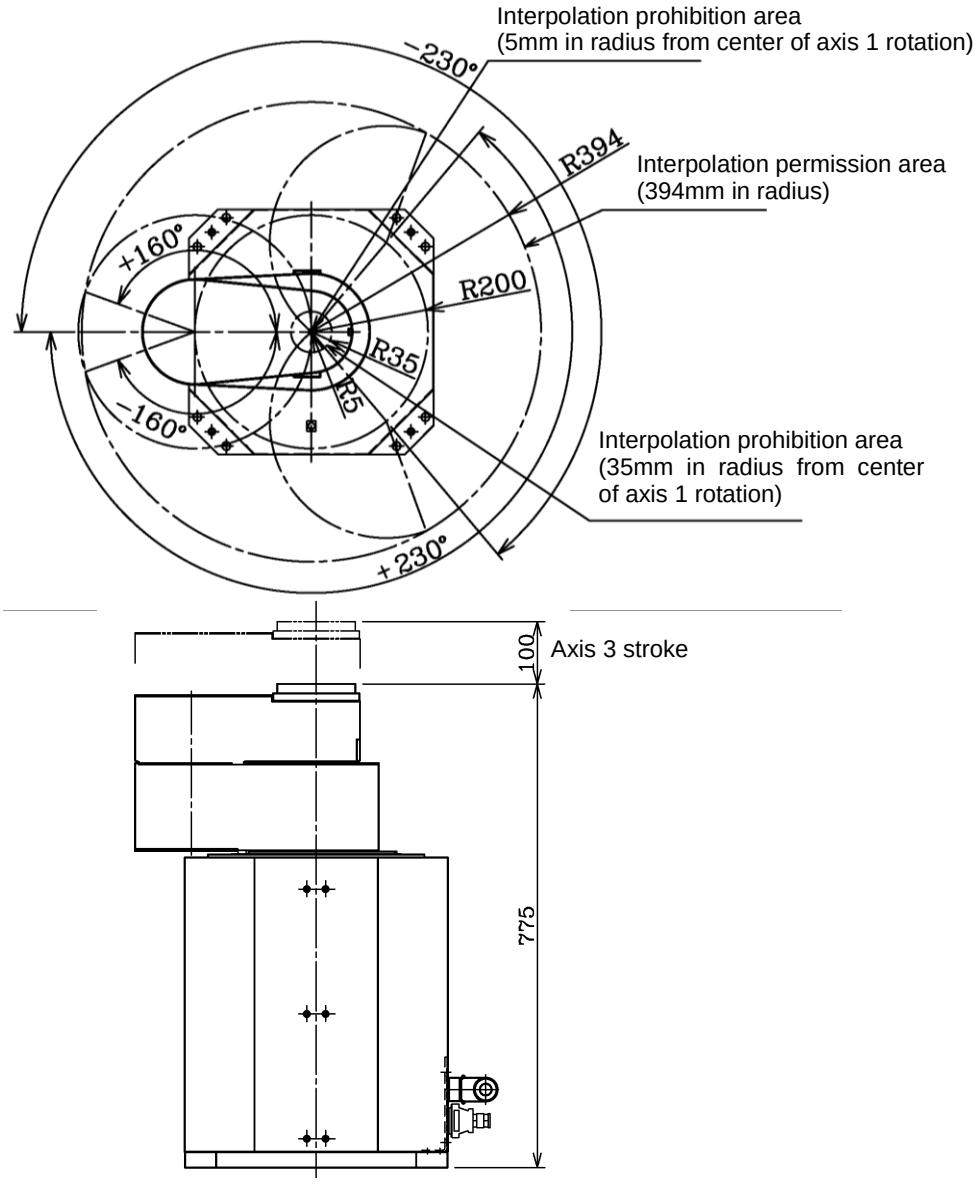


Fig. 5.4 External view (SR-405HC-L1530S-L)

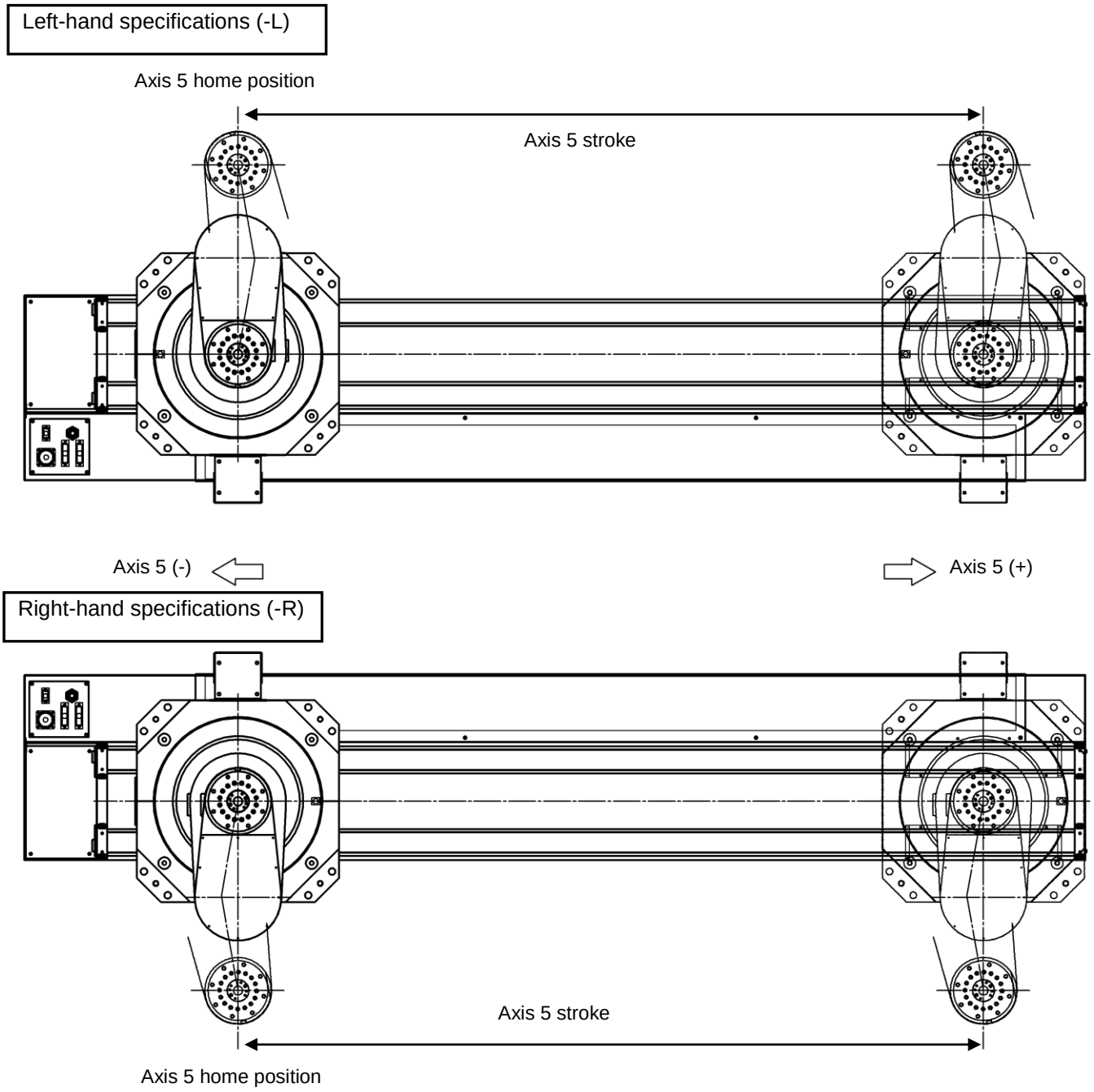
5.2.2 Working envelope

Fig. 5.5 show the working envelope of the standard robot.
The working envelope of the robot may vary depending on the robot specifications.
Be sure to see the external view of the robot attached to the specifications.
Axes 1, 2, 3 and 5 can operate within the working envelope. To prevent the robot from moving out of the working envelope by misoperation, the robot is equipped with mechanical stoppers outside the working envelope. Additionally, soft limits that can be set by the user are provided. For further information, see the “Instruction Manual: ST80725 TS3000 User Parameters” provided separately.



Axis 1: Angle where arm 1 motion is possible : -230° to 0° to +230°
Axis 2: Angle where arm 2 motion is possible : -160° to 0° to +160°
Axis 3: Working envelope : 0mm to 100mm
Axis 4: Angle where tool flange motion is possible : -230° to 0° to +230°

Fig. 4.5 Working envelope axes 1 to 4 (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)



・For SR-405HC-L1550-R/L
Axis 5: Working envelope: 0mm to 1550mm

・For SR-405HC-L1530S-R/L
Axis 5: Working envelope: 0mm to 1530mm

Fig. 5.6 Working envelope axis 5 (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

5.2.3 Changing Software Limits

For the software limit change procedure, see "4.2.3 Changing Software Limits".

5.2.4 Coordinate System

The coordinate system of the robot is exemplified in Fig. 5.7 and 5.8.

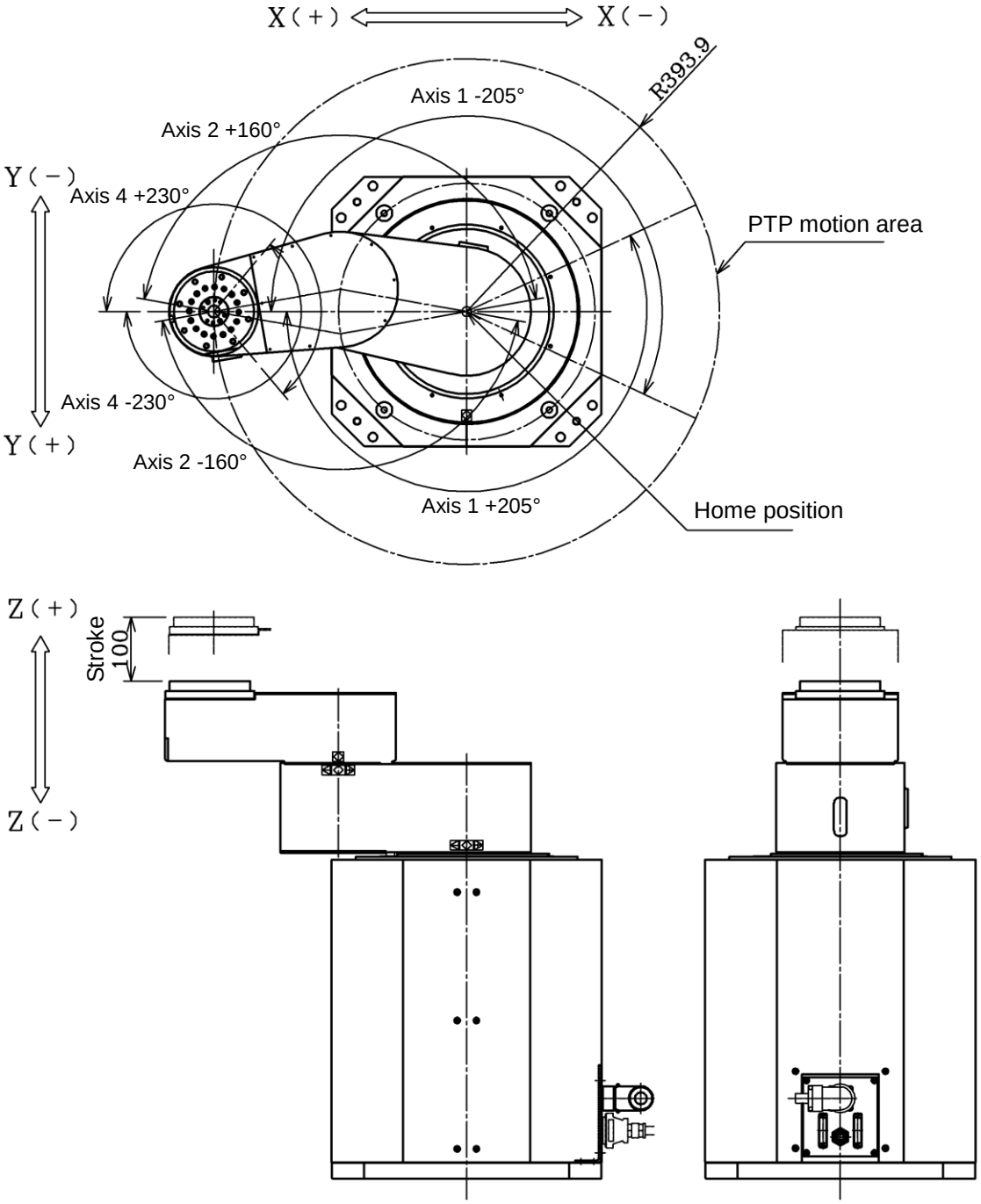


Fig. 5.7 Coordinate system (axes 1 to 4)

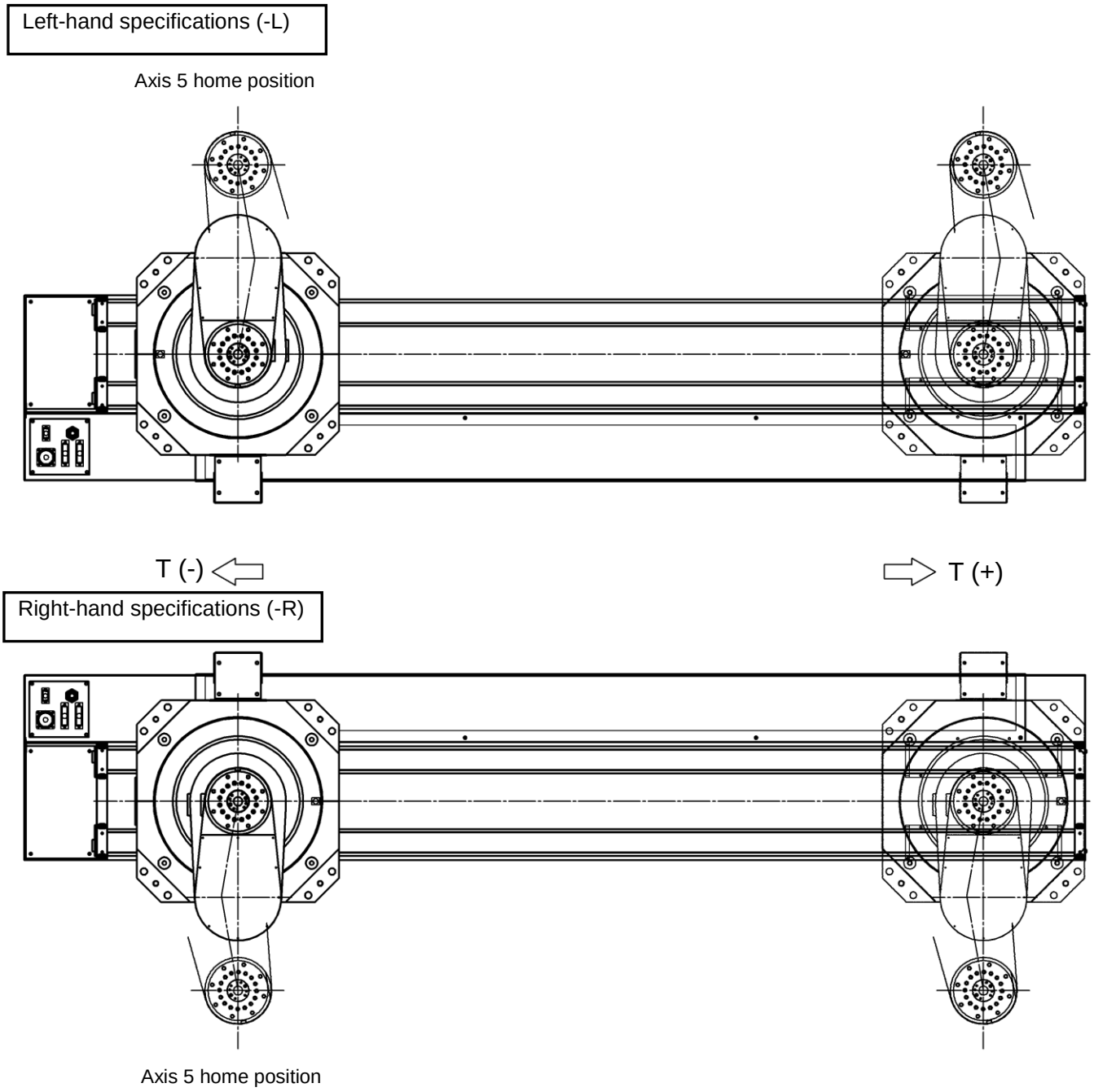


Fig. 5.8 Coordinate system (axis 5)

5.2.5 Installing the Robot

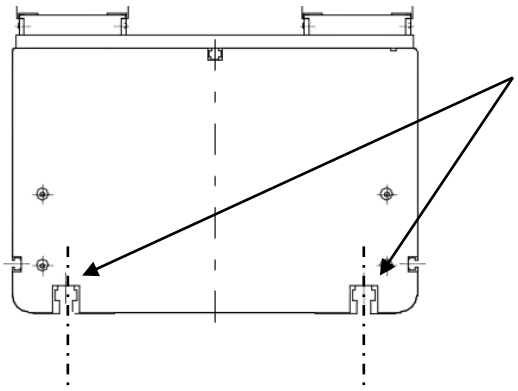
The robot is secured, using the T slots (for hexagon bolts M8) on the bottom of the traverse axis.
The robot base set areas should be rigid enough and their surfaces should be machined with flatness accuracy as small as 0.2 or less. Fig 5.4 shows

CAUTION

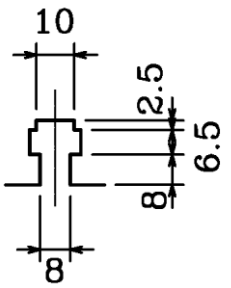
- The robot will suddenly accelerate and decelerate during operation. When installing it on a frame, make sure that the frame has sufficient strength and rigidity.

  If the robot is installed on a frame that does not have sufficient rigidity, vibration will occur while the robot is operating, and could lead to faults.

  When installing the robot on the floor, secure the robot with anchor bolts, etc.
- Install the robot on a level place. Failure to do so could lead to a drop in performance or faults.

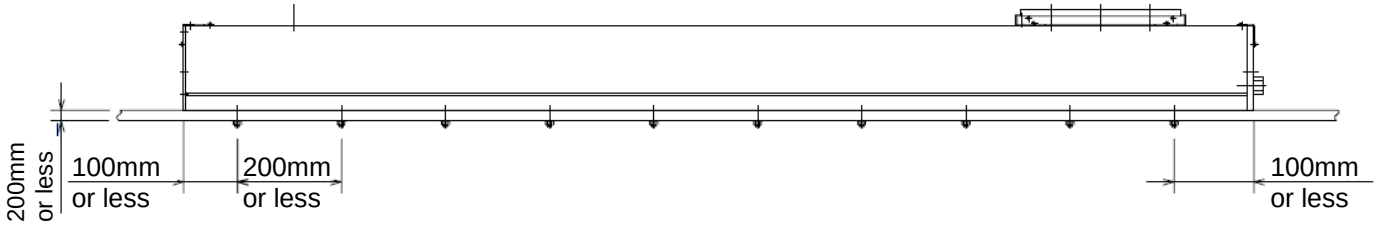


Side view of the traverse axis



Detail of dimension of T-slots of the bottom of traverse axis

Mount the bolts more than ten (10) into the T-slots (one side).
And the bolts of both edges of the robot body should be mounted at pitches of 100 mm or less from the frame edge.



Front view of the traverse axis

Fig. 5.9 Installation method (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

5.3 Precautions for Handling the Teach Pendant

For the teach pendant handing procedure, see "4.3 Precautions for Handling the Teach Pendant".

5.4 Safety Measures

For safety measures, see "4.4 Safety Measures".

6. Installing the Controller (TS3000 and TS3000E)

6.1 External Dimensions

External view of the controller TS3000 and TS3000E are shown in Fig. 6.1.

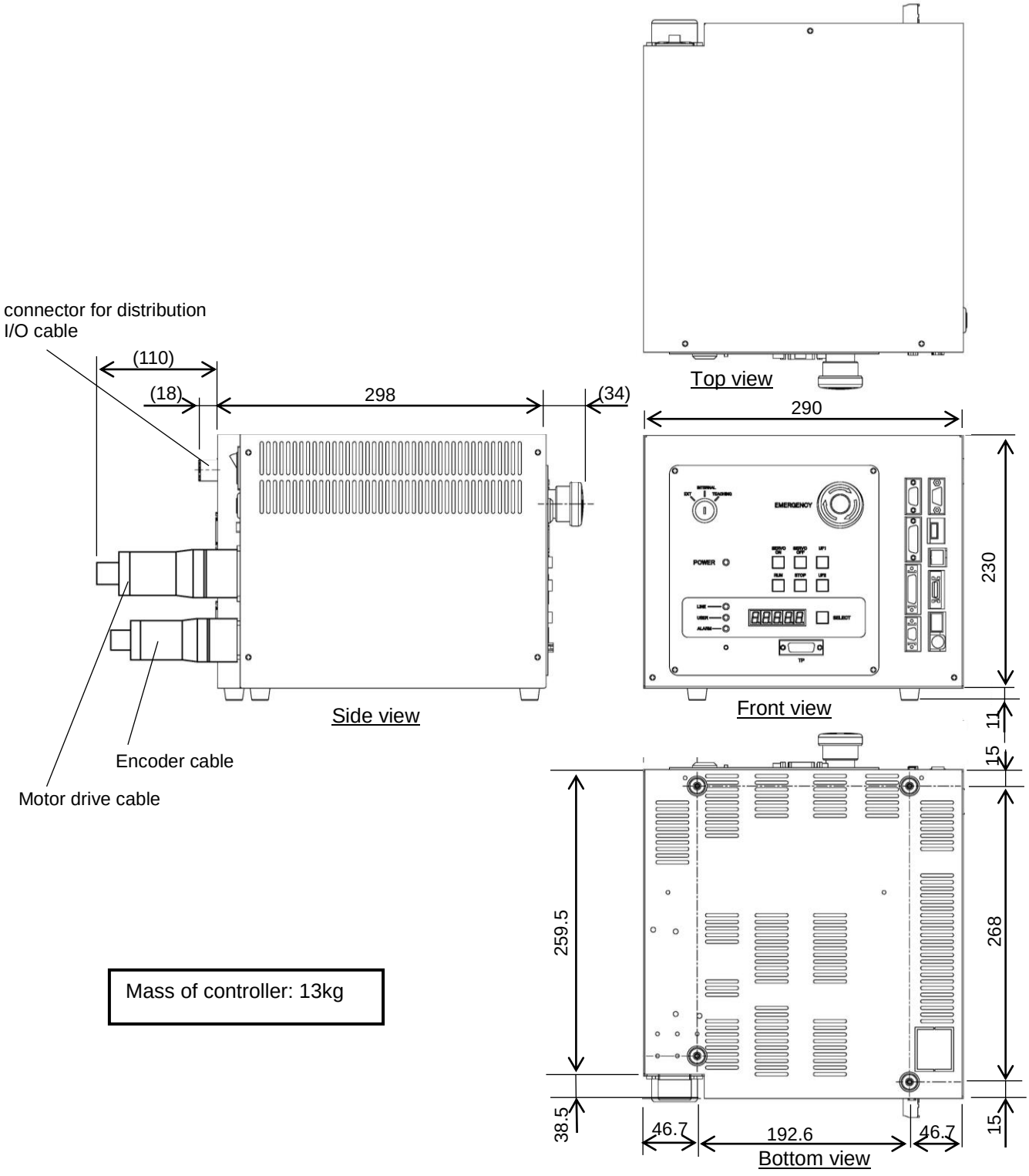


Fig. 6.1 External view of controller (TS3000 and TS3000E)

6.2 Precautions for Direct Installation

It is necessary to provide a clearance of 50 mm or more in the horizontal direction and a clearance of 100 mm or more in the upward direction near the controller.

CAUTION

- Provide a ventilation space at the side of the controller so that the air vent holes are not blocked. The space equal to the length of the legs should be kept below the lower surface.
  Failure to do so could cause the cooling performance to drop or to faults.
- DO NOT stack the controllers.
- DO NOT place any object on top of the controller.

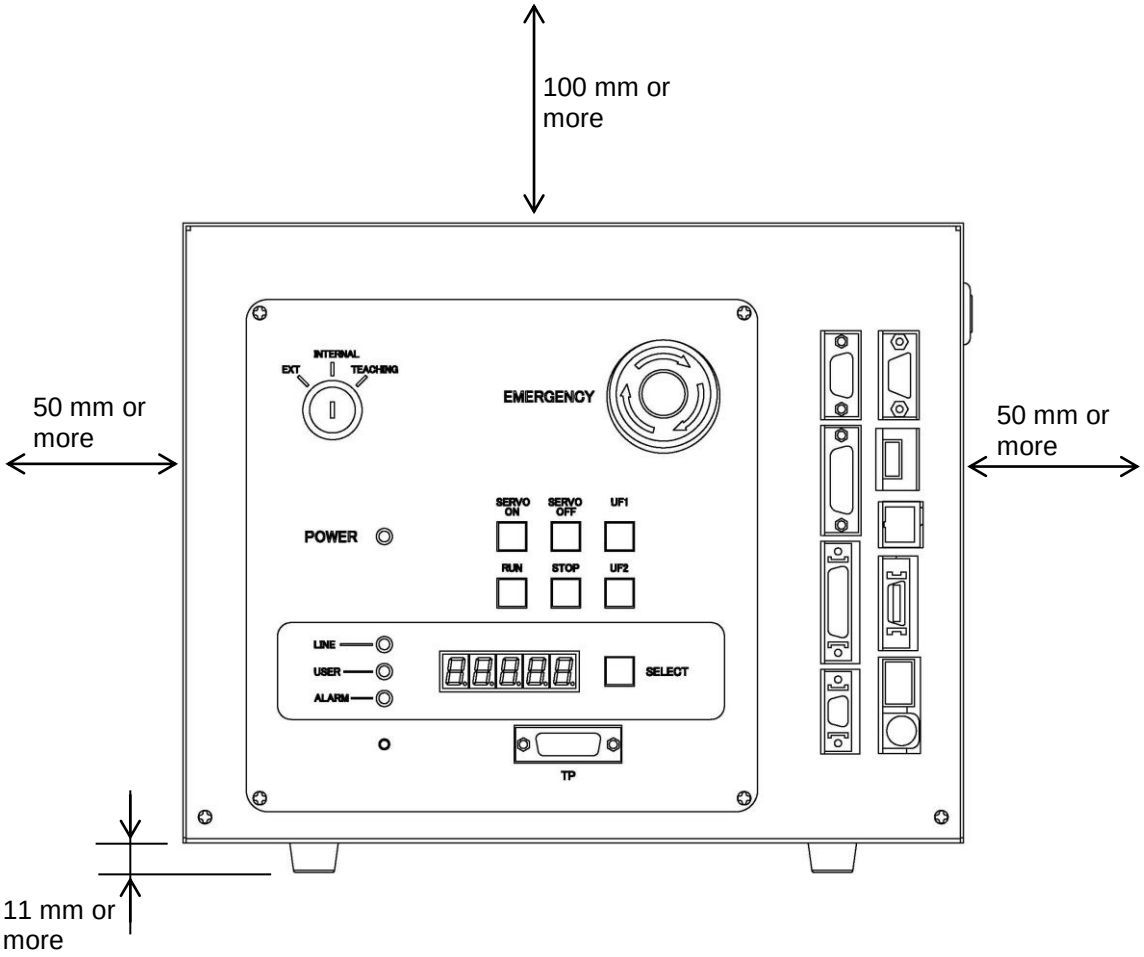


Fig. 6.2 Controller ventilation space (TS3000 and TS3000E)

6.3 Rack Mounting Dimensions

When mounting the robot controller TS3000 and TS3000E in a rack, mount the side brackets using the screw holes provided on both ends of the front panel, and secure the controller to the rack. The side brackets ① in Fig. 6.3 are optional.

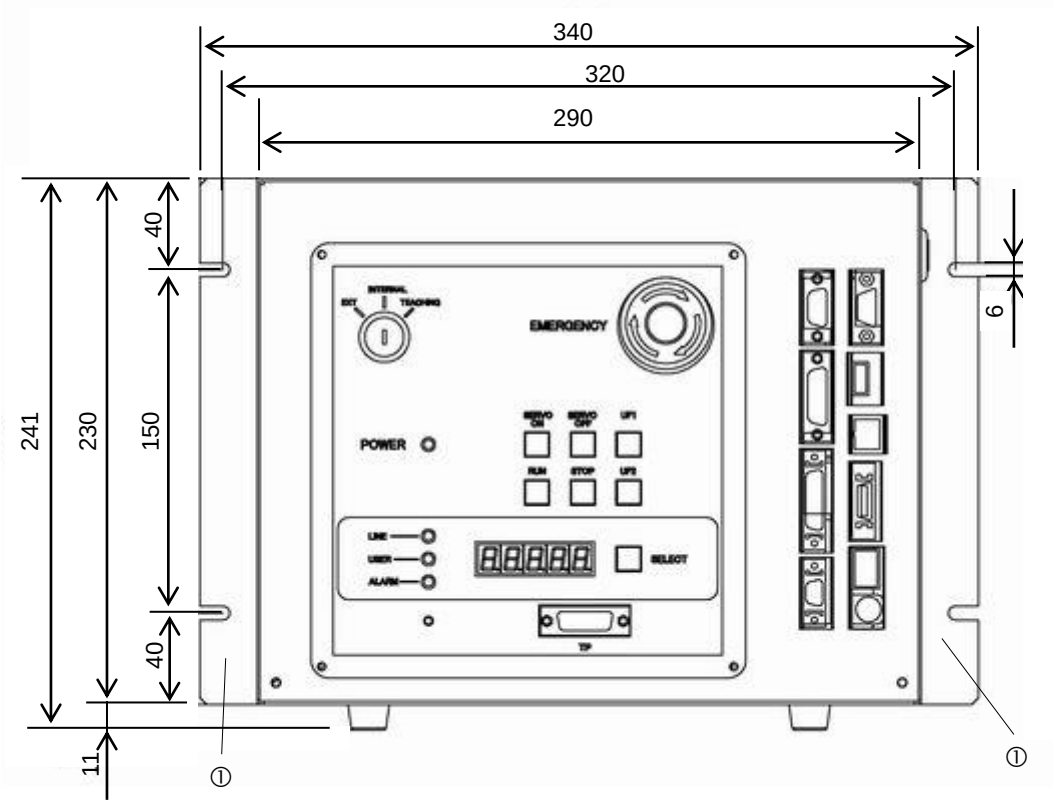


Fig. 6.3 Screw hole dimensions for securing controller (TS3000 and TS3000E)

6.4 Precautions for Rack Mounting

Pay attention to the following matters when rack-mounting the robot controller TS3000 and TS3000E.

a) When mounting the robot controller in a rack, use the screw holes provided on both ends of the front panel, and secure the controller. (Optional side brackets are required.)

! CAUTION

- If the rack is completely sealed, provide holes to allow the air to be let out, force-ventilate the rack with a fan, or cool it indirectly so that the heat will not be trapped in the rack.

  If the heat is trapped in the rack or controller, faults could occur.

For maintenance, the upper cover should be removed. (See Fig. 6.4.)

Install the controller for enabling easy access to the controller when conducting maintenance. In particular, when the controller is mounted in a rack, the controller must be removed from the rack during maintenance. Specifically, be careful of the following points.

1) Arrange the cables around the rear side of the controller (so that the controller can be removed).
2) Arrange the cables between the controller and control panel when the control panel is separated.
3) Connect all cables in such a manner that the robot can be operated even if the controller is removed from the rack.

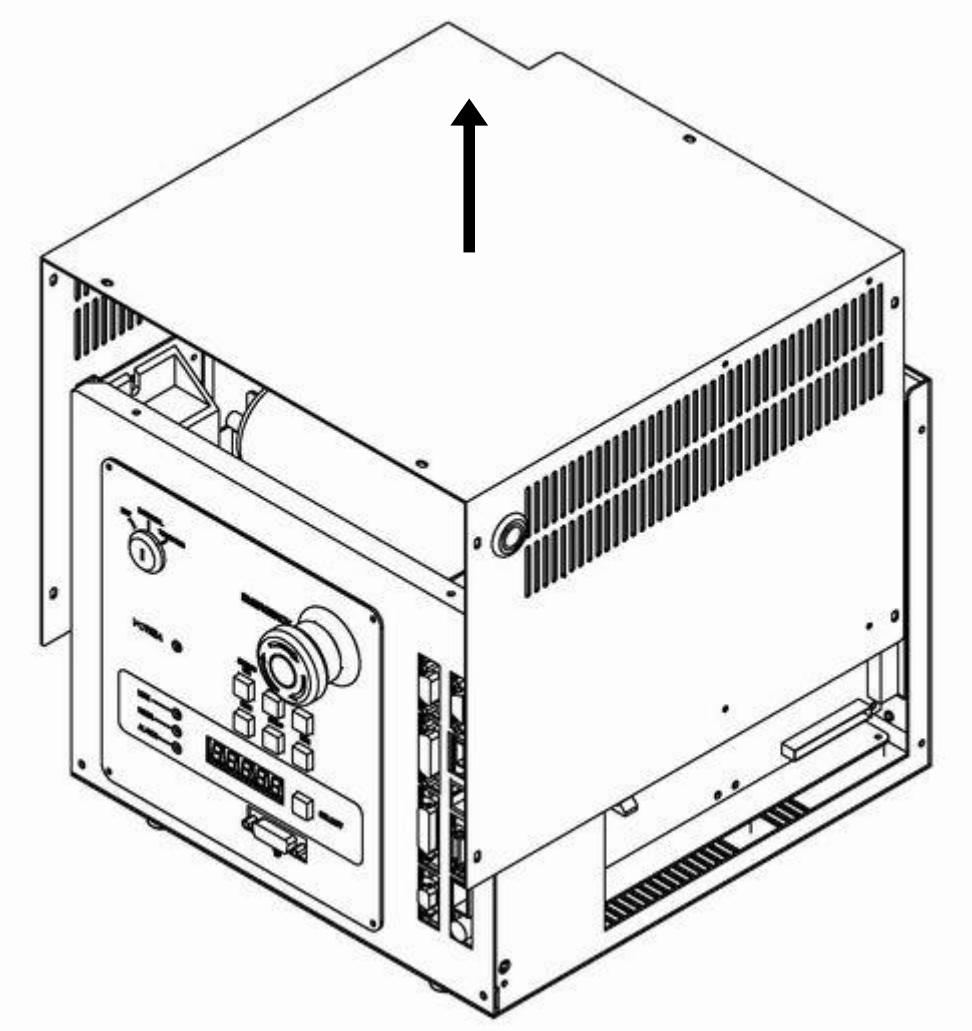


Fig. 6.4 Removing upper cover (TS3000 and TS3000E)

b) When the controller is mounted on the rack, the weight of the controller should be supported with the legs of the controller. Screw holes for rack-mounting the controller are used for securing the controller panel, and the weight of the controller cannot be supported only with these screws.

c) On the front of the controller, a clearance of approx. 90 mm should be provided for connecting the connector of the teach pendant. Even if the teach pendant is not used, a clearance of approx. 60 mm is required for connecting a dummy plug.

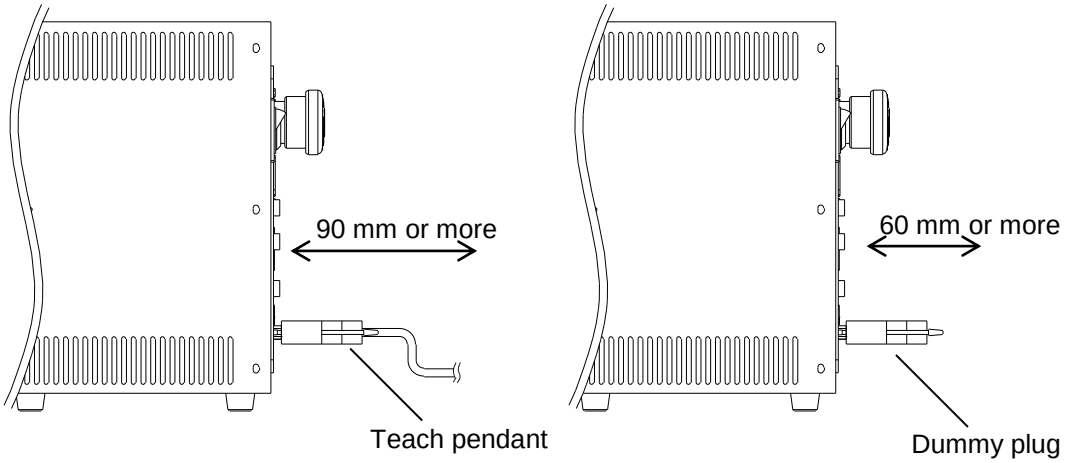


Fig. 6.5 Clearance of controller front side (TS3000 and TS3000E)

d) On the back of the controller, a clearance of approx. 110 mm should be provided for connecting the connector of the teach pendant.

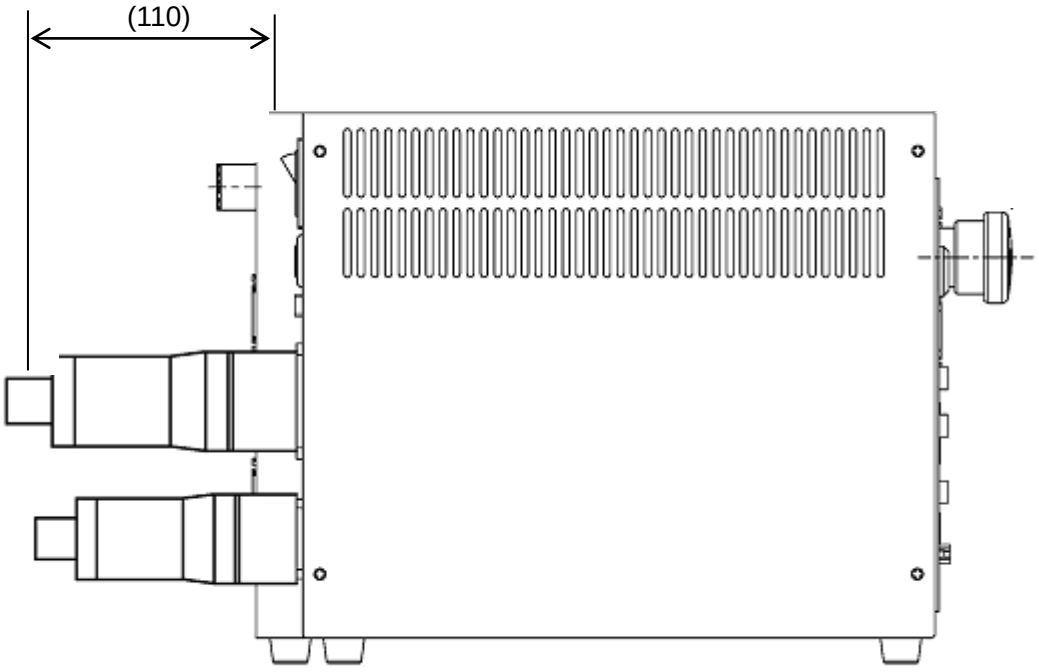
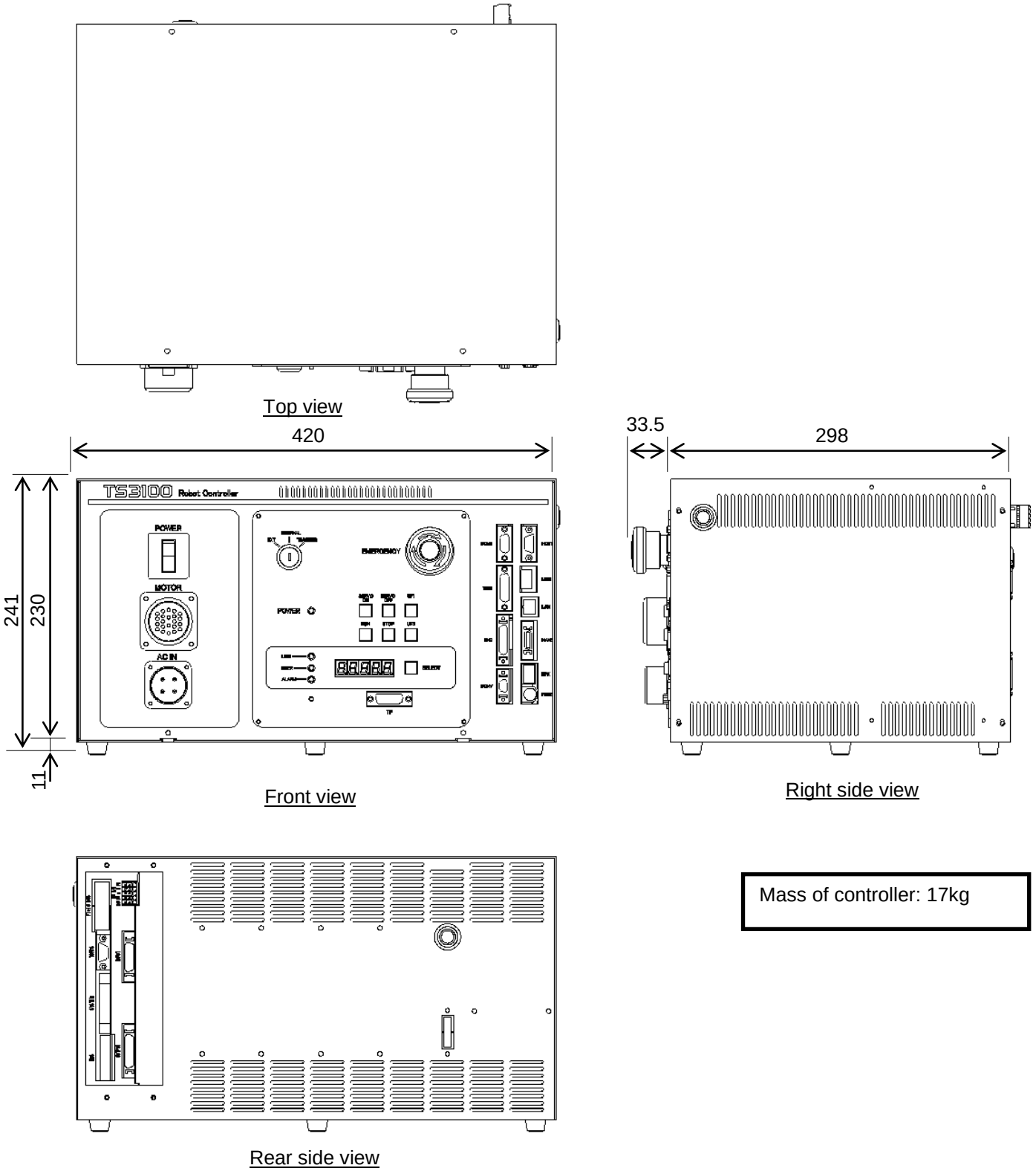


Fig. 6.6 Clearance of controller back side (TS3000 and TS3000E)

7. Installing the Controller (TS3100 and TS3100E)

7.1 External Dimensions

External view of the controller TS3100 and TS3100E are shown in Fig. 7.1.



Mass of controller: 17kg

Fig. 7.1 External view of controller (TS3100 and TS3100E)

7.2 Precautions for Direct Installation

It is necessary to provide a clearance of 50 mm or more in the horizontal direction and a clearance of 100 mm or more in the upward direction near the controller.
When installing the controller lengthwise, be sure to locate the left lateral side (side is indicated by arrows) for sideways installation at the bottom side. (The lengthwise installation is an option.)

CAUTION

- Provide a ventilation space at the side of the controller so that the air vent holes are not blocked. The space equal to the length of the legs should be kept below the lower surface.
  Failure to do so could cause the cooling performance to drop or to faults.
- DO NOT stack the controllers.
- DO NOT place any object on top of the controller.

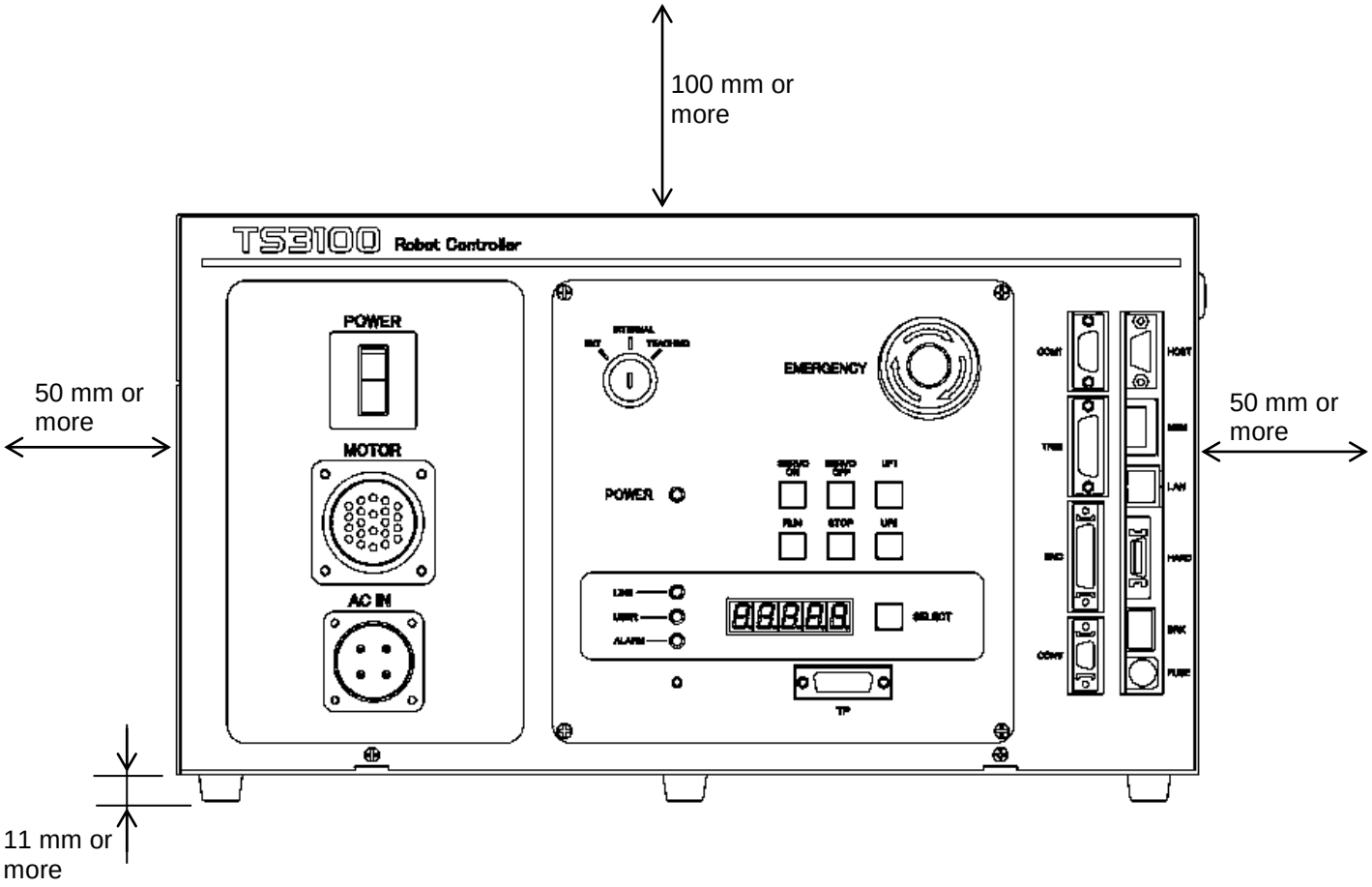


Fig. 7.2 Controller ventilation space (TS3100 and TS3100E)

7.3 Rack Mounting Dimensions

When mounting the robot controller TS3100 and TS3100E in a rack, mount the side brackets using the screw holes provided on both ends of the front panel, and secure the controller to the rack. The side brackets ① in Fig. 7.3 are optional.

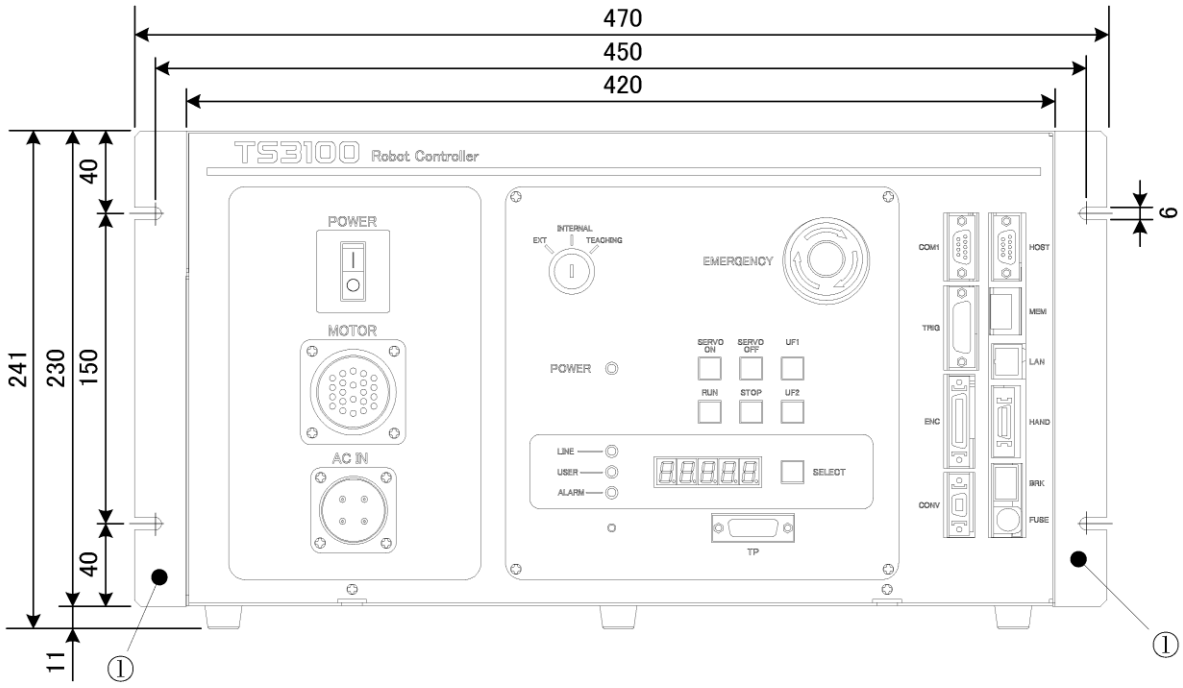


Fig. 7.3 Screw hole dimensions for securing controller (TS3100 and TS3100E)

7.4 Precautions for Rack Mounting

Pay attention to the following matters when rack-mounting the robot controller TS3100 and TS3100E.

a) When mounting the robot controller in a rack, use the screw holes provided on both ends of the front panel, and secure the controller. (Optional side brackets are required.)

! CAUTION

- If the rack is completely sealed, provide holes to allow the air to be let out, force-ventilate the rack with a fan, or cool it indirectly so that the heat will not be trapped in the rack.

  If the heat is trapped in the rack or controller, faults could occur.

For maintenance, the upper cover should be removed. (See Fig. 7.4.)

Install the controller for enabling easy access to the controller when conducting maintenance. In particular, when the controller is mounted in a rack, the controller must be removed from the rack during maintenance. Specifically, be careful of the following points.

1) Arrange the cables around the rear side of the controller (so that the controller can be removed).
2) Arrange the cables between the controller and control panel when the control panel is separated.
3) Connect all cables in such a manner that the robot can be operated even if the controller is removed from the rack.

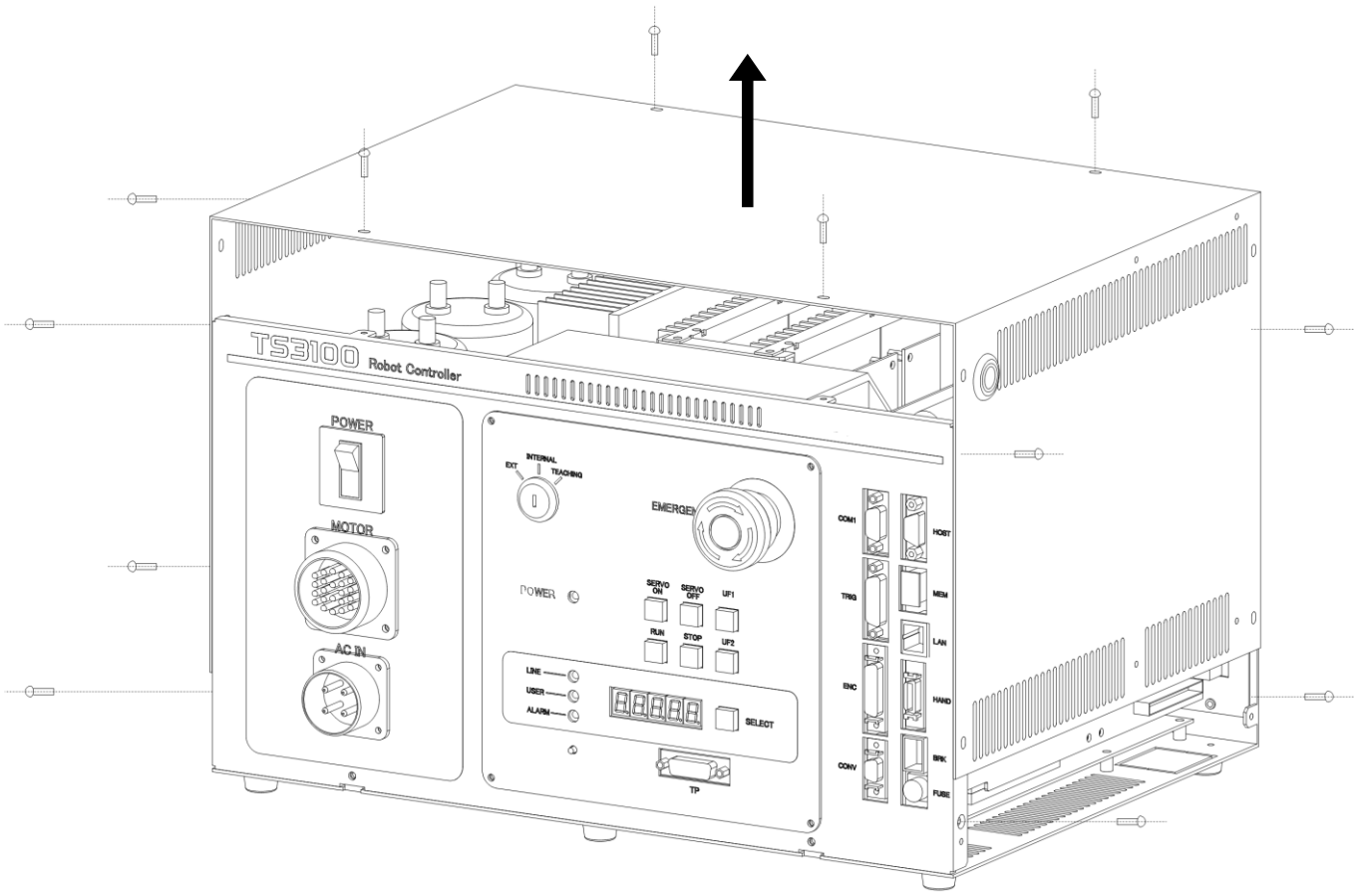


Fig. 7.4 Removing upper cover (TS3100 and TS3100E)

b) When the controller is mounted on the rack, the weight of the controller should be supported with the legs of the controller. Screw holes for rack-mounting the controller are used for securing the controller panel, and the weight of the controller cannot be supported only with these screws.

c) On the front of the controller, a clearance of approx. 90 mm should be provided for connecting the connector of the teach pendant. Even if the teach pendant is not used, a clearance of approx. 60 mm is required for connecting a dummy plug.

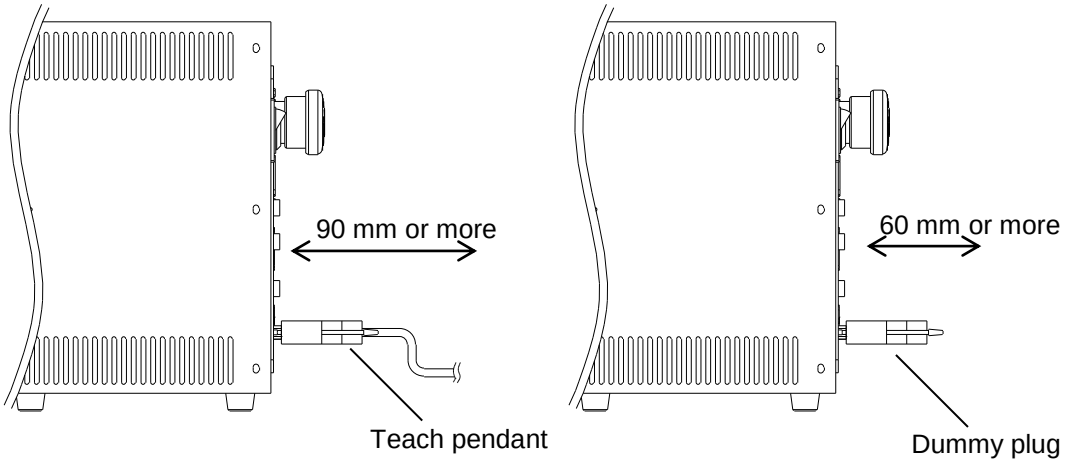


Fig. 7.5 Clearance of controller front side (TS3100 and TS3100E)

d) On the back of the controller, a clearance of approx. 110 mm should be provided for connecting the connector of the teach pendant.

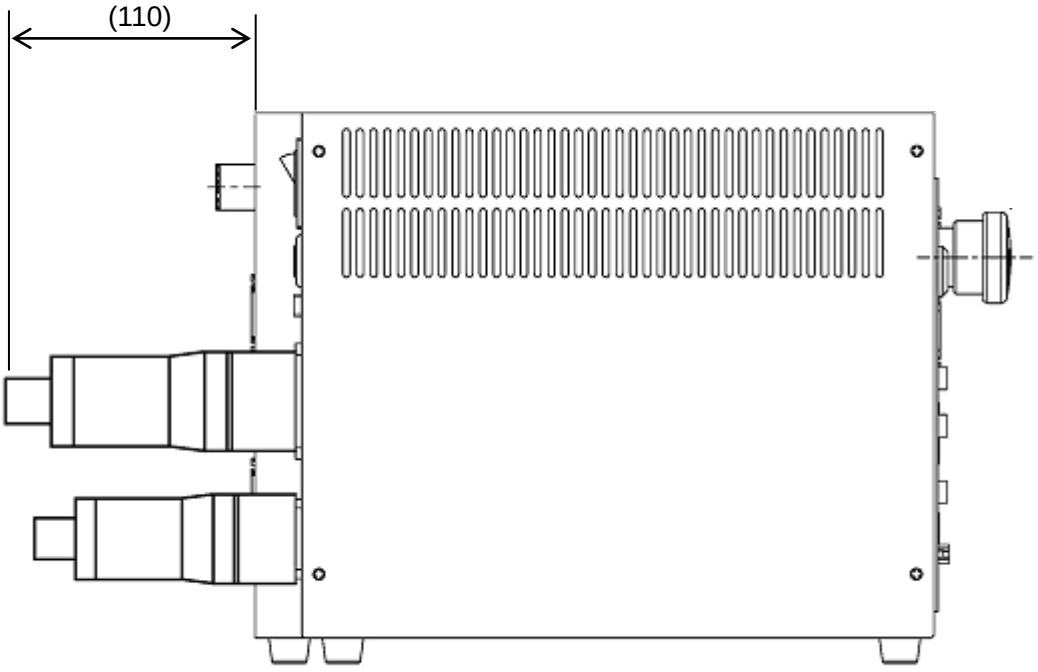


Fig. 7.6 Clearance of controller back side (TS3100 and TS3100E)

8. System Connections (SR-404HC and SR-624HC / TS3000 and TS3000E)

8.1 Cable Wiring

This section describes the various types of cables and connectors and explains how these are to be connected.

8.1.1 Connector Arrangement on the Controller

The cables connected to the robot controller are shown in Fig. 8.1.

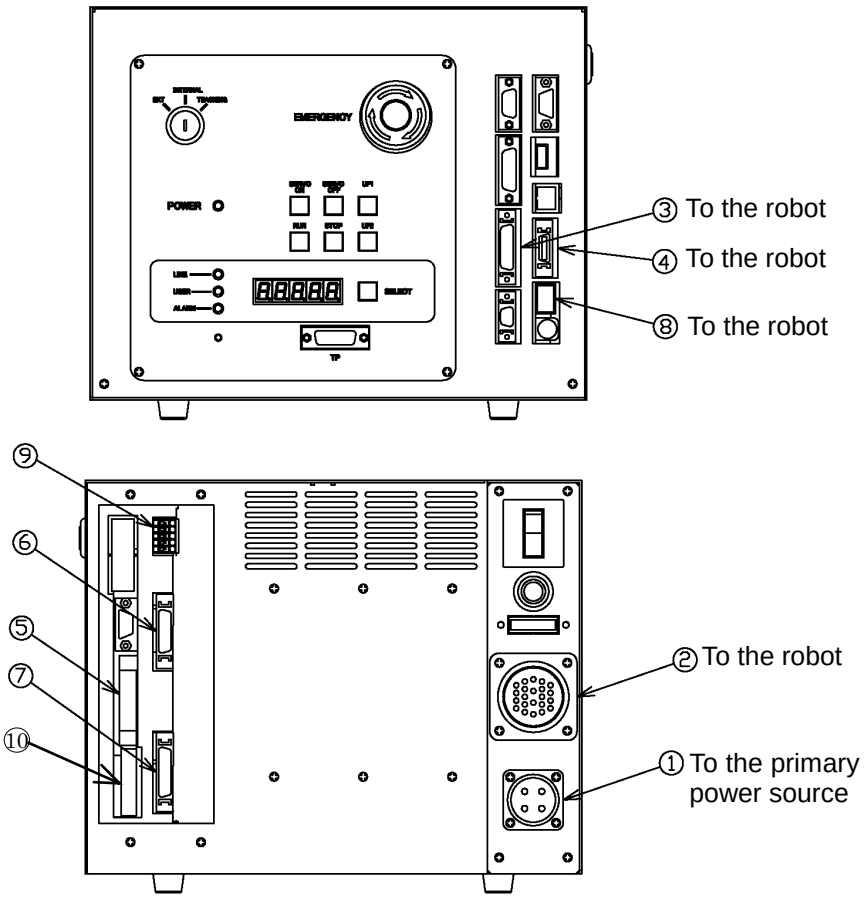


Fig. 8.1 Robot controller connector arrangement (TS3000 and TS3000E)

① Power cable (ACIN)
② Motor cable (MOTOR)
③ Encoder cable (ENC)
④ Robot control signal cable (HAND)
⑤ External operation input signal cables (SYSTEM)
⑥ External operation input signal cables (INPUT)
⑦ External operation input signal cables (OUTPUT)
⑧ Brake signal cable (BRK)
⑨ Distribution I/O cable (EXT-I/O)
⑩ External input/output power cable "EMS"

In the subsequent paragraphs, we explain how to connect cables ① to ④ and ⑩ inclusive. For information on how to connect cables ⑤ to ⑨, refer to the TS3000 Interface Manual STE80729.

8.1.2 Connecting the Power Cable "ACIN" (① of Fig. 8.1) (plug connector attached)

The power cable is used to supply the main AC power to the controller.

Table 8.1 Power supply specifications (TS3000 and TS3000E)

Item	Specifications
Power supply	SR-404HC, Single phase, 190 - 240 VAC, 50/60 Hz ±1 Hz SR-624HC, Single phase, 200 - 240 VAC, 50/60 Hz ±1 Hz
Power capacity	2.6 kVA
Instantaneous power failure	Within 40 msec
Grounding	JIS class D

The connector is ACIN (① of Fig. 8.1).

ACIN plug connector	Type:JL04V–6A18–10SE–EB	Maker: Japan Aviation Electronics Industry, Ltd
ACIN cable clamp	Type: JL04–18CK	Maker: Japan Aviation Electronics Industry, Ltd
Wire	3.5 mm^2 - 5.5 mm^2	

As the cable is not an accessory, use the attached plug connector connected to ACIN on the controller side to manufacture a cable.
Wires are to be soldered to the connector.

! DANGER

- Be sure to use the designated wire. Failure to do so could lead to fires or faults.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

For the terminal arrangement, see Para. 8.1.9.

! CAUTION

- Unless the main power is normally supplied to the controller due to phase defect or voltage drop, an error of "8–027 Slow Charge error" occurs. When this happens, make sure that the maser power voltage at the controller power connector satisfies the specified input power of the controller, and that the same voltage is stabilized.
- For details of the 8–027 error, see Para. 13.7 of the TS3000 Operator's Manual ST80729.

8.1.3 Connecting Motor Cable "MOTOR" (② of Fig. 8.1, Fig. 8.2 CN2; Cable included)

The motor cable connects the controller and robot, and supplies the power required to rotate the motor from the controller servo driver to each axis motor of the robot.
The controller connector is MOTOR; the robot connector is CN2.

8.1.4 Connecting Encoder Cable "ENC" (③ of Fig. 8.1, Fig. 8.2 CN3; Cable included)

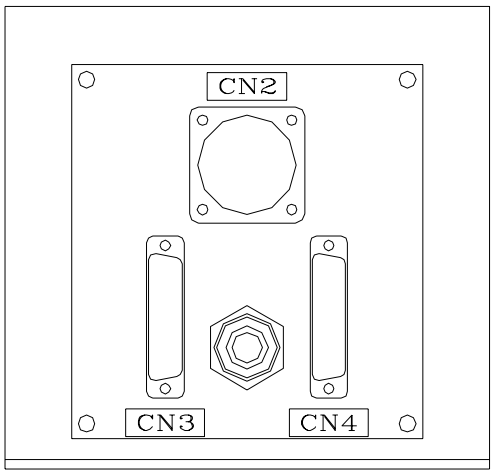
The encoder cable is a signal line used to transmit data from each axis encoder (i.e., device used to detect the revolution (and thereby the position) of each axis) to the controller.
The controller connector is ENC; the robot connector is CN3.

8.1.5 Connecting Robot Control Signal Cable "HAND" (④ of Fig. 8.1, Fig. 8.2 CN4; Cable included)

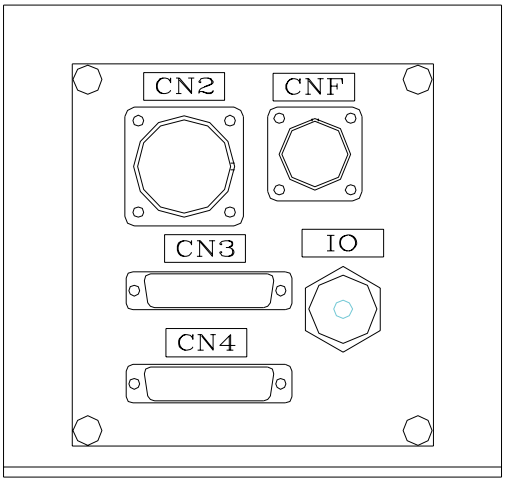
The robot control signal cable is used to input/output the robot control signal for robot operation (e.g., hand operation).
The controller connector is HAND; the robot connector is CN4.
The cable is not connected to the controller in the special specifications in Fig. 8.2.
See the specifications for the control signals.

8.1.6 Connecting Brake Signal Cable between "BRK" (⑧ of Fig. 8.1, Fig. 8.2 CN4; Cable included)

The brake signal cable is used to turn ON/OFF the brake for locking the motor shaft.
The controller connector is BRK; the robot connector is CN4.



Standard specifications



Special specifications
(The robot control signal cable is directly connected to the customer's device.)

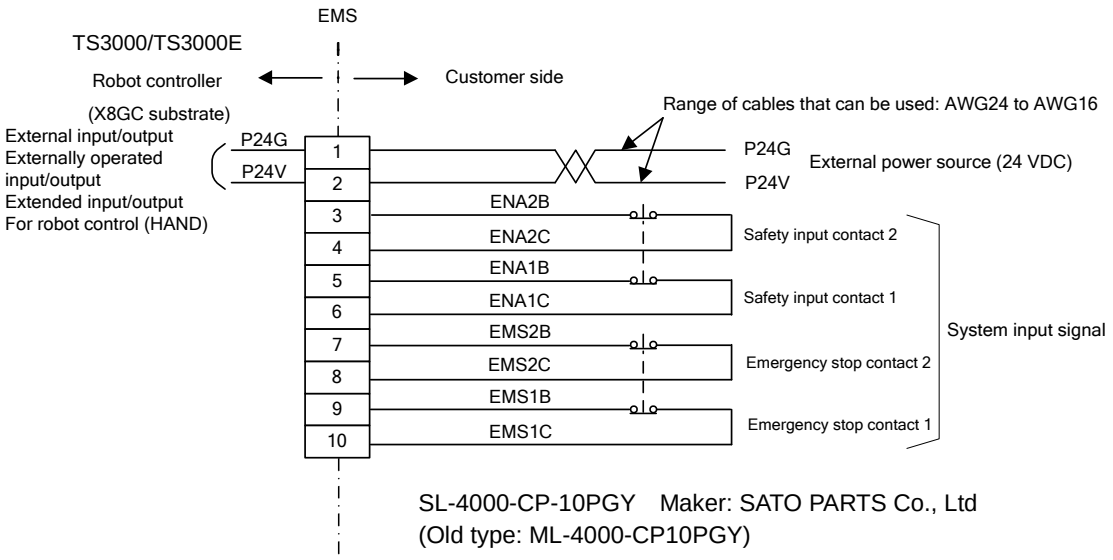
Fig. 8.2 Robot connector layout diagrams (SR-404HC and SR-624HC)

8.1.7 Connecting the External Input/Output Power Cable "EMS" (⑩ of Fig. 8.2)

Use the attached connector (SL-4000-CP-10PGYT) to connect the external input/output power cable, similarly to the case of connecting the input signal cable for safety. The P24V power is supplied from the connector "EMS" on the rear of the controller.
The following shows the I/O using the external supply power (24 VDC). Be sure to supply power (24 VDC) from the outside.

- External input/output
- Externally operated input/output
- Extended input/output
- Manual input/output

It should be noted that the controller internal power (24 VDC) is used for the trigger input and brake output.



SL-4000-CP-10PGY Maker: SATO PARTS Co., Ltd
(Old type: ML-4000-CP10PGY)

The connector EMS compatible cables are "AWG24 through 16").
Select the optimum external power source in conformity to the customer's system specifications (power capacity).
To connect and disconnect the connector EMS, see the description of "Interface Part".

! CAUTION

- Be sure to supply external power (24 VDC). If not, power cannot be turned on, independently of safety measures.

8.1.8 Connecting and Disconnecting Cables

CAUTION

- Before connecting or disconnecting any controller cable, be sure to turn off the main power ("POWER") switch.
- When disconnecting a cable, be sure to pull the plug and not the cord. Otherwise, you may damage the cable.

a) Circular connectors: ACIN, MOTOR
Firstly align the key position, and completely insert the connector on the cable side into the controller connector. Then turn the cable side lock screw to the right to clamp the cable. A loose screw can cause a contact failure or other accident. To avoid this, make sure that the screw is clamped completely.
To disconnect the connectors, turn the lock screw to the left and pull out the cable side connector.

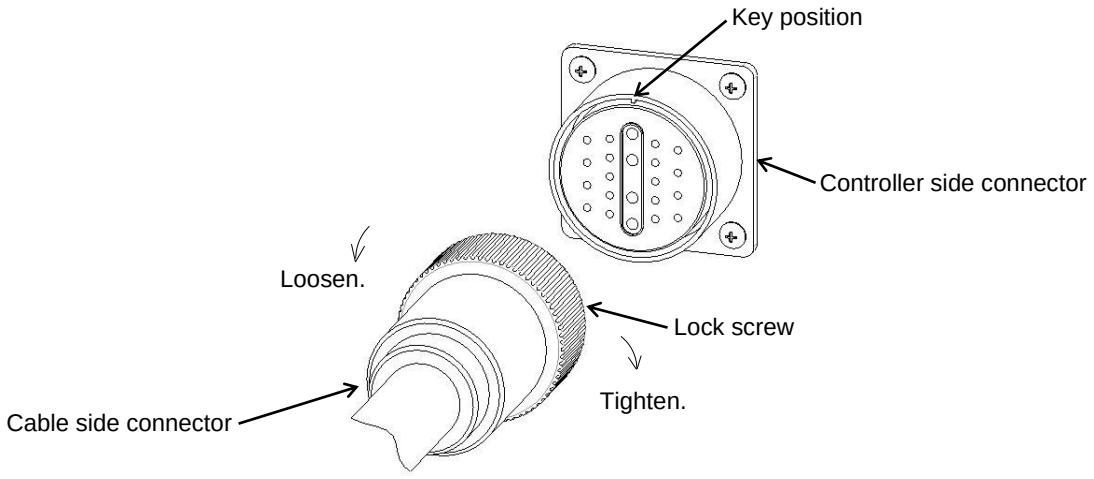


Fig. 8.3 Connecting and disconnecting a circular connector

b) Square connectors: ENC, HAND, SYSTEM, INPUT, OUTPUT, TRIG, CONV
Firstly, completely insert the cable side connector into the controller connector. Then tighten the lock screws on both ends of the cable side connector with a screwdriver. A loose screw can cause a contact failure or other accident. To avoid this, make sure that the screws are clamped completely.
To disconnect the connectors, first loosen the lock screws, then pull out the cable side connector. INPUT, OUTPUT is a quick-operated lock type connector, however.

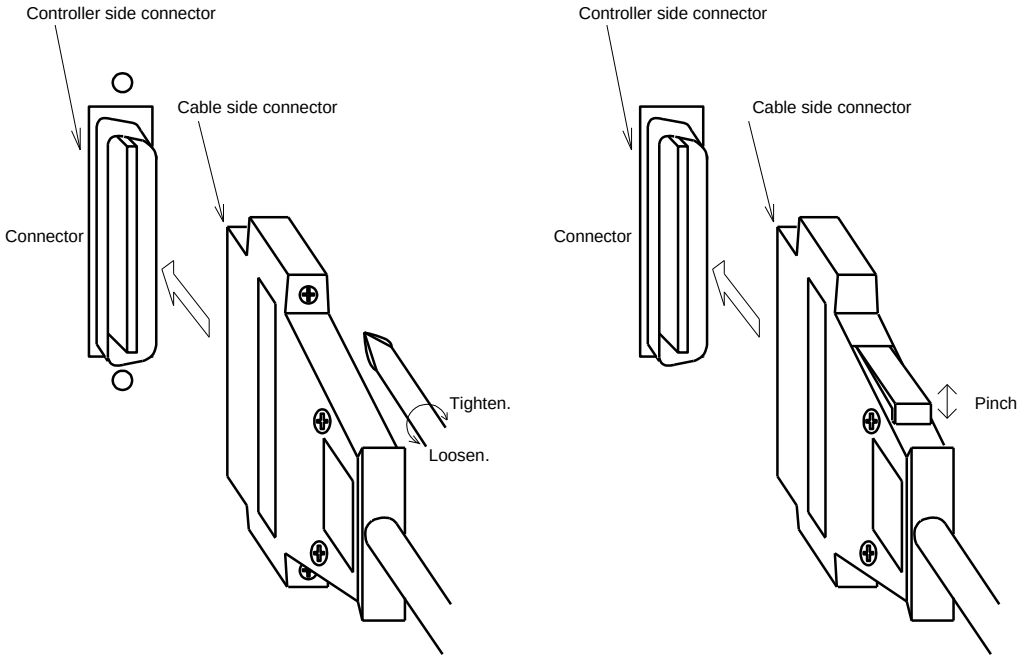
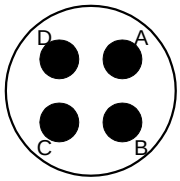


Fig. 8.4 Connecting and disconnecting a square connector

8.1.9 Examples of Connector Terminal Arrangement (Controller Side)

a) Power cable connector ACIN



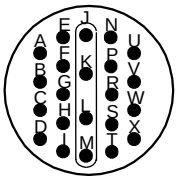
A
B
C
D

Connects to controller.
Type: JL04V–2E18–10PE–B
Manufacturer:
Japan Aviation Electronics Industry, Ltd
SR-404HC
Single phase, 190 - 240 VAC, 50/60 Hz
SR-624HC
Single phase, 200 - 240 VAC, 50/60 Hz
Grounding (Class D)

! DANGER

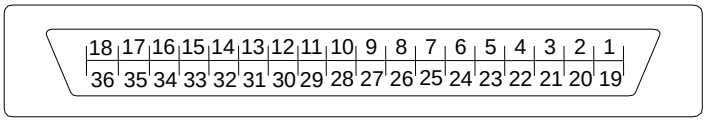
- Completely connect the grounding cable.
  Otherwise, an electric shock or fire may be caused if a fault or electric leak occurred. Or misoperation may be caused by noise.

b) Motor cable connector MOTOR



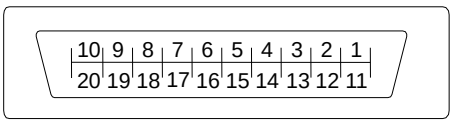
Connects to controller.
Type: JL04V-2A28-11SE
Manufacturer:
Japan Aviation Electronics Industry, Ltd

c) Encoder cable connector ENC



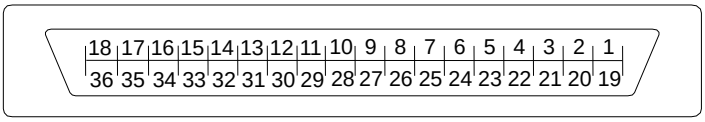
Connects to controller.
Type: 52986-3659
Manufacturer: MOLEX

d) Robot control signal cable connector HAND



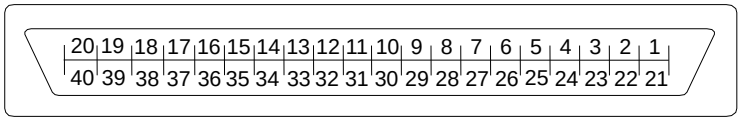
Connects to controller.
Type: 52986-2079
Manufacturer: MOLEX

e) General-purpose input signal cable connector INPUT



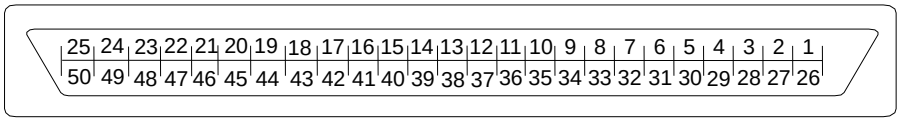
Connects to controller.
Type: DHA-RC36-R132N-FA
Manufacturer: DDK

f) General-purpose output signal cable connector OUTPUT



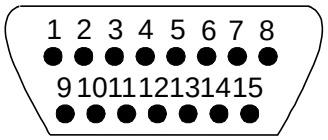
Connects to controller.
Type: DHA-RC40-R132N-FA
Manufacturer: DDK

g) System input/output signal cable connector SYSTEM



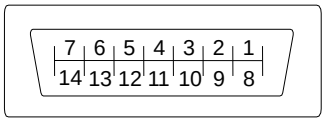
Connects to controller.
Type: 52986-5079
Manufacturer: MOLEX

h) Trigger input connector TRIG



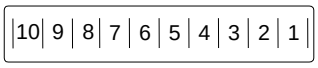
Connects to controller.
Type: XM2C-1542-112L
Manufacturer: OMRON

i) Encoder cable connector CONV



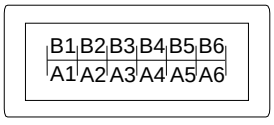
Connects to controller.
Type: 52986-1479
Manufacturer: MOLEX

j) Emergency stop, safety input, and external P24V supply connector EMS



Connects to controller.
Type: SL-4000CWJH-10PGY
Manufacturer: SATO PARTS

k) Brake connector BRK



Connects to controller.
Type: 1-1827876-6
Manufacturer:
Tyco Electronics AMP

8.2 Controller Connector Signals

8.2.1 Connector Signal Connection Diagrams

Diagrams showing which signals correspond to which terminals are shown in Section 2 of the TS3000 Interface Manual STE80729.

8.2.2 Jumpers for Safety Related Signals

The following system input signals are provided to serve for the safety purpose.

System input signals SYSTEM-12 (STOP)
SYSTEM-16 (SVOFF)
SYSTEM-14 (BREAK)
EMS-7, 8 (EMS2B - EMS2C)
EMS-9, 10 (EMS1B - EMS1C)
EMS-3, 4 (ENA2B - ENA2C)
EMA-5, 6 (ENA1B - ENA1C)

These signals are already jumpered for the connectors provided for the TS3000 and TS3000E robot controller. If you wish to use or change them, therefore, you should remove the jumpers and rewire as appropriate. If you plan to use the robot without using system input signals, be sure to connect the attached connectors to the controller side SYSTEM, EMS connectors.
Unless the following signals are used as the system signals, jumper them also.
SYSTEM-15 (LOW_SPD)
SYSTEM-13 (CYCLE)

Table 8.2 Connector jumpers

SYSTEM		EMS	
12-17 (18)	14-17 (18)	3-4	5-6
16-17 (18)	(13-17 (18))	7-8	9-10
(15-17 (18))	-		

! CAUTION

- Unless the signals of SVOFF and emergency stop contacts 1, 2 are jumpered, the controller servo power cannot be turned on.
- Unless the CYCLE signal is jumpered, the controller enters the cycle operation mode.
- Unless the LOW_SPD signal is jumpered, the robot is operated at low speed during automatic operation.
- Unless the STOP signal is jumpered, automatic operation of the robot is not possible.
- Unless the BREAK signal is jumpered, automatic operation of the robot is not possible.

8.3 Separating Control Panel from Controller

8.3.1 Removing Control Panel

Remove the control panel in the following manner.

a) Loosen the four (4) screws at the four (4) corners, which secure the control panel.
b) Remove these screws, then carefully draw out the control panel toward your side.
   Caution: Be careful of the cable connected on the rear side.

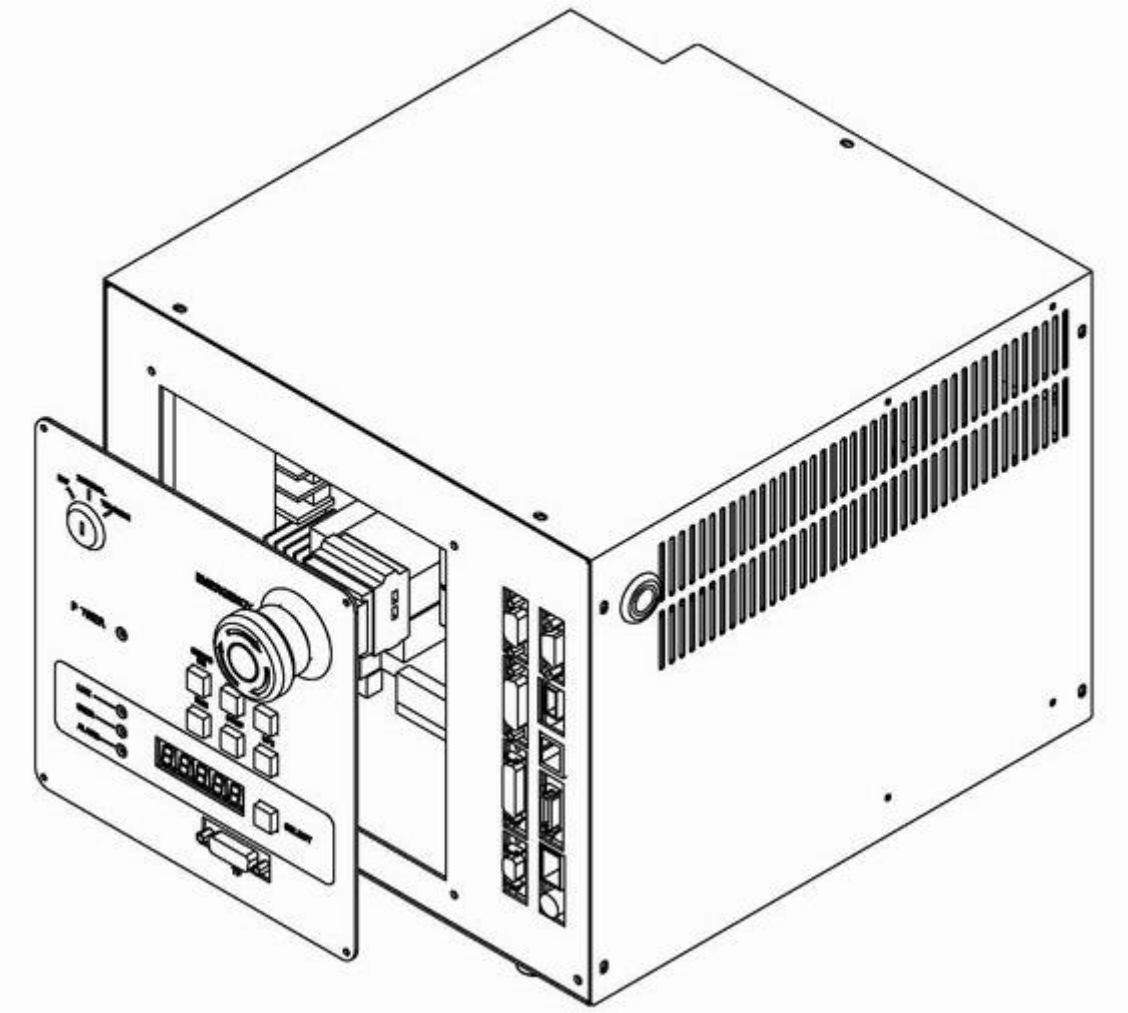


Fig. 8.5 Removing control panel (TS3000 and TS3000E)

8.3.2 Cable between Controller and Control Panel (option)

The cable required to connect the control panel and controller when they are installed separately can be provided optionally.

8.3.3 Control Panel Mounting Dimensions

The dimensions of mounting the control panel are shown in Fig. 8.6.
Cross truss head screws (ϕ3 x 6, ZN3–B) are used.

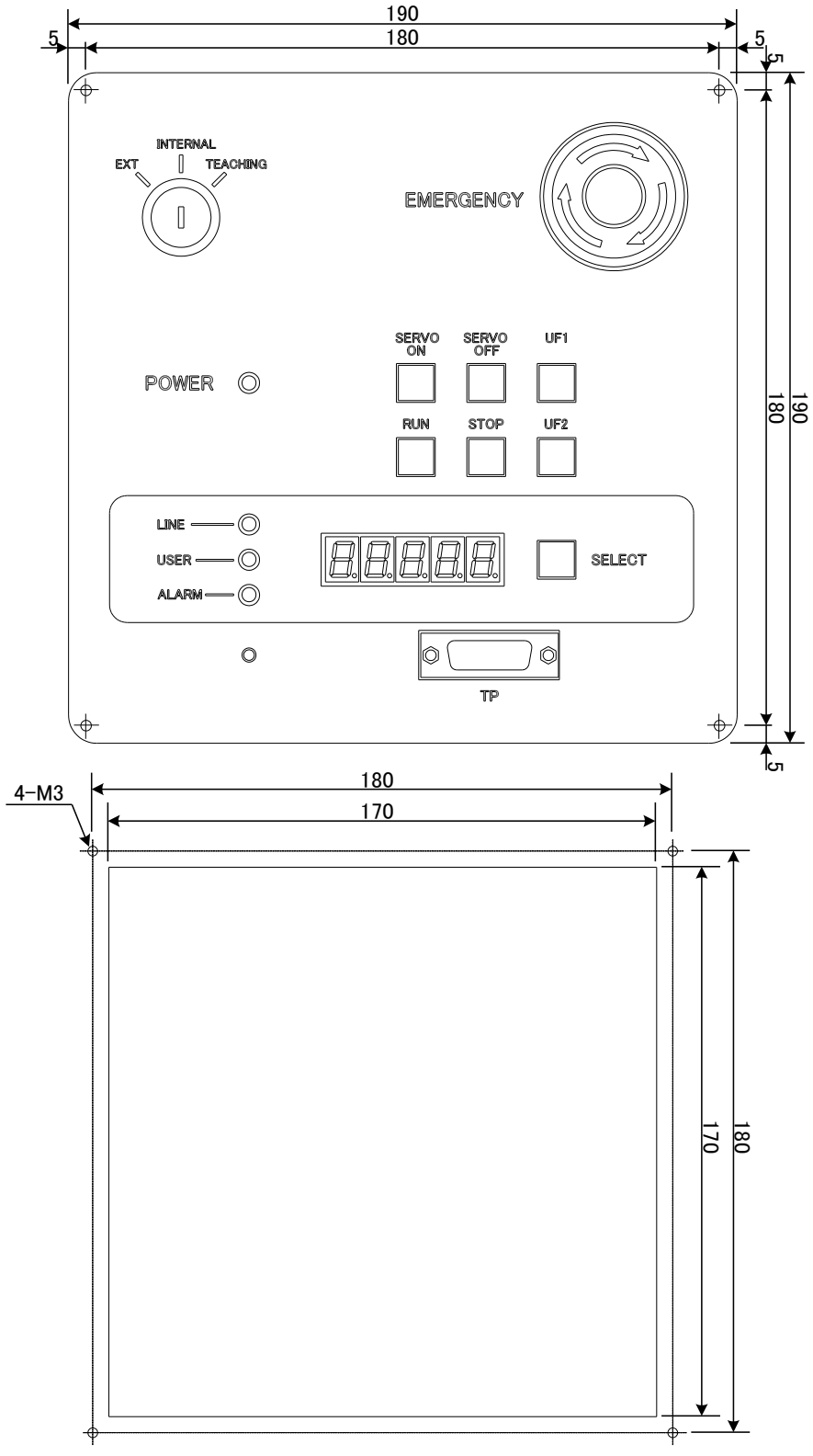


Fig. 8.6 Control panel mounting dimensions

8.3.4 Mounting Dummy Panel on Controller (option)

When the control panel has been disengaged from the controller, mount a dummy panel on the place where the control panel was set before, as shown in Fig. 8.7. The dummy panel, mounting parts, etc. are provided optionally.

a) Connect the cable connector which was disconnected when separating the controller from the control panel, to the rear side of the dummy panel, then screw both ends of the connector. When mounting the connector, use the cross truss screws (ϕ3 x 6, ZN3–B).

b) Screw the dummy panel into the controller.

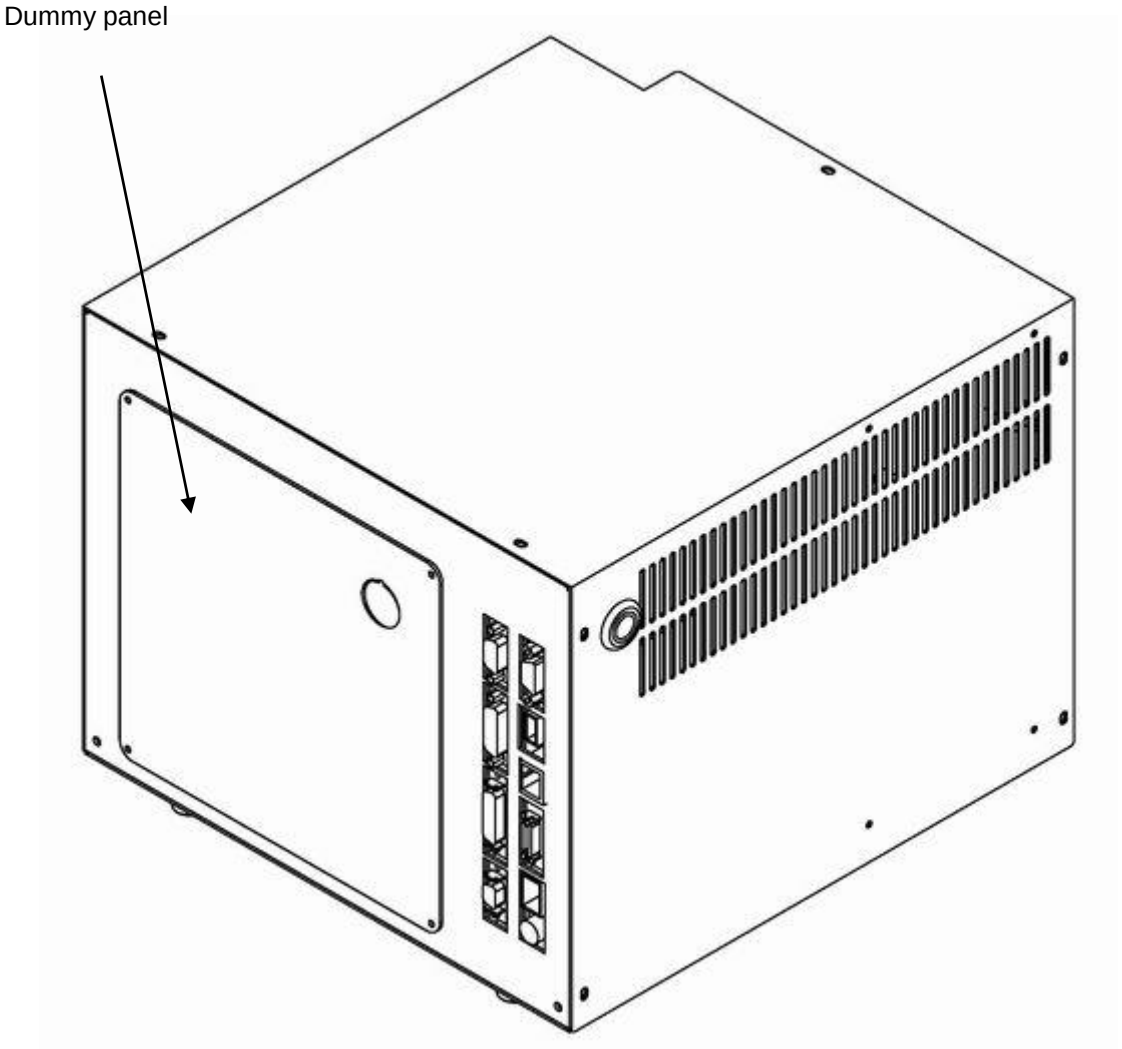


Fig. 8.7 Mounting dummy panel (TS3000 and TS3000E)

8.3.5 Dimensions when Separating Control Panel

Fig. 8.8 shows the connections of the control panel and dummy panel. Provide a clearance of 50 mm or more (with cover, 60 mm or more) on the rear side of the separated control panel.
When the cable is connected to the dummy panel of the controller, provide a clearance of 80 mm or so in front of the controller as the cable connector sticks out of the panel surface.

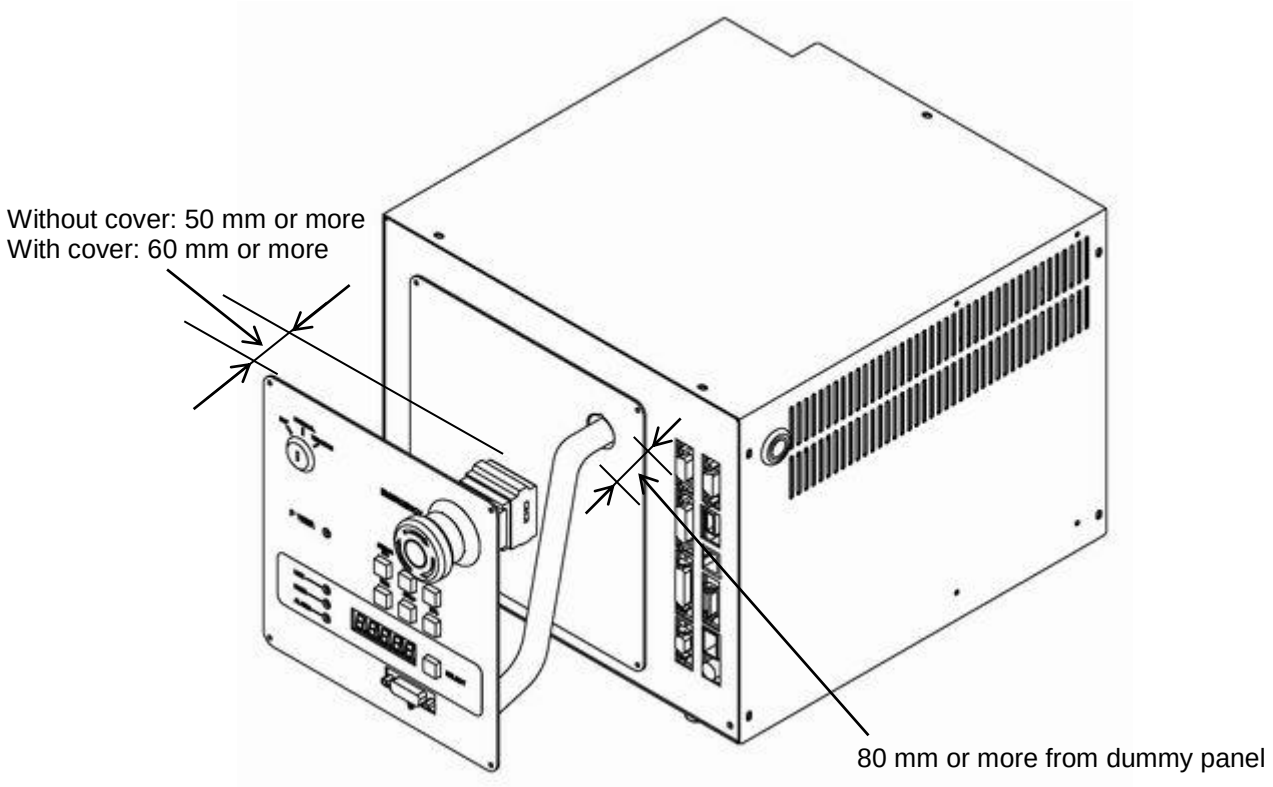


Fig. 8.8 Cable connections of dummy panel and control panel (TS3000 and TS3000E)

9. System Connections (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L / TS3100 and TS3100E)

9.1 Cable Wiring

This section describes the various types of cables and connectors and explains how these are to be connected.

9.1.1 Connector Arrangement on the Controller

The cables connected to the robot controller are shown in Fig. 9.1.

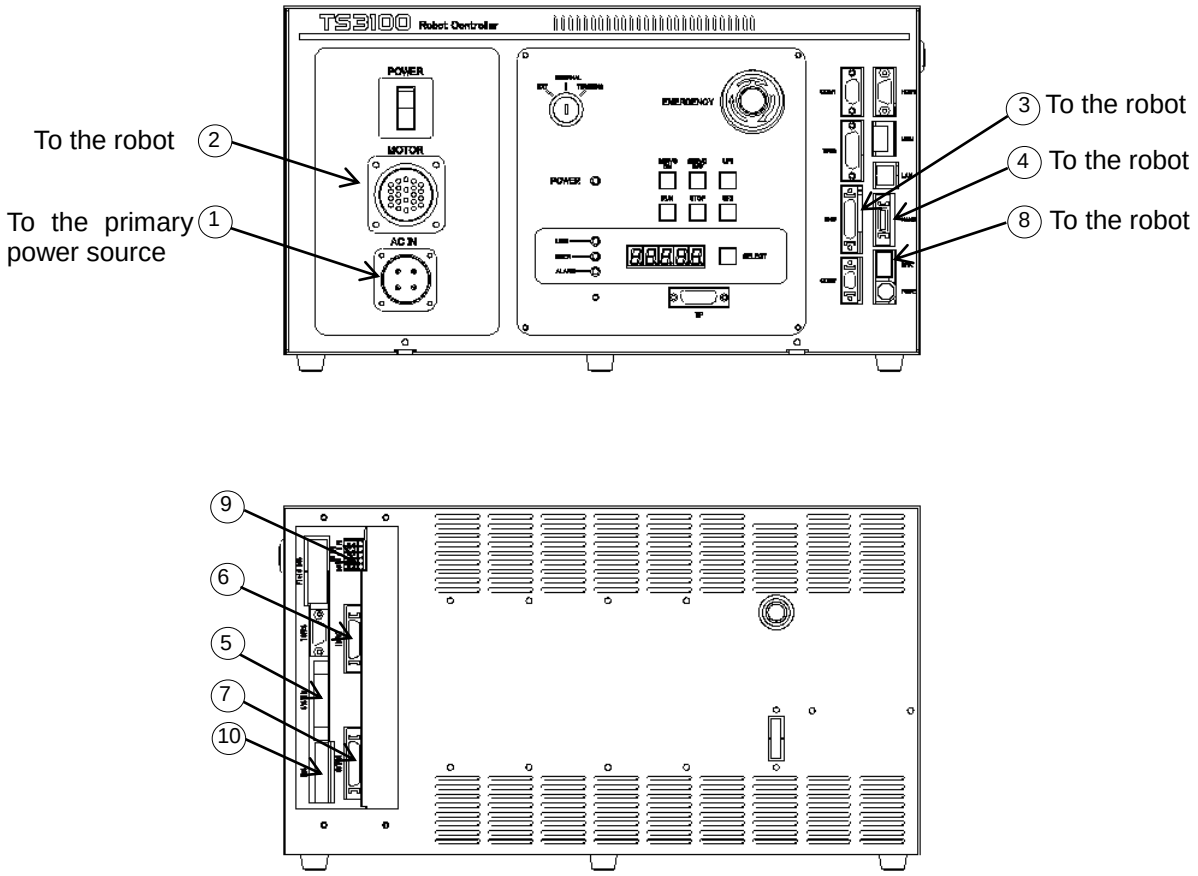


Fig. 9.1 Robot controller connector arrangement (TS3100 and TS3100E)

① Power cable (ACIN)
② Motor cable (MOTOR)
③ Encoder cable (ENC)
④ Robot control signal cable (HAND)
⑤ External operation input signal cables (SYSTEM)
⑥ External operation input signal cables (INPUT)
⑦ External operation input signal cables (OUTPUT)
⑧ Brake signal cable (BRK)
⑨ Distribution I/O cable (EXT-I/O)
⑩External input/output power cable "EMS"

In the subsequent paragraphs, we explain how to connect cables ① to ④ and ⑩ inclusive. For information on how to connect cables ⑤ to ⑨, refer to the TS3000 Interface Manual STE80729.

9.1.2 Connecting the Power Cable "ACIN" (① of Fig. 9.1) (plug connector attached)

The power cable is used to supply the main AC power to the controller.

Table 9.1 Power supply specifications (TS3100 and TS3100E)

Item	Specifications
Power supply	Single phase, 200 - 240 VAC, 50/60 Hz ±1 Hz
Power capacity	3.8 kVA
Instantaneous power failure	Within 40 msec
Grounding	JIS class D

The connector is ACIN (① of Fig. 9.1).

ACIN plug connector Type:JL04V–6A22–22SE–EB-R Maker: Japan Aviation Electronics Industry, Ltd

ACIN cable clamp Type: JL04–2022CK(14)-R Maker: Japan Aviation Electronics Industry, Ltd

Wire 3.5 mm^2 - 5.5 mm^2

As the cable is not an accessory, use the attached plug connector connected to ACIN on the controller side to manufacture a cable.
Wires are to be soldered to the connector.

! DANGER

- Be sure to use the designated wire. Failure to do so could lead to fires or faults.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

For the terminal arrangement, see Para. 9.1.9.

! CAUTION

- Unless the main power is normally supplied to the controller due to phase defect or voltage drop, an error of "8–027 Slow Charge error" occurs. When this happens, make sure that the maser power voltage at the controller power connector satisfies the specified input power of the controller, and that the same voltage is stabilized.
- For details of the 8–027 error, see Para. 13.7 of the TS3000 Operator's Manual ST80729.

9.1.3 Connecting Motor Cable "MOTOR" (② of Fig. 9.1, Fig. 9.2 CN2; Cable included)

The motor cable connects the controller and robot, and supplies the power required to rotate the motor from the controller servo driver to each axis motor of the robot.
The controller connector is MOTOR; the robot connector is CN2.

9.1.4 Connecting Encoder Cable "ENC" (③ of Fig. 9.1, Fig. 9.2 CN3; Cable included)

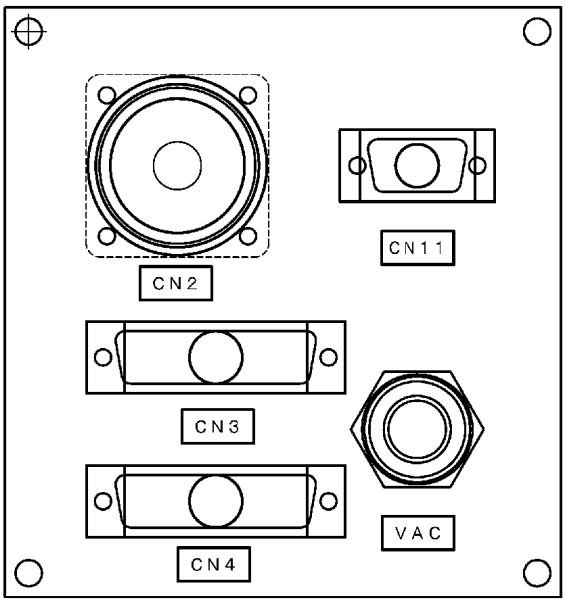
The encoder cable is a signal line used to transmit data from each axis encoder (i.e., device used to detect the revolution (and thereby the position) of each axis) to the controller.
The controller connector is ENC; the robot connector is CN3.

9.1.5 Connecting Robot Control Signal Cable "HAND" (④ of Fig. 9.1, Fig. 9.2 CN4; Cable included)

The robot control signal cable is used to input/output the robot control signal for robot operation (e.g., hand operation).
The controller connector is HAND; the robot connector is CN4.
See the specifications for the control signals.

9.1.6 Connecting Brake Signal Cable between "BRK" (⑧ of Fig. 9.1, Fig. 9.2 CN4; Cable included)

The brake signal cable is used to turn ON/OFF the brake for locking the motor shaft.
The controller connector is BRK; the robot connector is CN4.



Standard specifications

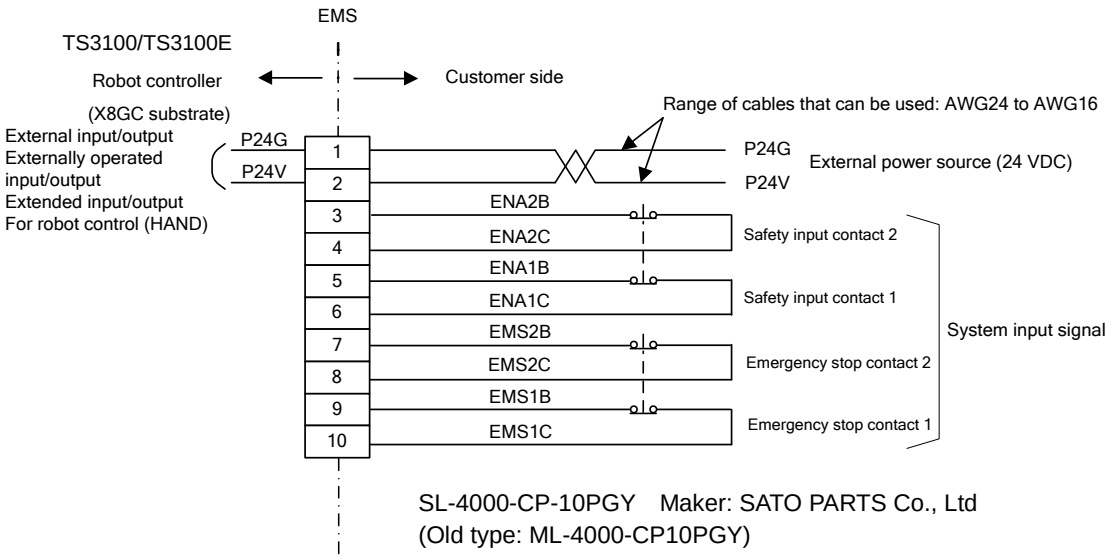
Fig. 9.2 Robot connector layout diagrams (SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)

9.1.7 Connecting the External Input/Output Power Cable "EMS" (⑩ of Fig. 9.1)

Use the attached connector (SL-4000-CP-10PGYT) to connect the external input/output power cable, similarly to the case of connecting the input signal cable for safety. The P24V power is supplied from the connector "EMS" on the rear of the controller.
The following shows the I/O using the external supply power (24 VDC). Be sure to supply power (24 VDC) from the outside.

- External input/output
- Externally operated input/output
- Extended input/output
- Manual input/output

It should be noted that the controller internal power (24 VDC) is used for the trigger input and brake output.



SL-4000-CP-10PGY Maker: SATO PARTS Co., Ltd
(Old type: ML-4000-CP10PGY)

The connector EMS compatible cables are "AWG24 through 16").
Select the optimum external power source in conformity to the customer's system specifications (power capacity).
To connect and disconnect the connector EMS, see the description of "Interface Part".

! CAUTION

- Be sure to supply external power (24 VDC). If not, power cannot be turned on, independently of safety measures.

9.1.8 Connecting and Disconnecting Cables

! CAUTION

- Before connecting or disconnecting any controller cable, be sure to turn off the main power ("POWER") switch.
- When disconnecting a cable, be sure to pull the plug and not the cord. Otherwise, you may damage the cable.

a) Circular connectors: ACIN, MOTOR
Firstly align the key position, and completely insert the connector on the cable side into the controller connector. Then turn the cable side lock screw to the right to clamp the cable. A loose screw can cause a contact failure or other accident. To avoid this, make sure that the screw is clamped completely.
To disconnect the connectors, turn the lock screw to the left and pull out the cable side connector.

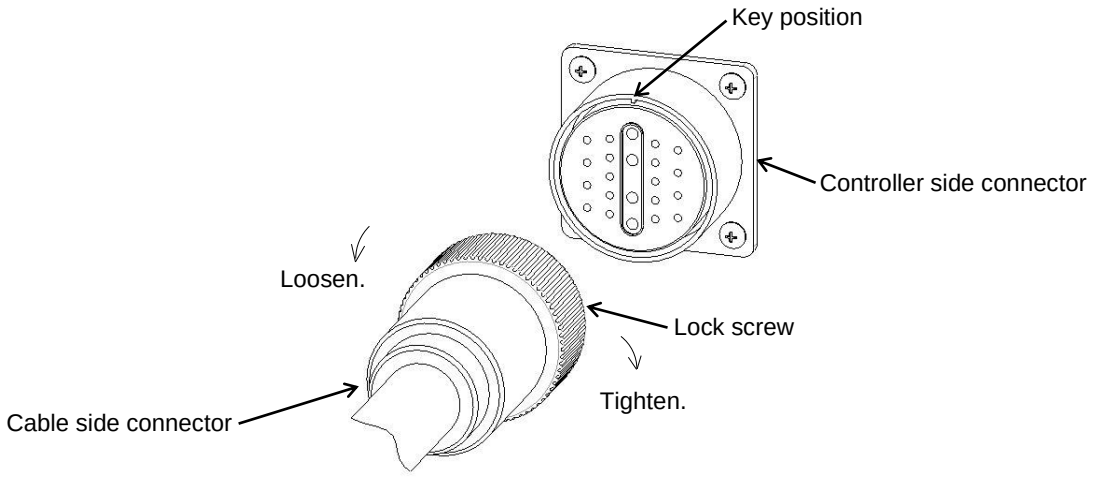


Fig. 9.3 Connecting and disconnecting a circular connector

b) Square connectors: ENC, HAND, SYSTEM, INPUT, OUTPUT, TRIG, CONV
Firstly, completely insert the cable side connector into the controller connector. Then tighten the lock screws on both ends of the cable side connector with a screwdriver. A loose screw can cause a contact failure or other accident. To avoid this, make sure that the screws are clamped completely.
To disconnect the connectors, first loosen the lock screws, then pull out the cable side connector. INPUT, OUTPUT is a quick-operated lock type connector, however.

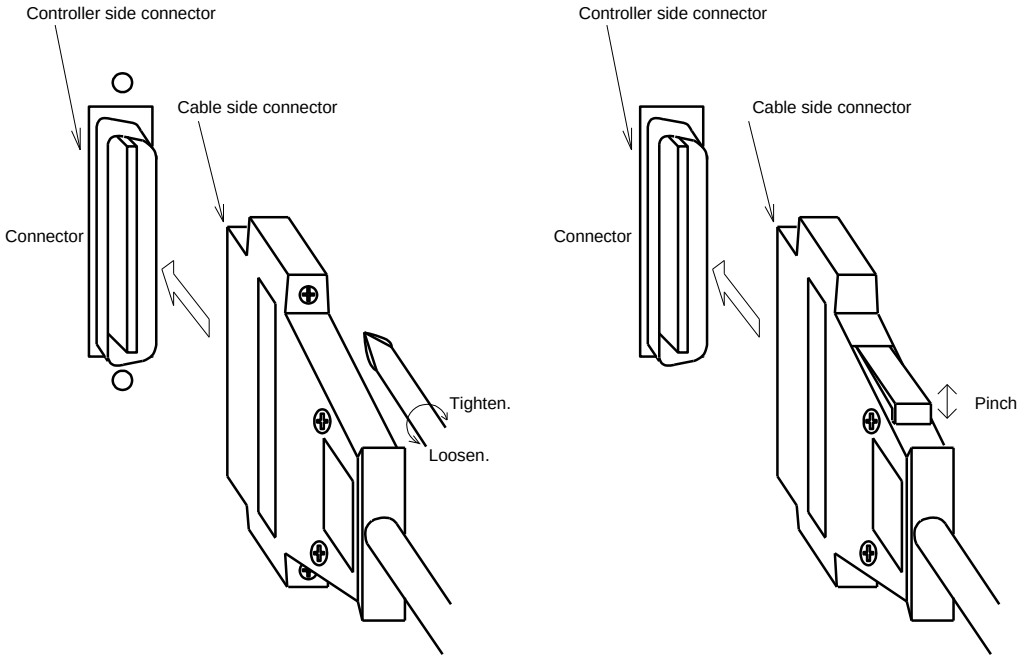
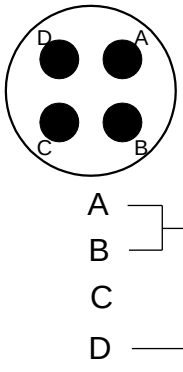


Fig. 9.4 Connecting and disconnecting a square connector

9.1.9 Examples of Connector Terminal Arrangement (Controller Side)

a) Power cable connector ACIN



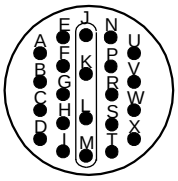
Connects to controller.
Type: JL04HV–2E22–22PE–B
Manufacturer:
Japan Aviation Electronics Industry, Ltd
Single phase, 200 - 240 VAC, 50/60 Hz

Grounding (Class D)

! DANGER

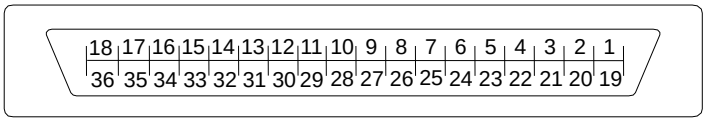
- Completely connect the grounding cable.
  Otherwise, an electric shock or fire may be caused if a fault or electric leak occurred. Or misoperation may be caused by noise.

b) Motor cable connector MOTOR



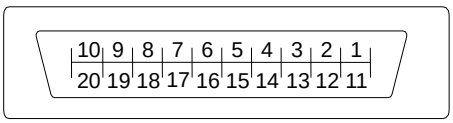
Connects to controller.
Type: JL04V-2A28-11SE
Manufacturer:
Japan Aviation Electronics Industry, Ltd

c) Encoder cable connector ENC



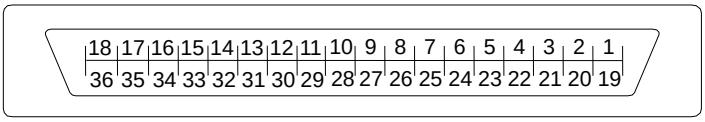
Connects to controller.
Type: 52986-3659
Manufacturer: MOLEX

d) Robot control signal cable connector HAND



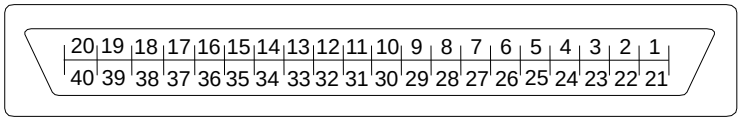
Connects to controller.
Type: 52986-2079
Manufacturer: MOLEX

e) General-purpose input signal cable connector INPUT



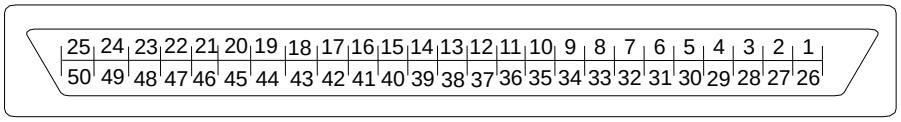
Connects to controller.
Type: DHA-RC36-R132N-FA
Manufacturer: DDK

f) General-purpose output signal cable connector OUTPUT



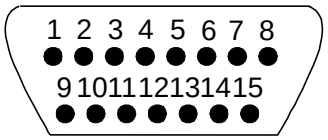
Connects to controller.
Type: DHA-RC40-R132N-FA
Manufacturer: DDK

g) System input/output signal cable connector SYSTEM



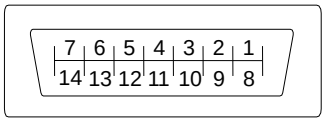
Connects to controller.
Type: 52986-5079
Manufacturer: MOLEX

h) Trigger input connector TRIG



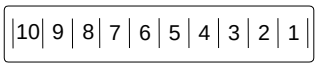
Connects to controller.
Type: XM2C-1542-112L
Manufacturer: OMRON

i) Encoder cable connector CONV



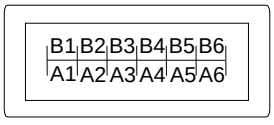
Connects to controller.
Type: 52986-1479
Manufacturer: MOLEX

j) Emergency stop, safety input, and external P24V supply connector EMS



Connects to controller.
Type: SL-4000CWJH-10PGY
Manufacturer: SATO PARTS

k) Brake connector BRK



Connects to controller.
Type: 1-1827876-6
Manufacturer:
Tyco Electronics AMP

9.2 Controller Connector Signals

9.2.1 Connector Signal Connection Diagrams

Diagrams showing which signals correspond to which terminals are shown in Section 2 of the TS3000 Interface Manual STE80729.

9.2.2 Jumpers for Safety Related Signals

The following system input signals are provided to serve for the safety purpose.

System input signals SYSTEM-12 (STOP)
SYSTEM-16 (SVOFF)
SYSTEM-14 (BREAK)
EMS-7, 8 (EMS2B - EMS2C)
EMS-9, 10 (EMS1B - EMS1C)
EMS-3, 4 (ENA2B - ENA2C)
EMA-5, 6 (ENA1B - ENA1C)

These signals are already jumpered for the connectors provided for the TS3100 and TS3100E robot controller. If you wish to use or change them, therefore, you should remove the jumpers and rewire as appropriate. If you plan to use the robot without using system input signals, be sure to connect the attached connectors to the controller side SYSTEM, EMS connectors.
Unless the following signals are used as the system signals, jumper them also.
SYSTEM-15 (LOW_SPD)
SYSTEM-13 (CYCLE)

Table 9.2 Connector jumpers

SYSTEM		EMS	
12-17 (18)	14-17 (18)	3-4	5-6
16-17 (18)	(13-17 (18))	7-8	9-10
(15-17 (18))	-		

! CAUTION

- Unless the signals of SVOFF and emergency stop contacts 1, 2 are jumpered, the controller servo power cannot be turned on.
- Unless the CYCLE signal is jumpered, the controller enters the cycle operation mode.
- Unless the LOW_SPD signal is jumpered, the robot is operated at low speed during automatic operation.
- Unless the STOP signal is jumpered, automatic operation of the robot is not possible.
- Unless the BREAK signal is jumpered, automatic operation of the robot is not possible.

9.3 Separating Control Panel from Controller

9.3.1 Removing Control Panel

Remove the control panel in the following manner.

a) Loosen the four (4) screws at the four (4) corners, which secure the control panel.
b) Remove these screws, then carefully draw out the control panel toward your side.
   Caution: Be careful of the cable connected on the rear side.

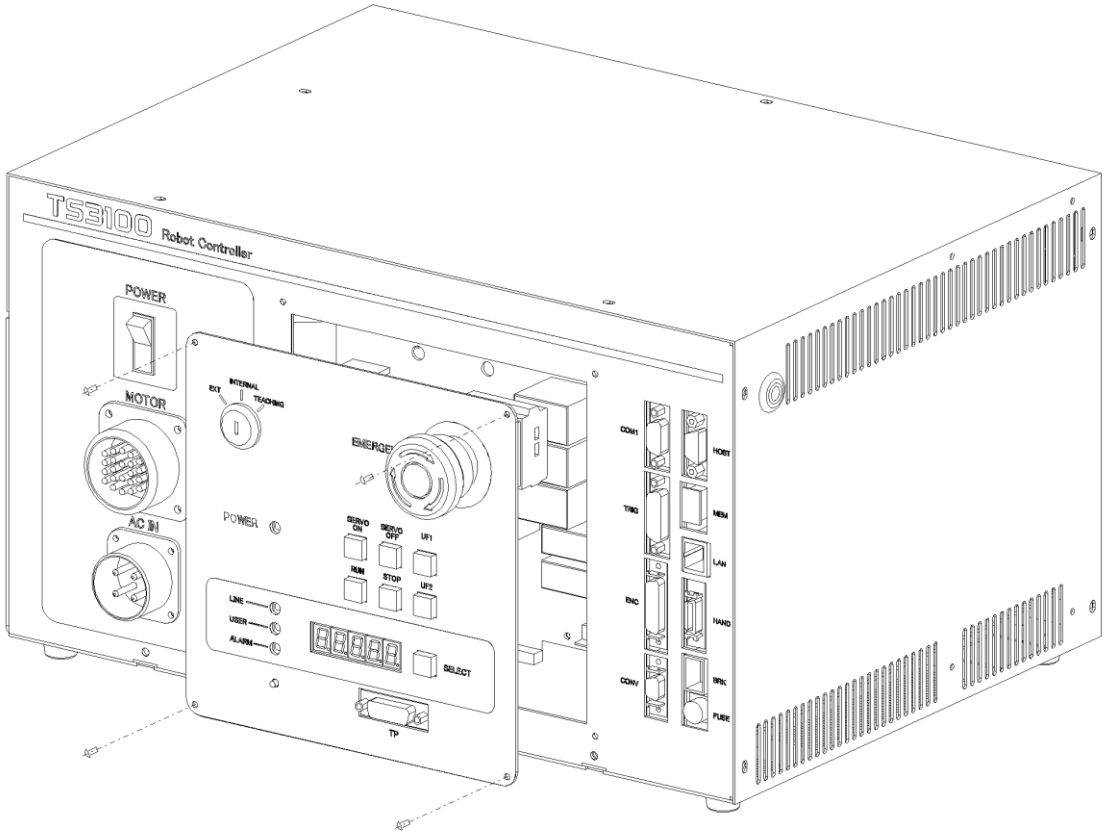


Fig. 9.5 Removing control panel (TS3100 and TS3100E)

9.3.2 Cable between Controller and Control Panel (option)

The cable required to connect the control panel and controller when they are installed separately can be provided optionally.

9.3.3 Control Panel Mounting Dimensions

The dimensions of mounting the control panel are shown in Fig. 9.6.
Cross truss head screws (ϕ3 x 6, ZN3–B) are used.

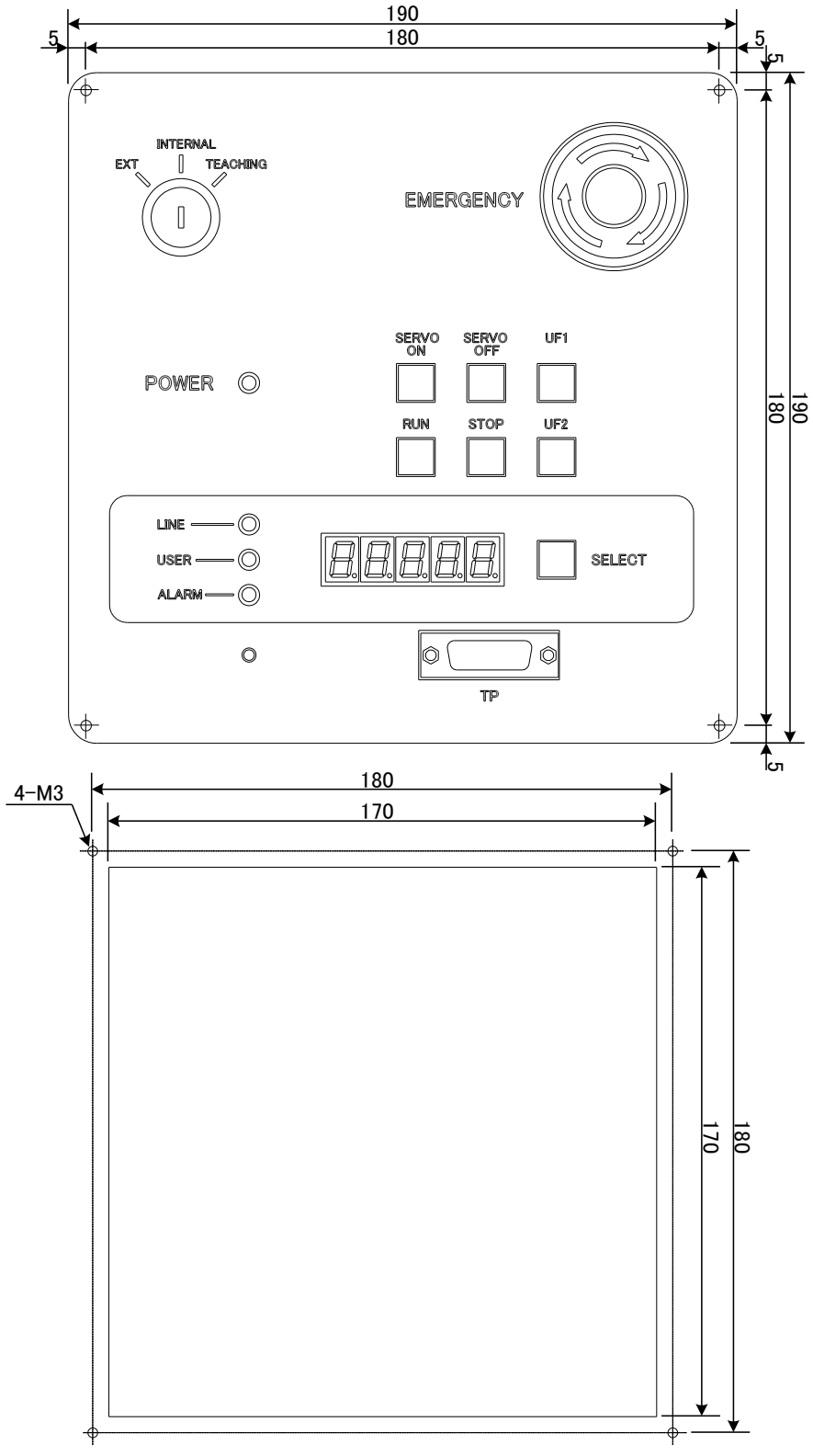


Fig. 9.6 Control panel mounting dimensions

9.3.4 Mounting Dummy Panel on Controller (option)

When the control panel has been disengaged from the controller, mount a dummy panel on the place where the control panel was set before, as shown in Fig. 9.7. The dummy panel, mounting parts, etc. are provided optionally.

a) Connect the cable connector which was disconnected when separating the controller from the control panel, to the rear side of the dummy panel, then screw both ends of the connector. When mounting the connector, use the cross truss screws (ϕ3 x 6, ZN3–B).

b) Screw the dummy panel into the controller.

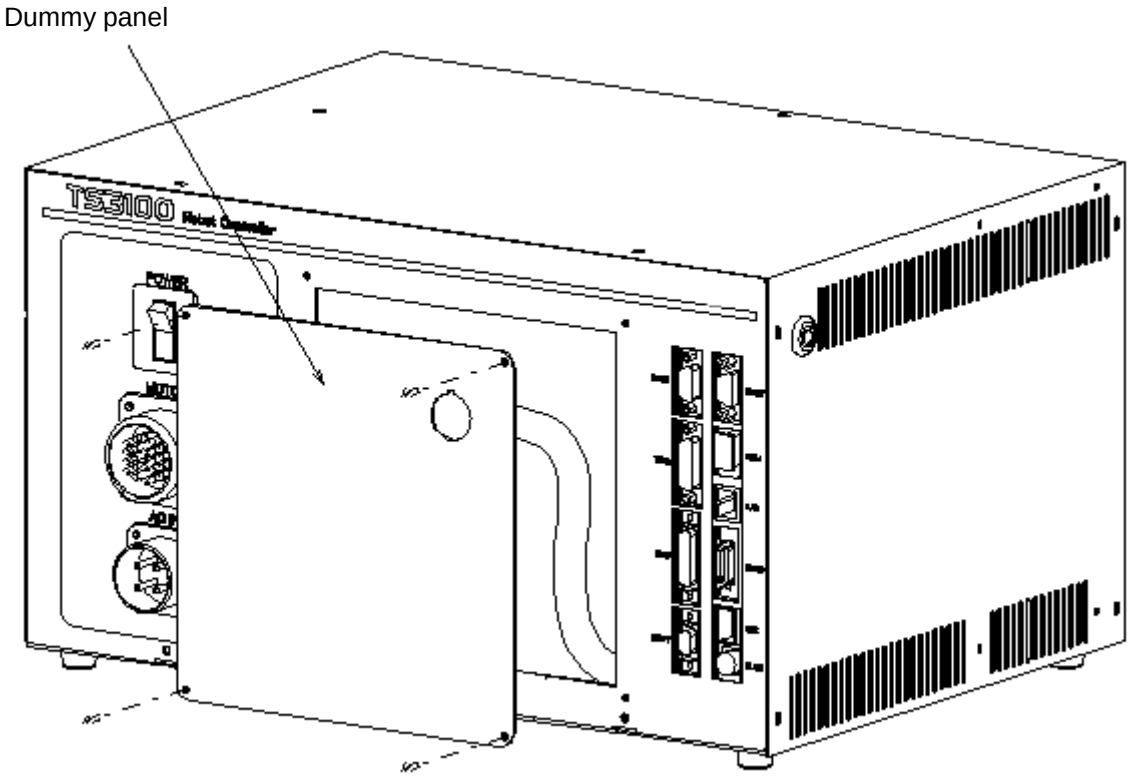


Fig. 9.7 Mounting dummy panel (TS3100 and TS3100E)

9.3.5 Dimensions when Separating Control Panel

Fig. 9.8 shows the connections of the control panel and dummy panel. Provide a clearance of 50 mm or more (with cover, 60 mm or more) on the rear side of the separated control panel.
When the cable is connected to the dummy panel of the controller, provide a clearance of 80 mm or so in front of the controller as the cable connector sticks out of the panel surface.

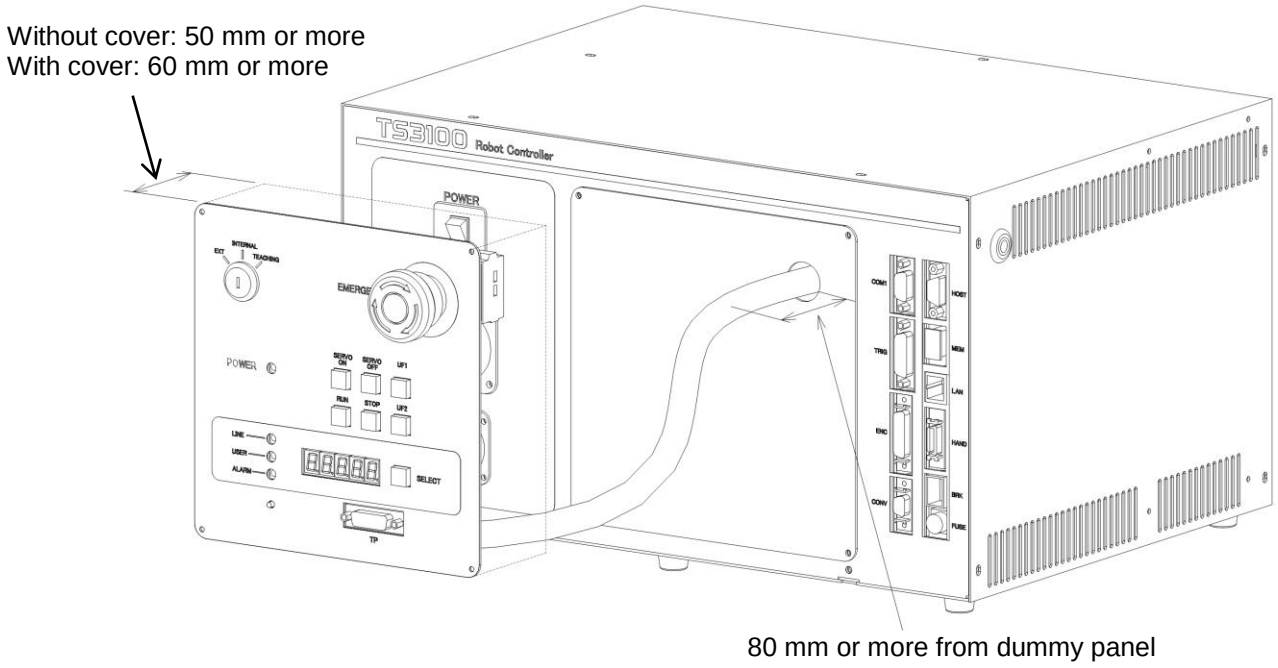
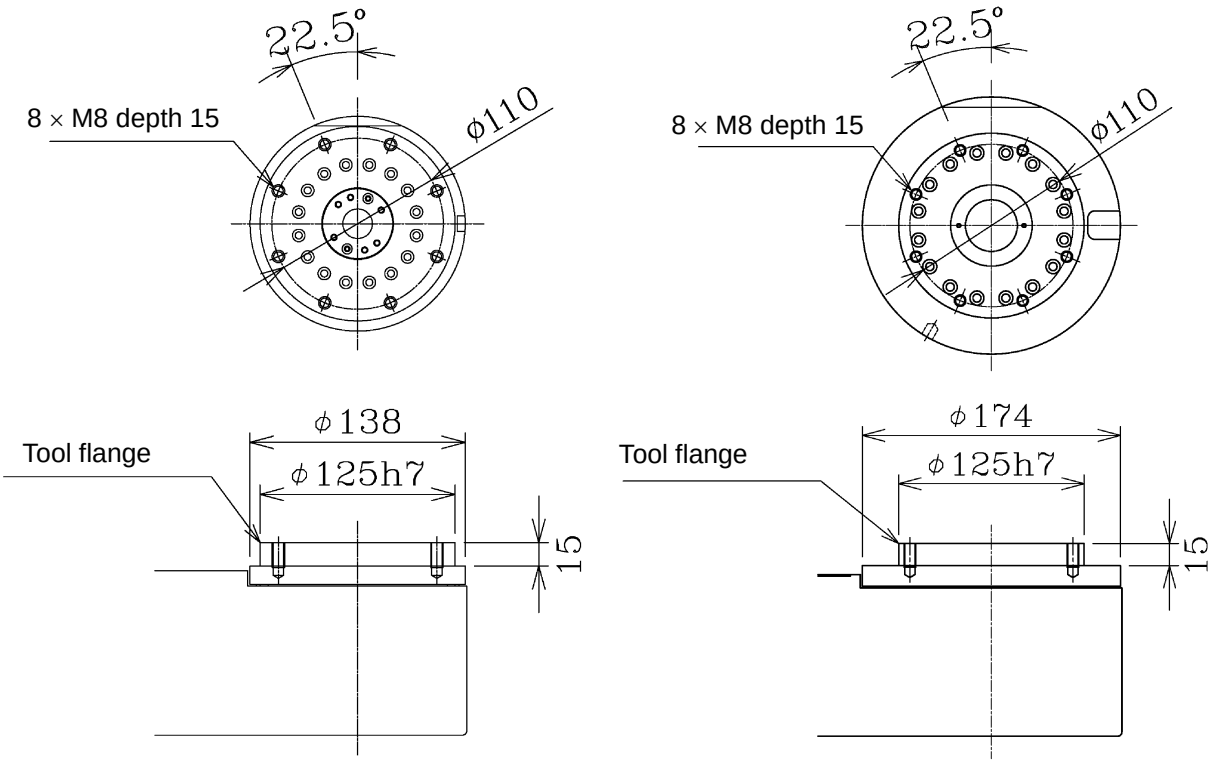


Fig. 9.8 Cable connections of dummy panel and control panel (TS3100 and TS3100E)

10. Hand Interface

10.1 Hand Mounting Section

Fig. 10.1 shows the dimensions of the hand mounting section.



SR-404HC, SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L

SR-624HC

Fig. 10.1 Dimensions of the hand mounting section

10.2 Hand Signal Cable

See the specifications for the hand signal cable.

10.3 Piping for Robot Scavenging

The inside of the robot needs to be scavenged in order to maintain the cleanness of the robot. To perform scavenging, provide the primary piping from the ϕ10 quick-operated joint on the robot panel in Figs. 10.2 and 10.3. The scavenging flow rate is about 250 liters per minute.

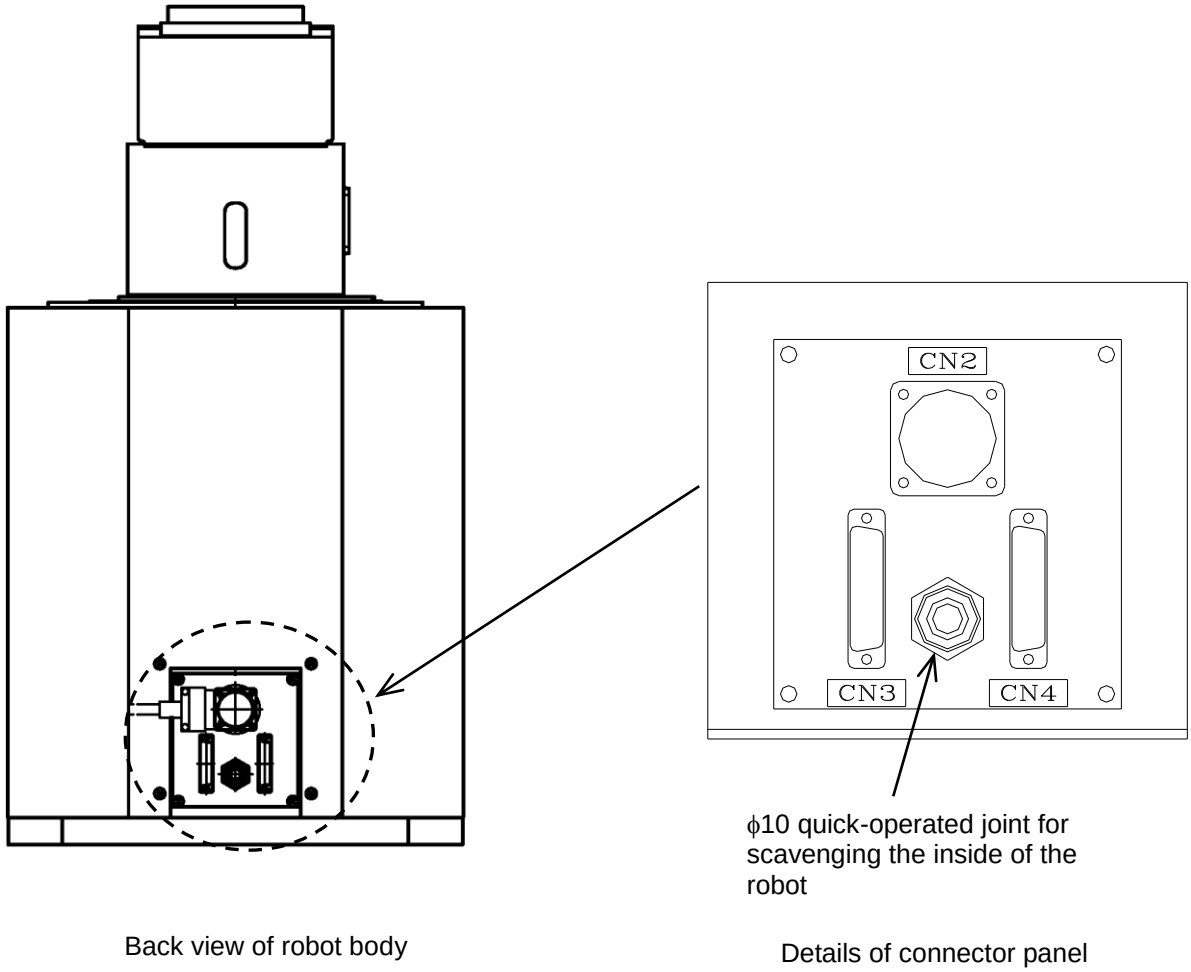


Fig. 10.2 Piping for Robot Scavenging (for SR-404HC and SR-624HC)

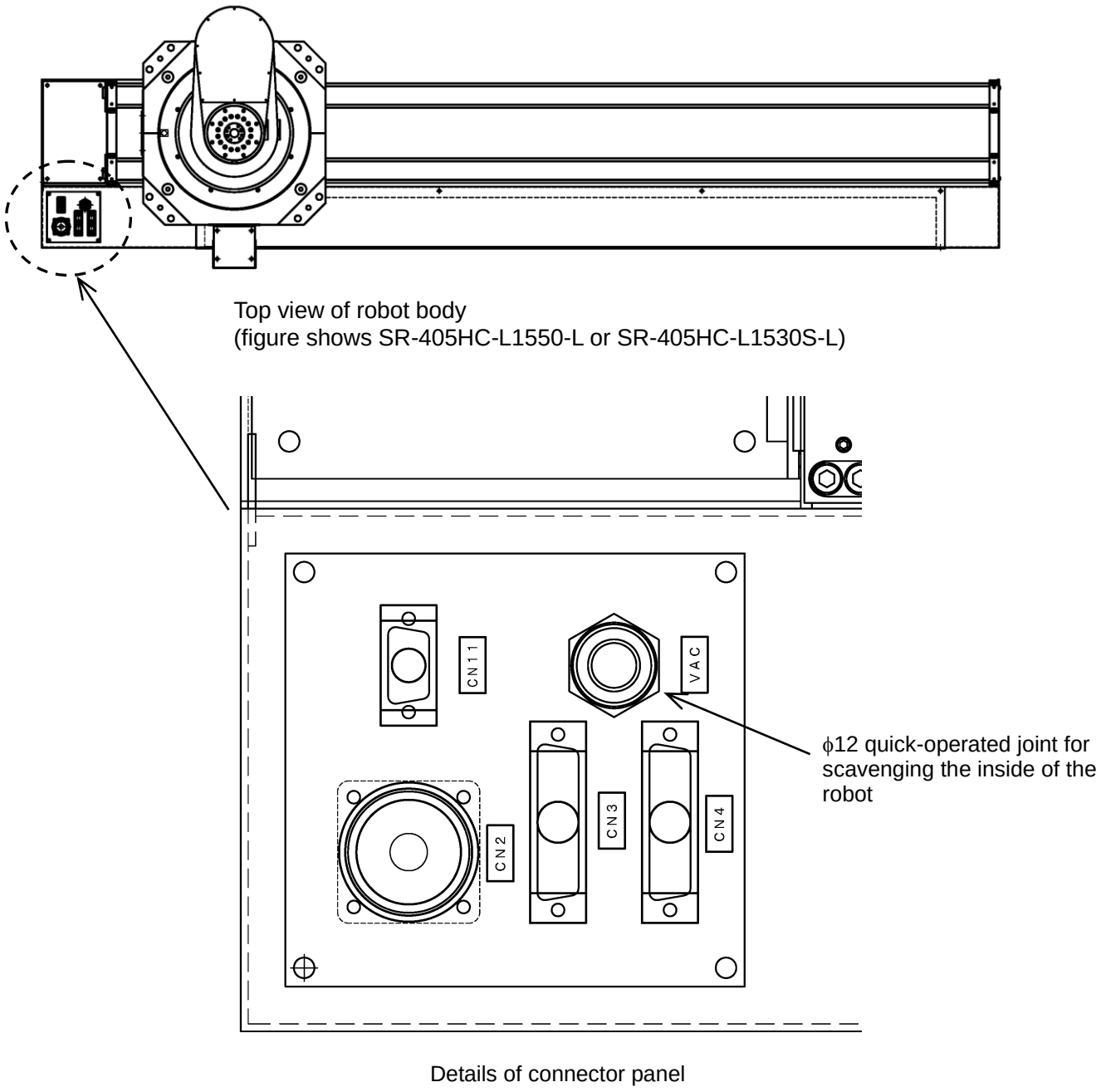


Fig. 10.3 Piping for Robot Scavenging (for SR-405HC-L1550-R/L and SR-405HC-L1530S-R/L)